

Unit - I

Internet Architecture and Network Layer

(marks 12)

- Structure of Internet, Intranet, Role of Internet Service Provider (ISP) and Internet Corporation for Assigned Names and Numbers (ICANN)
- IPv4-Header format, IPv6 -Header format
- Subnet, subnet addressing and address masking, supernetting
- Address Mapping- Address Resolution Protocol (ARP) - Mapping logical to physical addresses, working and message format , Reverse Address Resolution Protocol (RARP) – Mapping physical to logical addresses working and message format

Structure of Internet

Today an Internet has a multi-backbone structure and these backbone networks run by different private corporations today.

- Several backbone networks, operated by private communication companies, provide global internet connectivity. These backbones are interconnected through peering points, enabling data exchange between them.
- Below the backbone level, provider networks utilize these backbone connections to offer internet services to end-users and organizations.
- At the lowest level, customer networks access the internet through the services offered by provider networks.

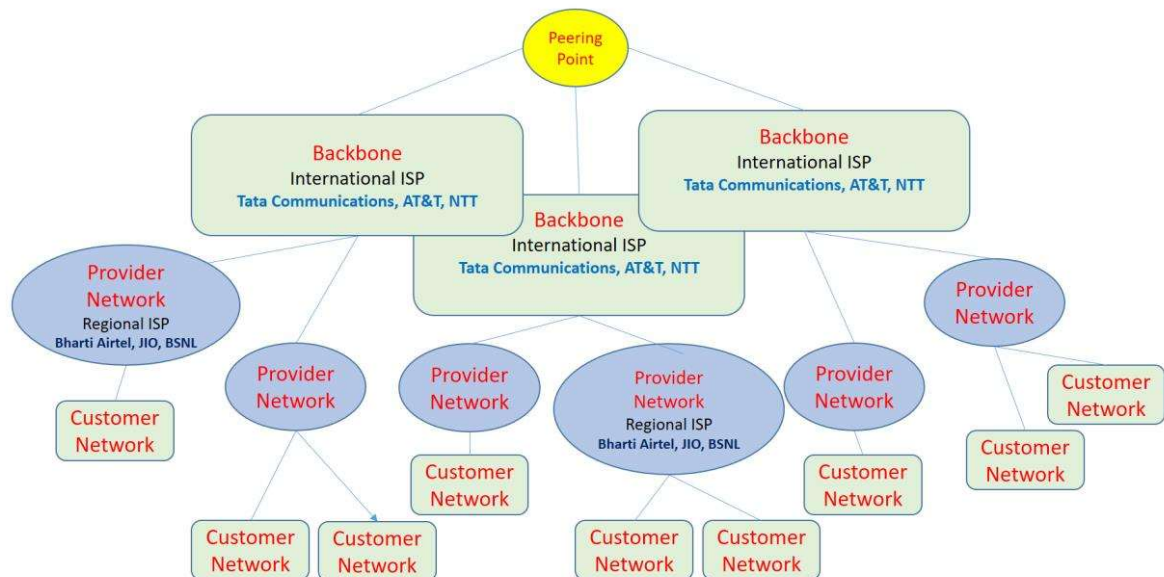


Fig1.1. Structure of Internet

Intranet

- A **private internal network within an organization** that uses TCP/IP protocol suite for communication.
- Uses internet technologies but restricted to internal access only.
- Provides services like internal email transfer, document sharing, internal employee chat system etc.

Role of ISP (Internet Service Provider)

- Provides internet access to users and businesses.
- Assigns public IP addresses.
- Offers services like email, domain hosting, and technical support.

- Any of these three entities (backbone, provider network, or customer network) can be called an Internet Service Provider or ISP. They provide services, but at different levels.

ICANN (Internet Corporation for Assigned Name and Numbers)

- Manages domain names, IP address allocation, and protocol identifiers.
- Ensures uniqueness and coordination of DNS across the internet.

Let's now understand the role of ICANN in the domain name registration process.

- Select a unique and relevant name for your website.
- Choose Top Level Domain based on your business type or region.
- Select a Domain Registrar
 - A domain registrar is a company accredited by ICANN to register domain names.
 - Popular registrars: GoDaddy, Namecheap, Google Domains, Bluehost.
- Go to the registrar's website.
 - Provide your contact details (name, email, phone, address).
 - These go into the WHOIS database (can be protected with privacy settings).
- Once payment is done, the registrar.
 - Registers the domain with ICANN
 - Updates the WHOIS record
 - Confirms registration via email

IPv4 (Internet Protocol version 4)

- Maximum size of IPv4 datagram : 65536B
- Maximum size of Header in IPv4: 60B
- Minimum size of Header in IPv4 : 20B

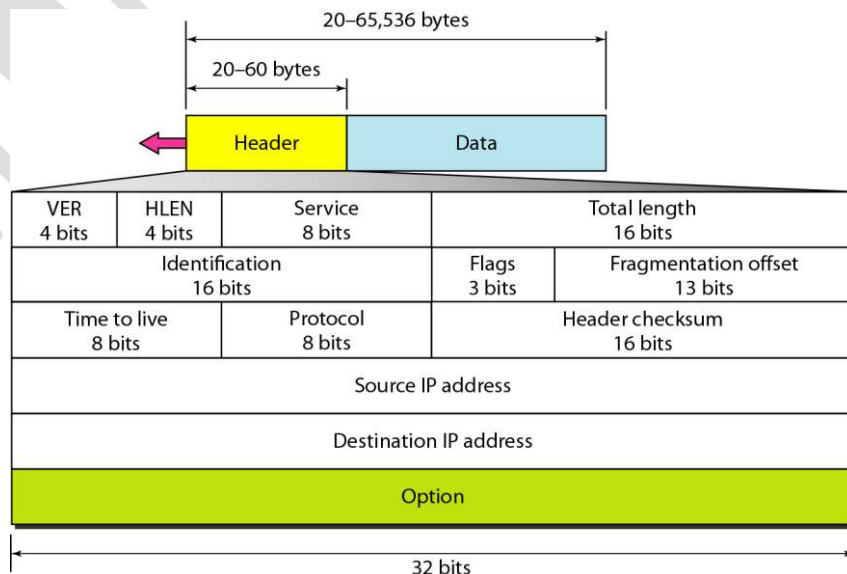


Fig. 1.2 IPv4 Datagram Header

IPV4 DATAGRAM HEADER FIELDS

VER (4 bits):

This 4-bit field defines the version of the IPv4 protocol. Currently the version is 4. However, version 6 (or IPng) may totally replace version 4 in the future.

HLEN (4 bits):

This 4-bit field defines the total length of the datagram header in 4-byte words. This field is needed because the length of the header is variable (between 20 and 60 bytes).

When there are no options, the header length is 20 bytes, and the value of this field is 5

($5 \times 4 = 20$). When the option field is at its maximum size, the value of this field is 15 ($15 \times 4 = 60$).

Total Length (16 bits):

Length in bytes of the datagram, including headers. Maximum datagram size is 65536 bytes.

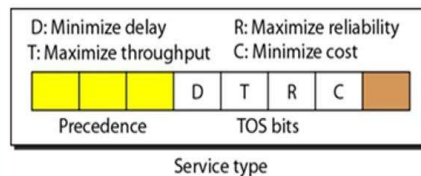
Service Type (8 bits):

IETF has changed the interpretation and name of this 8-bit field. This field, previously called *service type*, is now called *differentiated services*.

SERVICE TYPE:

- First three bits denote PRECEDENCE (priority) of a packet. Ranging from 0 (000) to 7 (111) higher number have highest priority. Packets having lowest priority discarded first in case of congestion.
- Next 4 bits denotes service type. Description for these 4 bits shown in following table
- Last one bit is not used.

TOS Bits	Description
0000	Normal (default)
0001	Minimize cost
0010	Maximize reliability
0100	Maximize throughput
1000	Minimize delay



value of 1111 (this might indicate low delay, high throughput, high reliability and minimized cost)

Time to Live (8 bits):

Prevents a packet from traveling forever in a loop. Senders sets a value, which is decremented at each hop. If it reaches zero, packet is discarded.

Protocol:

Defines the higher level protocol that uses the service of the IP layer

Header Checksum (16 bits):

Datagram is discarded in case of mismatch in checksum values

Source IP address (32 bits): Internet address of the sender.

Destination IP address (32 bits): Internet address of the destination.

Identification, Flags, Fragment Offset:

Used for handling fragmentation. (Discussed in detail later)

Options (variable width):

Can be used to provide more functionality to the IP datagram

IPv6 Packet Format

- The Ipv6 packet format is shown in figure-1.3. Each packet is composed of a mandatory base header followed by the payload.
- The payload consists of two parts: optional extension headers and data from an upper layer.
- The base header occupies 40 bytes, whereas the extension headers and data from upper layer contain up to 65,535 bytes of information.

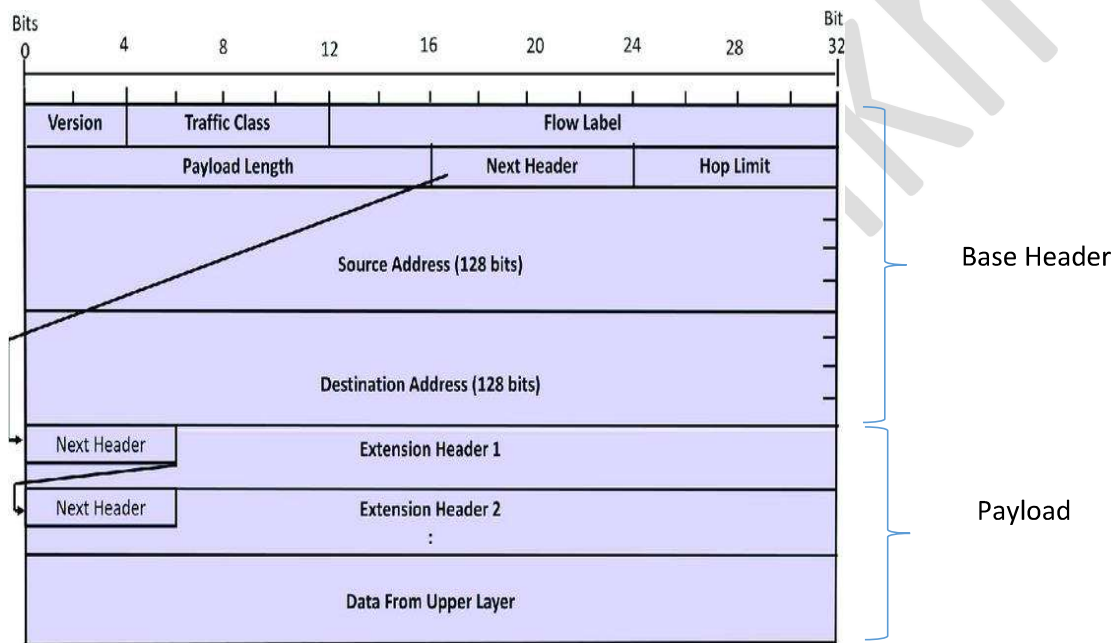


Fig. 1.3 IPv6 Packet Format

IPv6 Header Fields Description

Version (4-bits):

It represents the version of Internet Protocol, i.e. 0110.

Traffic Class (8-bits):

These 8 bits are divided into two parts. The most significant 6 bits are used for Type of Service to let the Router Known what services should be provided to this packet. The least significant 2 bits are used for Explicit Congestion Notification (ECN).

Flow Label (20-bits):

This label is used to maintain the sequential flow of the packets belonging to a communication. The source labels the sequence to help the router identify that a

particular packet belongs to a specific flow of information. This field helps avoid re-ordering of data packets. It is designed for streaming/real-time media.

Payload Length (16-bits):

This field is used to tell the routers how much information a particular packet contains in its payload. Payload is composed of Extension Headers and Upper Layer data. With 16 bits, up to 65535 bytes can be indicated; but if the Extension Headers contain Hop-by-Hop Extension Header, then the payload may exceed 65535 bytes and this field is set to 0.

Next Header (8-bits):

This field is used to indicate either the type of Extension Header, or if the Extension Header is not present then it indicates the Upper Layer PDU. The values for the type of Upper Layer PDU are same as IPv4's.

Hop Limit (8-bits):

This field is used to stop packet to loop in the network infinitely. This is same as TTL in IPv4. The value of Hop Limit field is decremented by 1 as it passes a link (router/hop). When the field reaches 0 the packet is discarded.

Source Address (128-bits):

This field indicates the address of originator of the packet.

Destination Address (128-bits):

This field provides the address of intended recipient of the packet.

Subnetting

Subnetting is the process of dividing a large network into smaller, more manageable subnetworks (subnets).

Why Subnetting ?

- To improve network performance by reducing congestion.
- To enhance security by isolating network segments.
- To efficiently use IP addresses and reduce wastage.
- To simplify network management and troubleshooting.

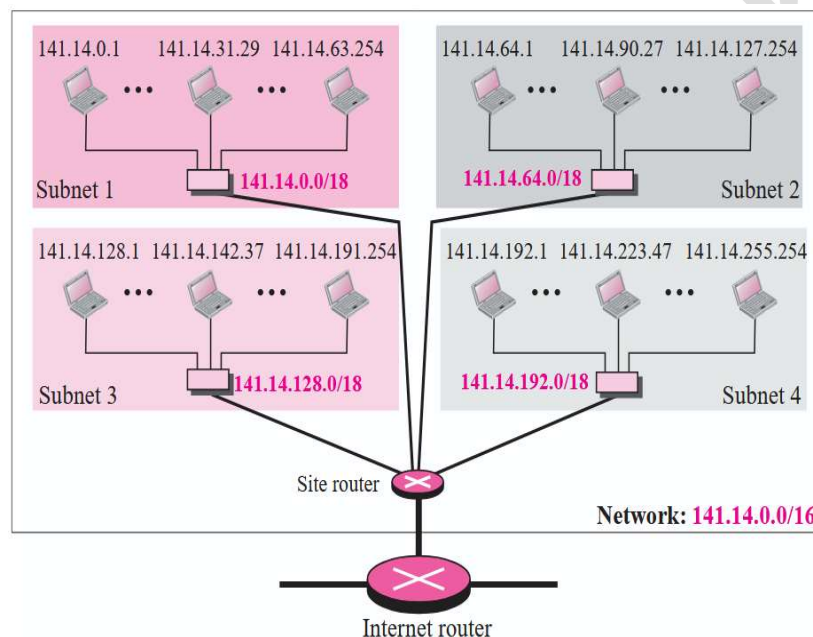


Fig. 1.4: Dividing a large network into small subnets (Subnetting)

Rules to be followed while Designing Subnets

1. The number of addresses in each subnetwork should be a power of 2. This means each subnet must contain 2, 4, 8, 16, 32, 64, 128, etc. IP addresses.
2. The prefix length (CIDR notation) for each subnetwork should be found using the formula:

$$\text{Prefix Length (n}_{\text{sub}}) = 32 - \log_2 (\text{N}_{\text{sub}})$$

where

number of addresses to each subnetwork is N_{sub}

prefix length for each subnetwork is n_{sub}

3. The starting address (last octet) of each subnetwork should be divisible by the number of addresses in that subnetwork.

Problem1

Design 04 equal subnets each of size 62 hosts. The given network address is 192.168.0.1/24.

SOLUTION

- /24 has 256 IPs
- 4 subnets = need 2 bits ($2^2 = 4$)
- New mask = /26 $\rightarrow$ 255.255.255.192
- Each subnet = 64 IPs (62 usable)

Subnet	Network Address	First IP	Last IP	Broadcast Address
1	192.168.1.0/26	192.168.1.1	192.168.1.62	192.168.1.63
2	192.168.1.64/26	192.168.1.65	192.168.1.126	192.168.1.127
3	192.168.1.128/26	192.168.1.129	192.168.1.190	192.168.1.191
4	192.168.1.192/26	192.168.1.193	192.168.1.254	192.168.1.255

Problem 2 (based on VLSM – Variable Sized Subnet Masks)

Design a Network with 3 Subnets (100, 30, and 10 Hosts), Design a network to support :

Subnet A: 100 hosts, Subnet B: 30 hosts, Subnet C: 10 hosts. Use 192.168.1.0/24 as the base network.

SOLUTION

100 hosts

Minimum 7 bit required for hosts addressing, therefore subnet mask will be 192.168.1.0/25 (total 128 IPs)

Network ID : 192.168.1.0 (Starts from the base)

First Valid IP : 192.168.1.1

:

Last Valid IP : 192.168.1.126

Broadcast Address: 192.168.1.127

30 hosts

Minimum 5 bit required for hosts addressing , therefore subnet mask will be 192.168.1.0/27 (total 32 IPs)

Network ID : 192.168.1.128 (Starts right after the previous subnet's broadcast address)

First Valid IP : 192.168.1.129

:

Last Valid IP : 192.168.1.158

Broadcast Address: 192.168.1.159

10 hosts

Minimum 4 bit required for hosts addressing , therefore subnet mask will be 192.168.1.0/28

Network ID : 192.168.1.160 (Starts right after the previous subnet's broadcast address)

First Valid IP : 192.168.1.161

:

Last Valid IP : 192.168.1.174

Broadcast Address: 192.168.1.175

Problem 3

For given below IP Addresses find out

- Network ID
 - First Valid IP
 - Last Valid IP
 - Broadcast Address
 - Block Size
1. 193.168.0.90/25
 2. 180.0.0.80/30
 3. 1.1.1.1/9

SOLUTION

1. 193.168.0.90/25

193.168.0.90/25

net = 25 host = 7

11111111.11111111.11111111.10000000

255.255.255.128

no. of networks = $2^1 = 2$

Block Size = $2^7 = 128$

193.168.0.0

193.168.0.128

Network ID

193.168.0.1

193.168.0.129

First Valid IP

193.168.0.126

193.168.0.254

Last Valid IP

193.168.0.127

193.168.0.255

Broadcast Address

193.168.0.90/25

Net ID = 193.168.0.0

B. Add = 193.168.0.127

193.168.0.140/25

Net ID = 193.168.0.128

B. Add = 193.168.0.255

2. 180.0.0.80/30

180.0.0.1/30

net = 30

host = 2

11111111.11111111.11111111.11111100

255.255.255.252

no. of networks = $2^{14} = 16,384$

Block Size = $2^2 = 4$

180.0.0.80/30

Net ID = 180.0.0.80

B. Add = 180.0.0.83

145.0.0.15/30

Net ID = 145.0.0.12

B. Add = 145.0.0.15

180.0.0.0	180.0.0.4	...	180.0.0.248	180.0.0.252	Net ID
180.0.0.1	180.0.0.5	...	180.0.0.249	180.0.0.253	FVI
180.0.0.2	180.0.0.6	...	180.0.0.250	180.0.0.254	LVI
180.0.0.3	180.0.0.7	...	180.0.0.251	180.0.0.255	Broadcast Add

3. 1.1.1.1/9

Class A Classless Subnetting:
DSM: 11111111.00000000.00000000.00000000

1.1.1.1/9
net = 9, host = 23

11111111.10000000.00000000.00000000
255.128.0.0

no. of networks = $2^1 = 2$
2nd octet block size = $2^7 = 128$

1.0.0.0	1.128.0.0	Net ID
1.0.0.1	1.128.0.1	FVI
1.127.255.254	1.255.255.254	LVI
1.127.255.255	1.255.255.255	Broadcast Add

1.1.1.1/9
net ID = 1.0.0.0
B.Add = 1.127.255.255

9.9.9.9/9
Net ID = 9.0.0.0
B.Add = 9.127.255.255

NOTE: Additional subnetting problems can be found on page 217.

(Reference Book: **Data Communication and Networking with TCP/IP Protocol Suite**)

Address Mapping

- Devices in a network are identified using two types of addresses: logical addresses (IP address) and physical addresses (MAC address in case of Ethernet Protocol). Physical addresses are used at the Data Link layer for node-to-node communication, while logical addresses are used at the Network layer for host-to-host communication.
- This means that delivering a packet to a host or router requires two levels of addressing: logical and physical. To achieve this, we must map a logical address to its corresponding physical address and vice versa, It can be done either by using static mapping or dynamic mapping methods.

Static Mapping

Static mapping involves creating a table that links each logical address to its corresponding physical address. This table is stored on every machine within the network. However, static mapping has certain limitations because physical addresses can change in the following situations:

- A device may replace its Network Interface Card (NIC), which results in a new physical address.
- In some LAN technologies, such as LocalTalk, the physical address may change every time the computer is restarted.

Dynamic Mapping

In dynamic mapping, when a computer knows another computer's logical address (IP address), it **uses a special protocol** to find its physical address (MAC address) or vice versa.

There are two main protocols used for this purpose:

- ARP (Address Resolution Protocol)
- RARP (Reverse Address Resolution Protocol)

Address Resolution Protocol (ARP)

ARP accepts a logical address from the IP protocol maps the address to the corresponding physical address and pass it to the data link layer.

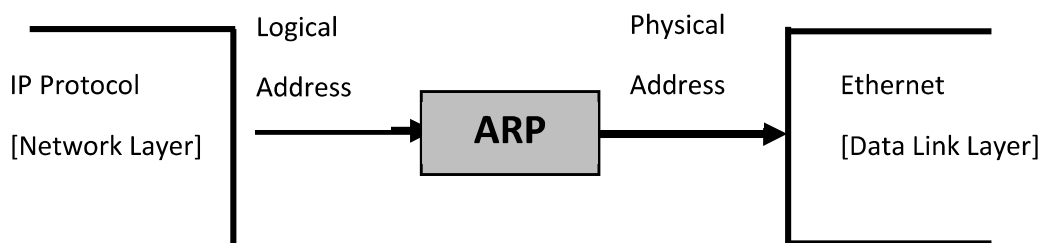


Fig. 1.5: ARP Position

Working

1. ARP Query Packet Broadcast

ARP Query Packet contains

- Senders IP Address
- Senders Physical Address
- Receivers IP Address (Whom Physical Address to be found)

2. Every host or router on the network receives and examines the ARP request packet. However, only the device with the matching IP address recognizes the request and replies by sending an ARP response packet.

3. ARP Response Packet Unicast

The response packet includes the IP address and physical (MAC) address of the replying device. This packet is sent directly (unicast) to the original sender using the physical address provided in the ARP request.

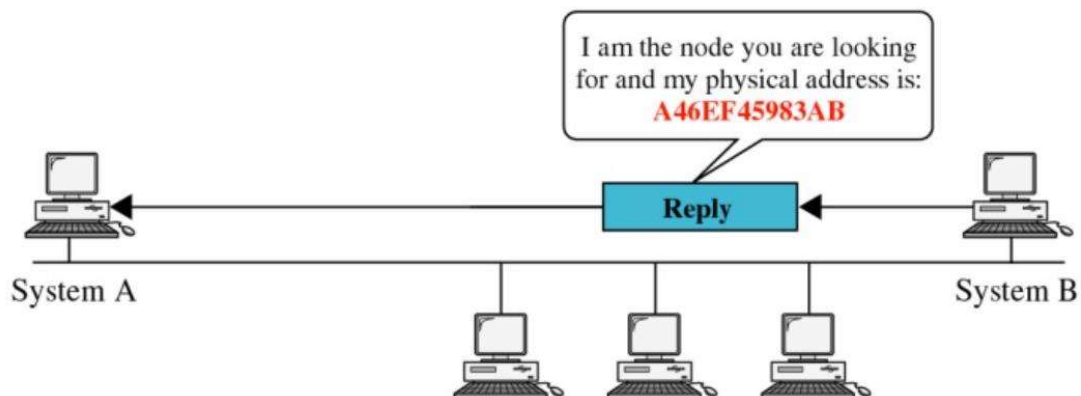
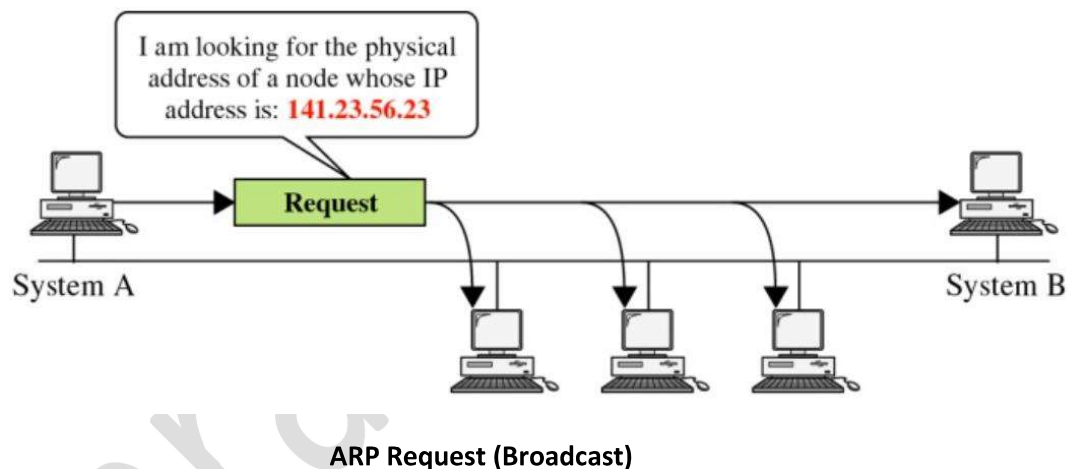


Fig. 1.6 ARP Working

ARP packet format

Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

Fig. 1.7 ARP packet format

Hardware Type (16bit)

It define type of the network ARP is running, e.g. Ethernet and the integer value assigned is 1. Similarly, other hardware types as well assigned with integer number.

Protocol Type (16bit)

This field defines the protocol value, e.g. for IP protocol it is 0800₁₆.

Hardware Length (8bit)

This field defines length of physical address in terms of bytes, e.g. for Ethernet type network it is 48bit means 6 bytes.

Protocol Length (8bit)

This field defines length of logical address in terms of bytes, e.g. for IPv4 protocol it is 32bit means 4 bytes.

Operation Request (16bit)

This field defines the type of message is either a ARP Request (value is 1) or a ARP Response (value is 2).

Sender Hardware Address (variable length)

This field contains sender's physical address. For an Ethernet network, this field is 6 byte long.

Senders Protocol Address (variable length)

This field contains sender's logical address. For an IP protocol, this field is 32bit (4Byte) long.

Target Hardware Address (variable length)

This field contains target's physical address. For an Ethernet network, this field is 6 byte long. In case of ARP Request this field value is all 0's.

Target Protocol Address (variable length)

This field contains target's logical address. For an IPv4 protocol, this field is 4 byte long.

Imran Shaikh

Reverse Address Resolution Protocol

- RARP is a network layer protocol used to map a physical address (MAC address) to a logical address (IP address). RARP is the reverse of ARP.
- Mainly used by diskless computers (which don't have storage to save IP addresses).
- It helps these computers automatically get their IP address from a RARP server at boot time.
- RARP is no longer used today for two main reasons:
 - RARP uses broadcasting, so a RARP server must be present on every local network, which is not efficient.
 - RARP only provides the IP address, but modern computers need more information to connect to a network—such as the subnet mask, default gateway, and DNS server.

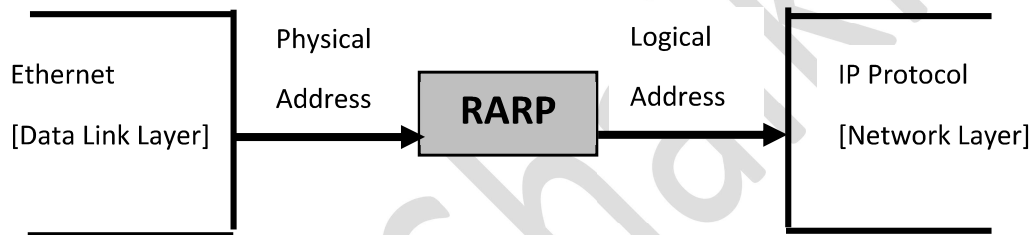
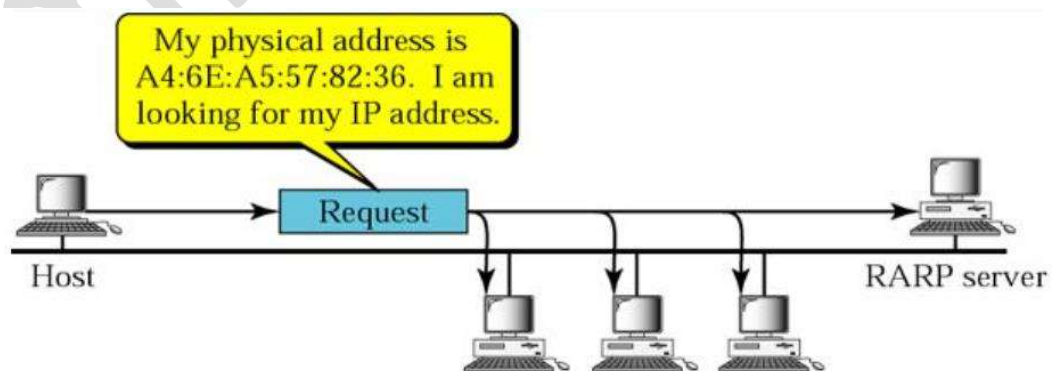


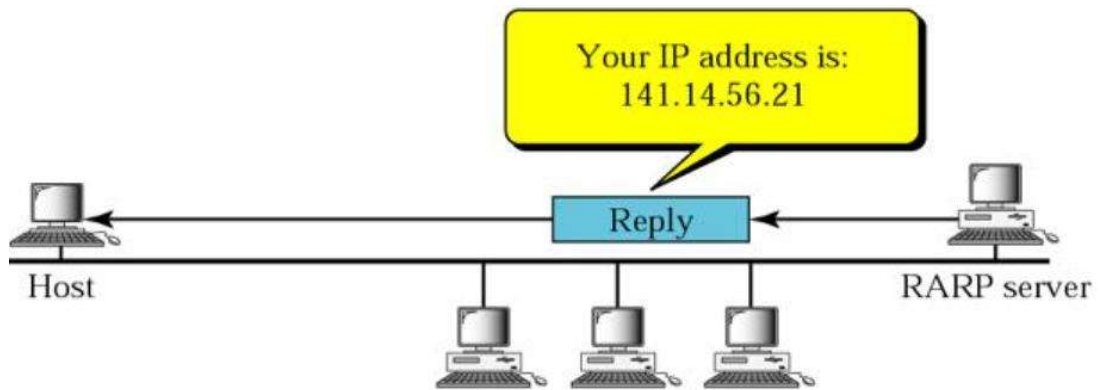
Fig. 1.8 RARP Position

RARP Working

1. A diskless machine knows only its MAC address, not its IP address.
2. When the machine boots up, it sends a RARP request as a broadcast on the network.
3. A RARP server (configured on the same LAN) receives the request and looks up the IP address that matches the MAC address.
4. The server then sends a RARP reply with the correct IP address to the requesting machine.



RARP Request



RARP Response

Fig. 1.9 RARP Working

RARP Packet Format

Hardware type		Protocol type
Hardware length	Protocol length	Operation Request 3, Reply 4
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP) (It is not filled for request)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled for request)		
Target protocol address (For example, 4 bytes for IP) (It is not filled for request)		

Fig. 1.10 RARP packet format

Note:

The description of the fields in the RARP packet is the same as in ARP. The only difference is in the operation (opcode) field, where the value 3 indicates a RARP request and 4 indicates a RARP reply.

Unit - II

Routing Protocols

(Marks 16)

- Router architecture, routing table, queueing and switching
- Routing protocols- Intra domain routing- Distance vector routing-Creating distance vector routing tables, Initialization, Sharing, Updating- Routing Information Protocol (RIPv2), Link State Routing Open Shortest Path First (OSPF)-Types of links, Graphical representation, Inter domain Routing-Path Vector Routing- Border Gateway Protocol (BGPv4)
- Internet Control Message Protocol (ICMP)-Types of messages, Message format, Error reporting messages

Router Architecture

A router is a network device that connects multiple networks and directs data packets between them. Its architecture consists of:

1. **Input Ports**
 - Receives incoming data packets from different networks.
 - Performs error checking, packet classification, and buffering.
2. **Routing Processor**
 - Maintains the routing table and makes forwarding decisions.
 - Handles control plane functions like routing protocols (e.g. OSPF, BGP).
3. **Switching Fabric**
 - Connects input ports to output ports.
 - Transfers packets quickly from one port to another.
 - Can be implemented using buses, shared memory, or crossbar switches.
4. **Output Ports**
 - Queues packets before sending them to the next hop.
 - Handles forwarding, error detection, and flow control.

Routing Table

The routing table stores information about how to reach different network destinations.

It typically includes:

- **Destination Network** – The target network or IP address.
- **Next Hop** – The address of the next router to which the packet should be sent.
- **Metric** – Cost or distance associated with reaching the destination (lower is better).
- **Interface** – The output port through which the packet should be forwarded.

Queueing in Routers

Queueing refers to the process of temporarily storing packets in memory buffers before they are transmitted over an output link. It is an essential function in routers because packets do not always arrive at the same rate as they can be forwarded. If too many packets arrive at once, they need to wait their turn before being sent, which is managed through queues.

Switching in Routers

Switching is the process by which a router transfers incoming packets from the input port to the appropriate output port so that they can be sent toward their destination. It plays a crucial role in ensuring that packets move efficiently and quickly through the router.

Switching is performed using the switching fabric, which connects the input ports and output ports and allows packets to be forwarded based on decisions made by the routing processor.

Autonomous System (AS)

- An autonomous system (AS) is a group of networks and routers under the authority of a single administration.
- Routing **inside an autonomous system** is referred to as **intra-domain routing**.
- Routing **between autonomous systems** is referred to as **inter-domain routing**.
- Each autonomous system can choose one or more intra-domain routing protocols to handle routing inside the autonomous system. However, only one inter-domain routing protocol handles routing between autonomous systems.

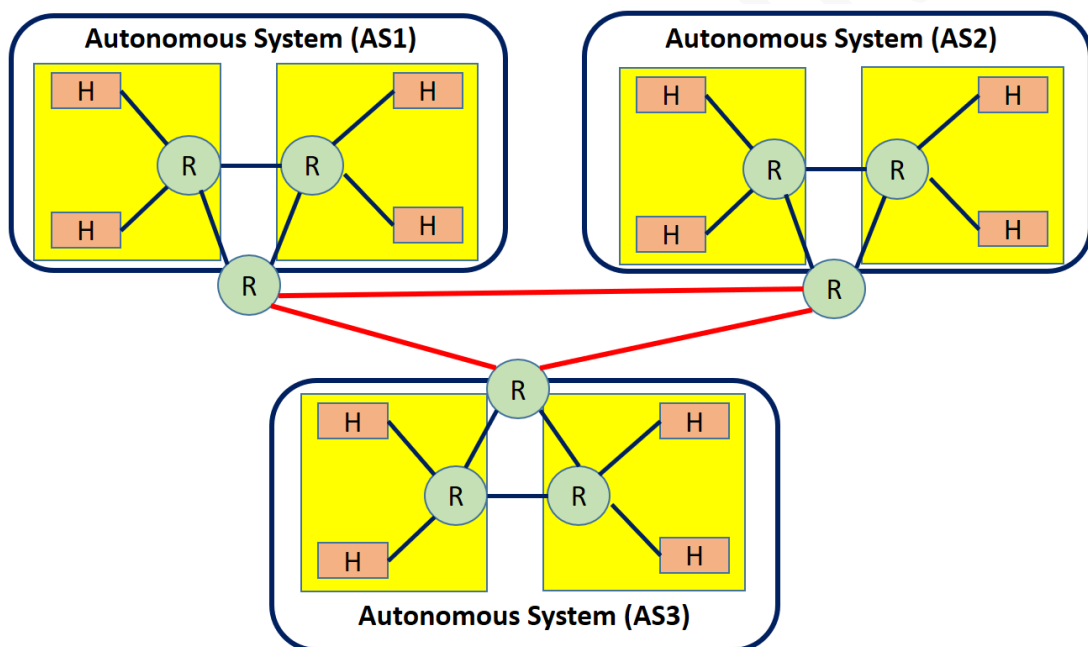


Fig. 2.1 Autonomous System

Intra-domain routing

Routing **inside an autonomous system** is referred to as **intra-domain routing**. There are different protocols for routing inside an autonomous system. These protocols are based on Intra-domain routing algorithms.

Intra-domain routing algorithm

- Distance Vector Routing algorithm
Routing Information Protocol (RIP) is based on this algorithm.
- Link State Routing algorithm
Open Shortest Path First (OSPF) is based on this algorithm.

Distance Vector Routing algorithm (Intra-domain Routing Algorithm)

- Autonomous System considered as a graph, in which each router is node and these nodes connected via links – every pair of node have a cost, represented above a link. (refer fig. 2.2)
- In distance vector routing, the ***least-cost route between any two nodes is the route*** with minimum distance.
- In this algorithm, as the name implies, each node maintains a vector (table) of minimum distances to every node.
- The table at each node also guides the packets to the desired node by showing the next stop in the route (next-hop routing).
- Steps to build Routing Table for each node.
 - Initialization
 - Sharing
 - Updating

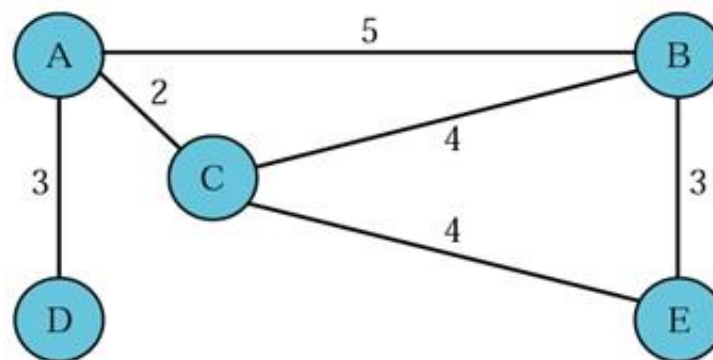


Fig. 2.2 Autonomous System (Represented as a graph)

Consider above network in fig 2.2 of nodes (A,B,C,D,E) with cost. We will apply initialization step to this network.

Initialization

At the beginning, however, this is not the case. Each node can know only the distance between itself and its immediate neighbours, those directly connected to it. So for the moment, we assume that each node can send a message to the immediate neighbours and find the distance between itself and these neighbours. Figure 3.2 shows initial vector table for each node. The distance for any entry that is not a neighbour is marked as infinite (unreachable).

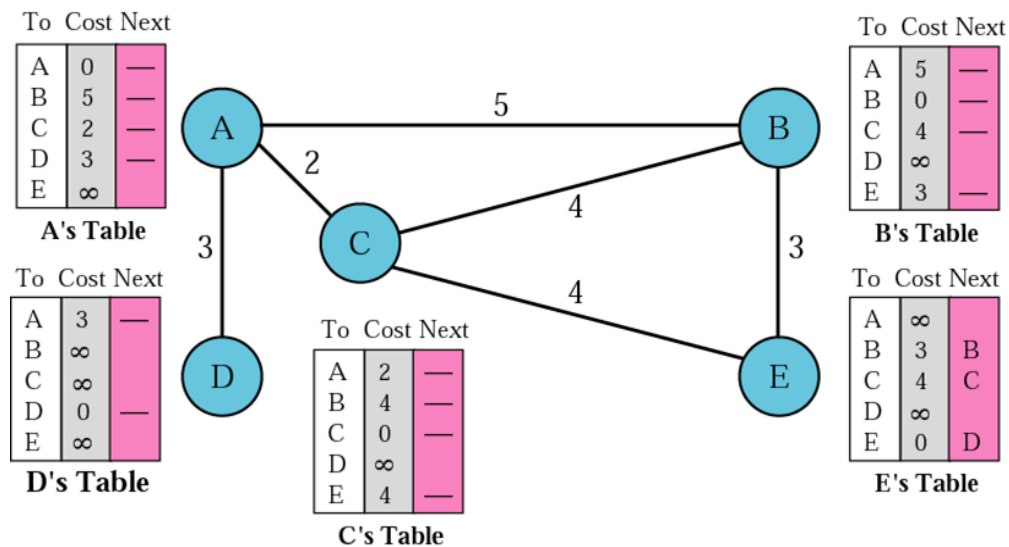


Fig. 2.3 Initial Vector tables for nodes

Sharing

In distance vector routing, each node shares its routing table with its immediate neighbours periodically and when there is a change. The whole idea of distance vector routing is the sharing of information between neighbours. Although node A does not know about node E, node C does. Therefore, if node C shares its routing table with A, node A can also know how to reach node E.

Updating in distance vector routing

When a node receives a two-column table from a neighbour, it needs to update its routing table. Updating takes three steps:

1. The receiving node needs to add the cost between itself and the sending node to each value in the second column. The logic is clear. If node C claims that its distance to a destination is x , and the distance between A and C is y , then the distance between A and that destination, via C, is $x + y$.
2. The receiving node needs to add the name of the sending node to each row as the third column if the receiving node uses information from any row. The sending node is the next node in the route.
3. The receiving node needs to compare each row of its old table with the corresponding row of the modified version of the received table.

- a) If the next-node entry is different, the receiving node chooses the row with the smaller cost. If there is a tie, the old one is kept.
- b) If the next-node entry is the same, the receiving node chooses the new row. For example, suppose node C has previously advertised a route to node X with distance 3. Suppose that now there is no path between C and X; node C now advertises this route with a distance of infinity. Node A must not ignore this value even though its old entry is smaller. The old route does not exist any more. The new route has a distance of infinity.

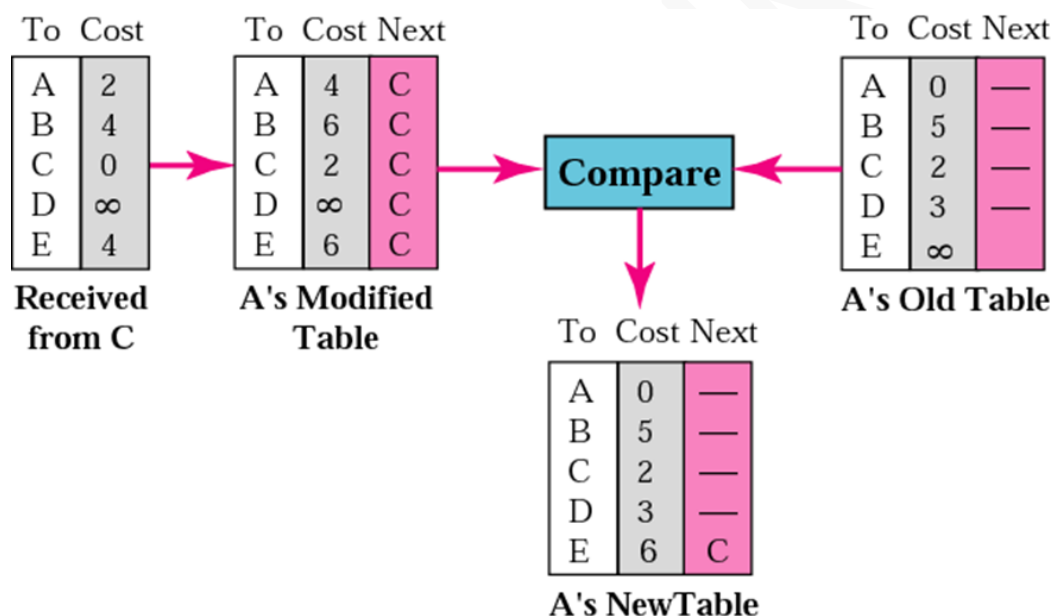


Fig. 2.4 Example for Updating Vector Table for Node A

Routing Information Protocol (RIP v2)

- RIP protocol is based on Distance vector routing algorithm.
- RIP is an Intra-domain routing protocol.
- RIP requires information about directly connected networks. We add this information and activate the RIP routing protocol on routers of the network.

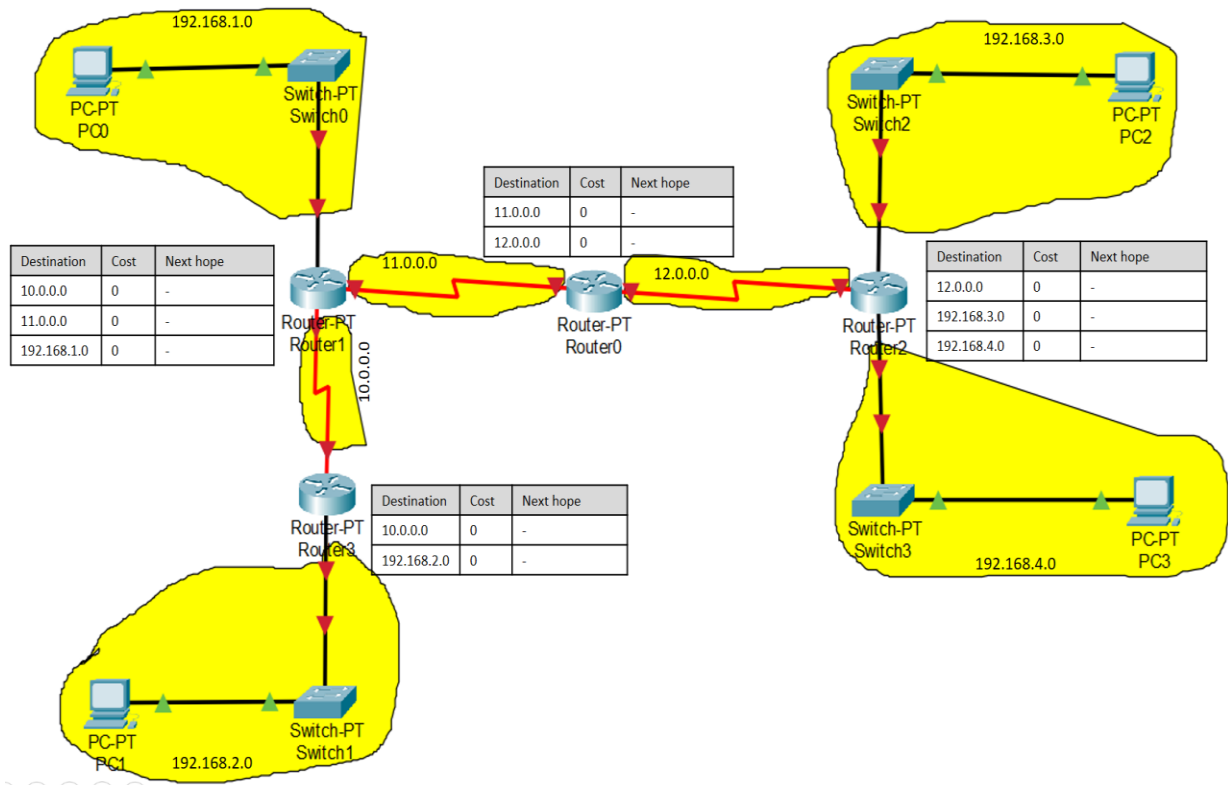


Fig. 2.5 Directly connected networks

- Once configured and activated, each router sends the routing update out of all active interfaces every 30 seconds.
- Each router also receives routing updates from its neighbouring routers.
- A routing update contains the entire routing table of the sending router. Routers compare the received routing tables with their routing tables. If they find any new route in the received routing tables, they add them to their routing tables.
- In the next routing update, routers advertise the updated routing tables. Over time, as each router learns more routes, they advertise about those routes as well. By the end of the process, all routers know about all routes.

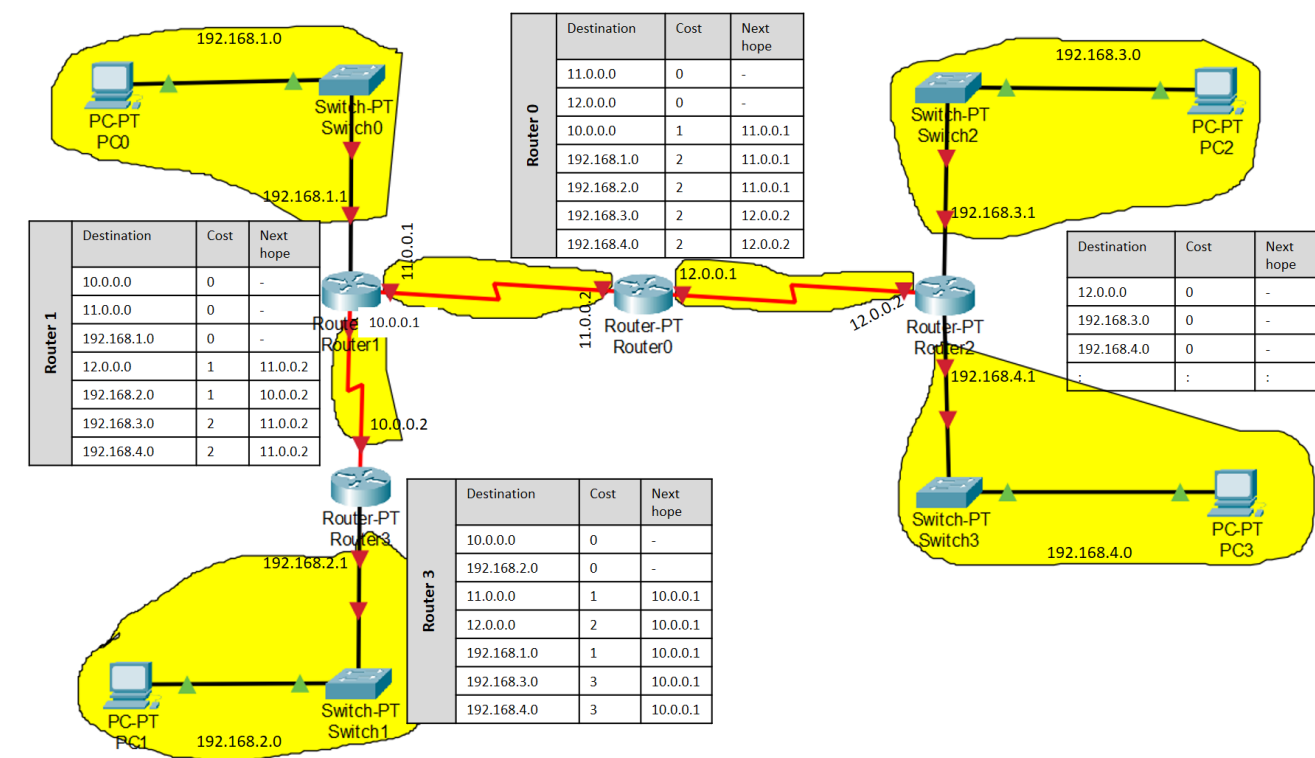


Fig. 2.5 Updated Routing Tables using RIP

RIP message format (RIP v2)

Command	Version	Reserved
Family		Tag
Network Address		
Subnet Mask		
Next Hope Address		
Distance		

Repeated

Fig. 2.6 RIP Packet format version2

Command (8bit) - type of message: request (1) or response (2).

Version (8bit) - current Version 2

Family (16bit) - Family of protocol, for TCP/IP value is 2

Tag (16bit) - Information about autonomous system

Network Address (32bit) - Destination address

Subnet Mask (32bit) – To identify prefix bits

Next Hope Address (32bit) – Defines via which interface packet can reach destination

Distance (32bit) - Number of hops to the destination

LINK STATE ROUTING - Building Routing Tables

In link state routing, four sets of actions are required to ensure that each node has the routing table showing the least-cost node to every other node.

1. Creation of the states of the links by each node, called the link state packet (LSP).
2. Dissemination of LSPs to every other router, called flooding, in an efficient and reliable way.
3. Formation of a shortest path tree for each node.
4. Calculation of a routing table based on the shortest path tree.

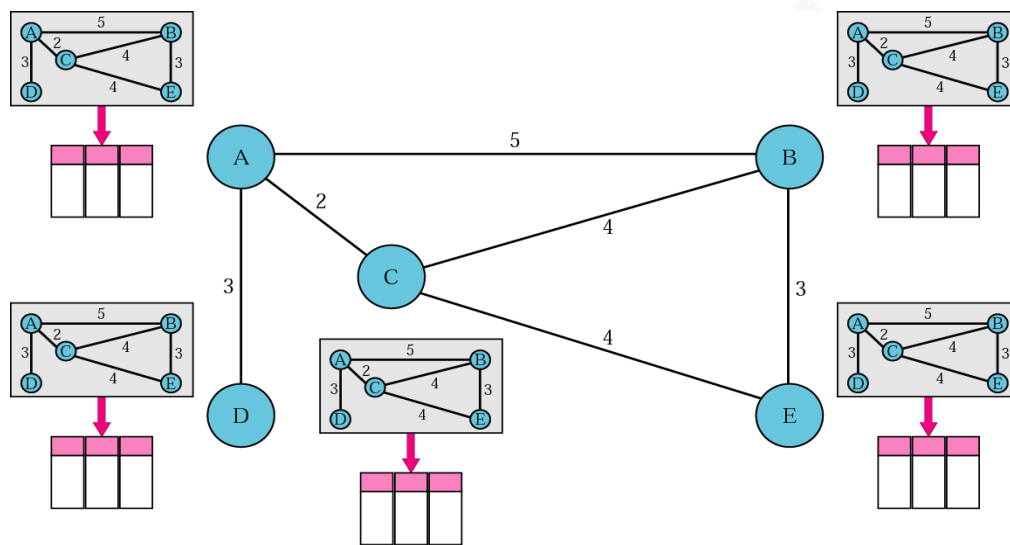


Fig. 2.7 Concept of Link State Routing (Every node have domain topology)

Creation of LINK STATE PACKET (LSP)

- A link state packet can carry a large amount of information. For the moment, however, we assume that it carries a minimum amount of data:

the node identity, the list of links, a sequence number, and age.

- The first two, node identity and the list of links, are needed to make the topology.
- The third, sequence number, facilitates flooding and distinguishes new LSPs from old ones.
- The fourth, age, prevents old LSPs from remaining in the domain for a long time.

Flooding of LSPs (Help to know complete Topology of N/W)

Flooding of LSPs After a node has prepared an LSP, it must be disseminated to all other nodes, not only to its neighbours. The process is called flooding and based on the following:

1. The creating node sends a copy of the LSP out of each interface.
2. A node that receives an LSP compares it with the copy it may already have. If the newly arrived LSP is older than the one it has (found by checking the sequence number), it discards the LSP. If it is newer, the node does the following:
 - a. It discards the old LSP and keeps the new one.
 - b. It sends a copy of it out of each interface except the one from which the packet arrived. This guarantees that flooding stops somewhere in the domain (where a node has only one interface).

Formation of Shortest path tree

A shortest path tree is a tree in which the path between the root and every other node is the shortest. What we need for each node is a shortest path tree with that node as the root.

The Dijkstra algorithm creates a shortest path tree from a graph. The algorithm divides the nodes into two sets: tentative and permanent.

It finds the neighbours of a current node, makes them tentative, examines them, and if they pass the criteria, makes them permanent. We can informally define the algorithm by using the flowchart in figure

Let us apply the algorithm to node A of our sample graph in Figure 2.8

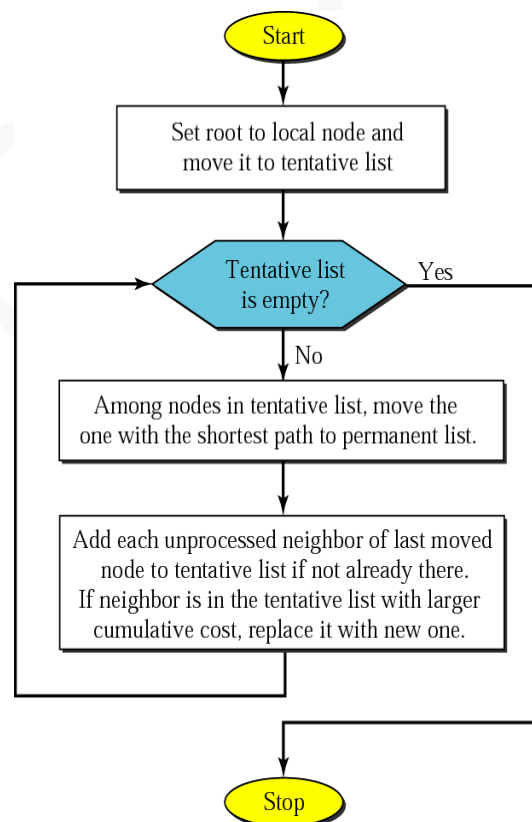


Fig. 2.8 Dijkstra Algorithm for Shortest path tree

Example to form Shortest path tree for Node A (refer a figure 2.9)

1. We make node A the root of the tree and move it to the tentative list. Our two lists are

Permanent list: empty**Tentative list: A(0)**

2. Node A has the shortest cumulative cost from all nodes in the tentative list. We move A to the permanent list and add all neighbours of A to the tentative list. Our new lists are

Permanent list: A(0)**Tentative list: B(5), C(2), D(3)**

3. Node C has the shortest cumulative cost from all nodes in the tentative list. We move C to the permanent list. Node C has three neighbours(A,B,E), but node A (self node) is already processed, which makes the unprocessed neighbours just B and E. However, B is already in the tentative list with a cumulative cost of 5. Node A could also reach node B through C with a cumulative cost of 6. Since 5 is less than 6, we keep node B with a cumulative cost of 5 in the tentative list and do not replace it. Our new lists are

Permanent list: A(0), C(2)**Tentative list: B(5), D(3), E(6)**

4. Node D has the shortest cumulative cost of all the nodes in the tentative list. We move D to the permanent list. Node D has no unprocessed neighbour to be added to the tentative list. Our new lists are

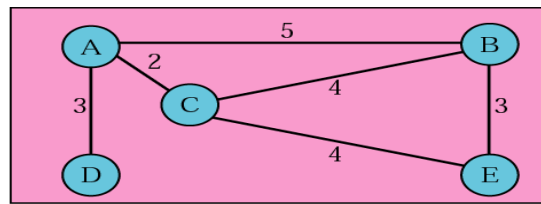
Permanent list: A(0), C(2), D(3)**Tentative list: B(5), E(6)**

5. Node B has the shortest cumulative cost of all the nodes in the tentative list. We move B to the permanent list. We need to add all unprocessed neighbours of B to the tentative list (this is just node E). However, E(6) is already in the list with a smaller cumulative cost. The cumulative cost to node E, as the neighbour of B, is 8. We keep node E(6) in the tentative list. Our new lists are

Permanent list: A(0), B(5), C(2), D(3)**Tentative list: E(6)**

6. Node E has the shortest cumulative cost from all nodes in the tentative list. We move E to the permanent list. Node E has no neighbour. Now the tentative list is empty. We stop; our shortest path tree is ready. The final lists are

Permanent list: A(0), B(5), C(2), D(3), E(6) Tentative list: empty



Topology

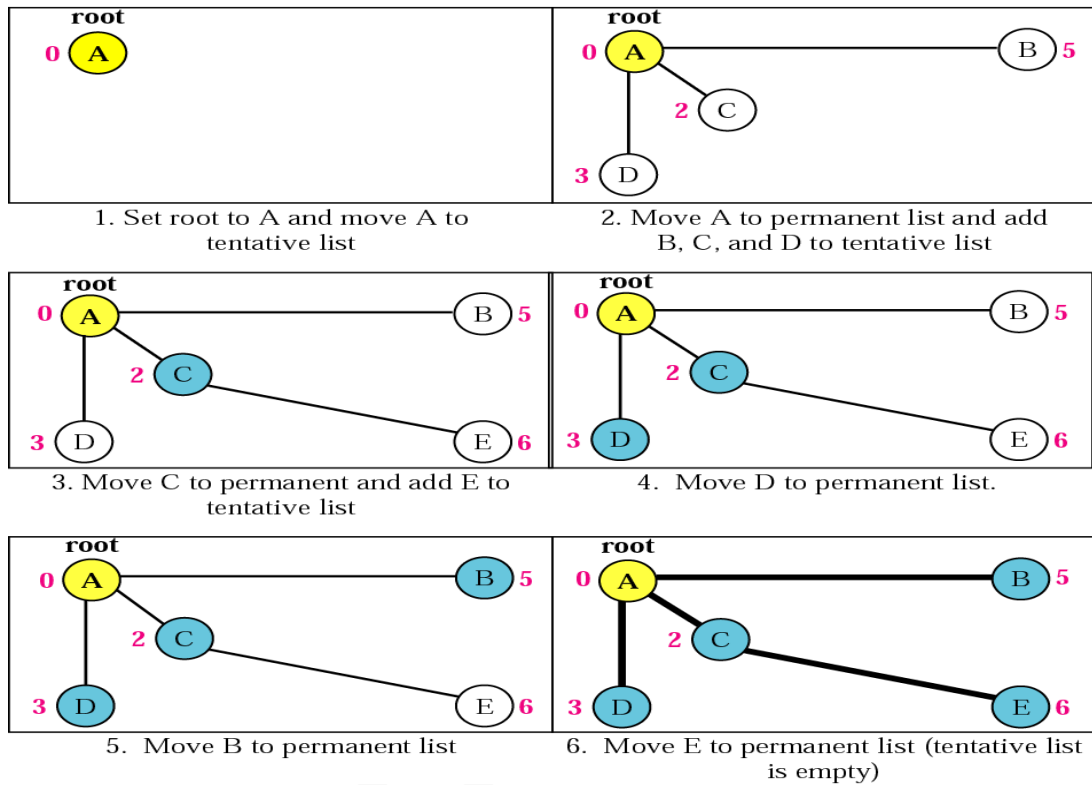


Fig. 2.9 Example of formation of shortest path tree

Calculation of Routing Table

Calculation of Routing Table from Shortest Path Tree Each node uses the shortest path tree protocol to construct its routing table. The routing table shows the cost of reaching each node from the root. Table 2.1 shows the routing table for node A.

Node	Cost	Next Router
A	0	—
B	5	—
C	2	—
D	3	—
E	6	C

Table 2.1

Open Shortest Path First (OSPF)

- It is an interior gateway protocol that has been designed within a single autonomous system.
- It is based on a link-state routing algorithm in which each router contains the information of every domain, and based on this information, it determines the shortest path.
- Every router contains the same information about the network. The way the router learns this information by sending LSA (Link State Advertisements).
- Once the LSAs have been flooded, the OSPF stores the information in a link-state database known as LSDB.

Areas

- Compared with RIP, which is normally used in small ASs, OSPF was designed to be able to handle routing in a small or large autonomous system. However, the formation of shortest-path trees in OSPF requires that all routers flood the whole AS with their LSPs to create the global LSDB.
- It may have created a huge volume of traffic in a large AS. To prevent this, the AS needs to be divided into small sections called areas.
- However, each router in an area needs to know the information about the link states not only in its area but also in other areas. For this reason, one of the areas in the AS is designated as the backbone area, responsible for connecting the areas together.
- By using Backbone network, a router in an area can receive all LSPs generated in other areas.

Autonomous System (AS)

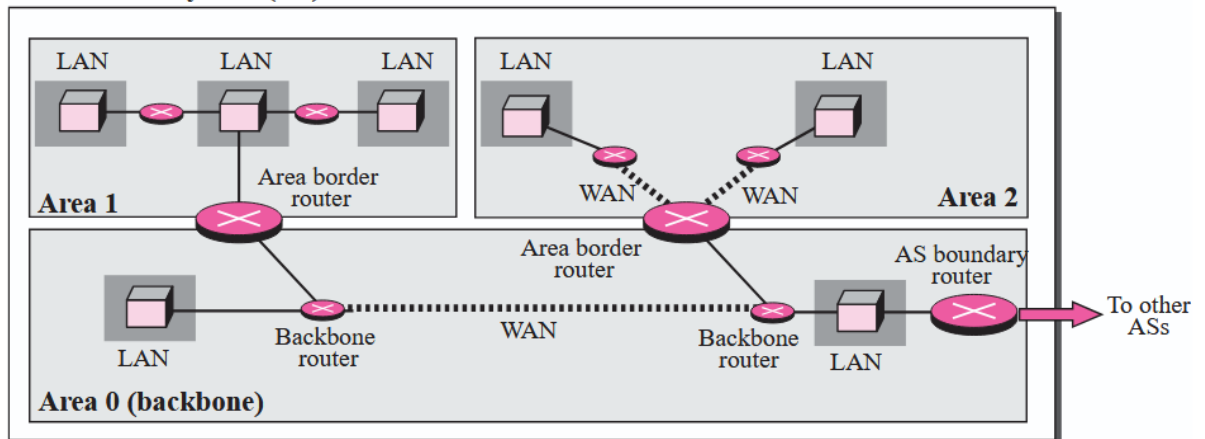


Fig. 2.10 Areas in an autonomous system

Types of Links

- Point to Point
- Transient
- Stub
- Virtual

Point to Point

A point-to-point link connects two routers without any other host or router in between.



Fig. 2.11a Point to Point Network

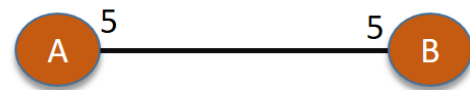


Fig.2.11b Graphical Representation

Transient link

A transient link is a *network*, where multiple routers are connected to the same physical network (like Ethernet). The data can enter through any of the routers and leave through any router.

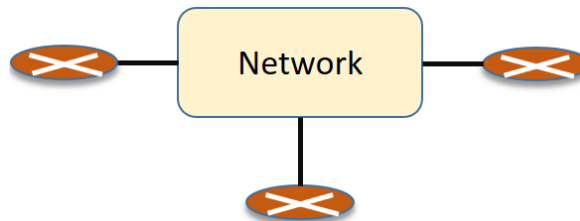


Fig. 2.12 Transient Link

Stub Link

A network segment with only one router connected. Traffic can only enter or leave through that single router. There are no OSPF neighbors on the other side.

Example: A router's loopback interface or a network connected to just one router.

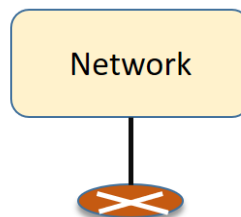


Fig. 2.13 Stub Link

Virtual Link

In OSPF, all areas must connect to Area 0 (the backbone area). Sometimes, due to network design or changes, an area gets separated from Area 0. In such a situation, the administration may create a virtual link between separated area and Area 0 (the backbone area) by using a longer path that probably goes through several routers.

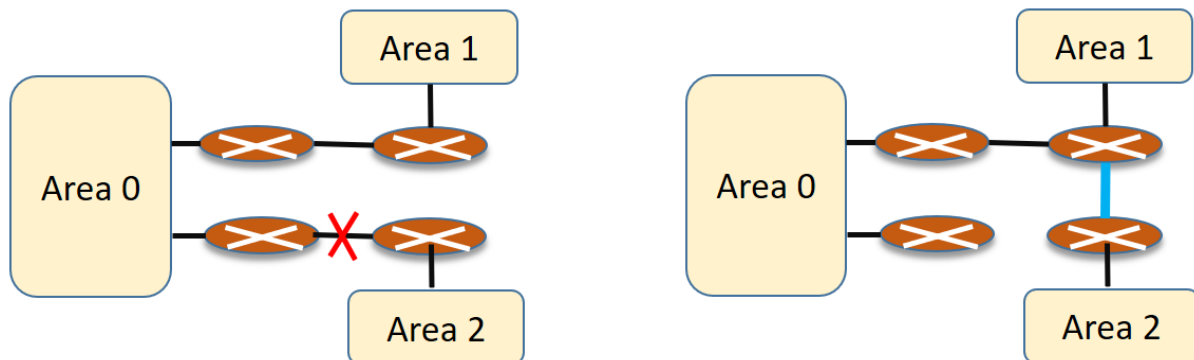


Fig. 2.14 Virtual Link

Path Vector Routing (Inter-domain Routing Algorithm)

- Path vector routing is used to route packets from one Autonomous System to another.
- The principle of path vector routing is similar to that of distance vector routing (with steps Initialization, Sharing, Update). In path vector routing, we assume that there is one node in each autonomous system that acts on behalf of the entire autonomous system. Called as **speaker node**.
- The speaker node in an AS creates an initial routing table and advertises it to speaker nodes in the neighbouring ASs. The idea is the same as for distance vector routing except that only speaker nodes in each AS can communicate with each other.
- However, what is advertised is different. A speaker node **advertises the path, not the metric (distance)** of the nodes, in its autonomous system or other autonomous systems.
- We can summarize the fields that advertise by DVR and PVR as
 - DVR - *Destination, Cost, Next Hope IP*
 - PVR - *Destination, Path(based on policies), Next Hope IP*

Initialization

At the beginning, each speaker node can know only the reachability of nodes inside its autonomous system. Figure 3.7 shows the initial tables for each speaker node in a system made off our ASs.

Sharing

A path vector routing table for each router can be created if ASs share their reachability list with each other. In Figure 2.15, router R1 in AS1 can send its

reachability list to router R2. Router R2, after combining its reachability list, can send the result to both R1 and R3. Router R3 can send its reachability list to R2, which in turn improves its routing table, and so on.

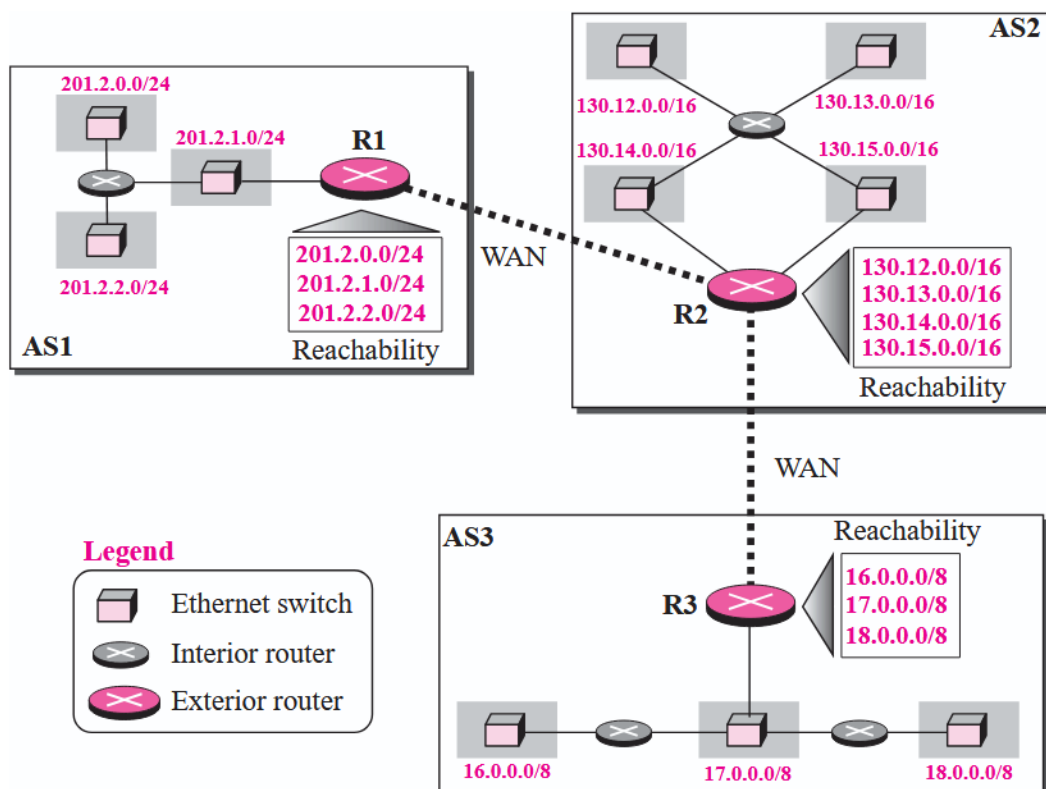


Fig. 2.15 PVR - Initialization

Updating

Figure 2.16 shows the routing table for each router after all three routers have updated their routing table.

R1		R2		R3	
Network	Path	Network	Path	Network	Path
201.2.0.0/24	AS1 (This AS)	201.2.0.0/24	AS2, AS1	201.2.0.0/24	AS3, AS2, AS1
201.2.1.0/24	AS1 (This AS)	201.2.1.0/24	AS2, AS1	201.2.1.0/24	AS3, AS2, AS1
201.2.2.0/24	AS1 (This AS)	201.2.2.0/24	AS2, AS1	201.2.2.0/24	AS3, AS2, AS1
130.12.0.0/16	AS1, AS2	130.12.0.0/16	AS2 (This AS)	130.12.0.0/16	AS3, AS2
130.13.0.0/16	AS1, AS2	130.13.0.0/16	AS2 (This AS)	130.13.0.0/16	AS3, AS2
130.14.0.0/16	AS1, AS2	130.14.0.0/16	AS2 (This AS)	130.14.0.0/16	AS3, AS2
130.15.0.0/16	AS1, AS2	130.15.0.0/16	AS2 (This AS)	130.15.0.0/16	AS3, AS2
16.0.0.0/8	AS1, AS2, AS3	16.0.0.0/8	AS2, AS3	16.0.0.0/8	AS3 (This AS)
17.0.0.0/8	AS1, AS2, AS3	17.0.0.0/8	AS2, AS3	17.0.0.0/8	AS3 (This AS)
18.0.0.0/8	AS1, AS2, AS3	18.0.0.0/8	AS2, AS3	18.0.0.0/8	AS3 (This AS)

Path-Vector Routing Table Path-Vector Routing Table Path-Vector Routing Table

Fig. 2.15 PVR – Updated Routing tables with Path

Border Gateway Protocol (BGP)

- It is an Exterior gateway protocol that has been designed for routing between multiple autonomous system (Inter-domain Routing).
- It is based on a path vector routing algorithm.
- It runs over TCP port number 179, support reliability.
- Separate TCP Connection is established between BGP Speaker routers to exchange Information regarding reachability to different AS.

BGP works for routing between different autonomous systems, Autonomous systems may be of different type.

Types of Autonomous System

Stub AS

It has only one connection to another AS. Data traffic cannot pass through a stub AS. A Stub AS is either a source or sink.

Multihomed AS

It has more than one connection to other ASs but it is still only a source or sink for data traffic.

Transit AS

It is a Multihomed AS that allows transient traffic.

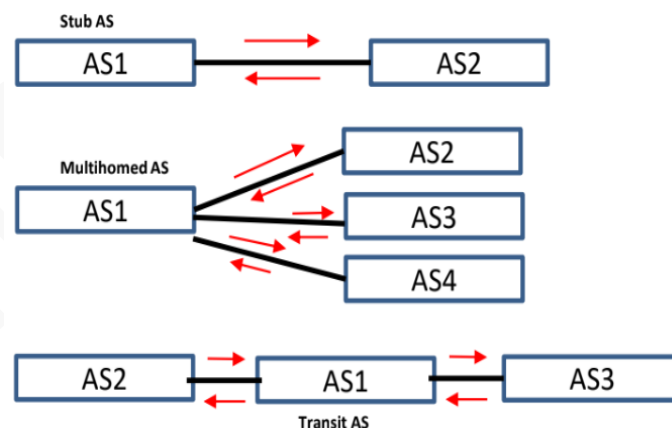


Fig. 2.16 Types of Autonomous System

Types of BGP Sessions

E - BGP:

Exchange Information between two speaker nodes

I - BGP:

Exchange information between routers of same AS.

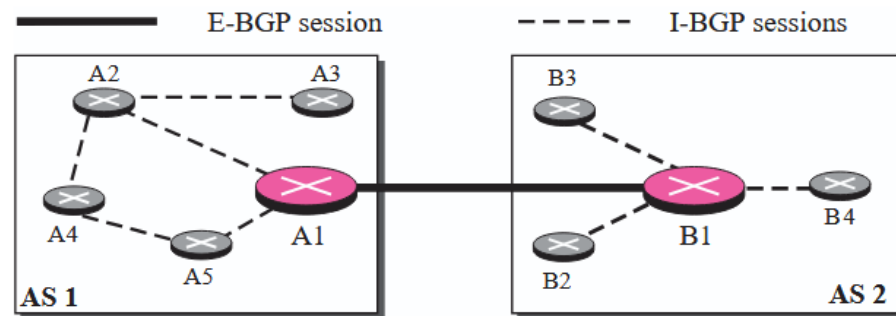


Fig. 2.17 Types of BGP Sessions

Types of BGP packets/message

Open Message

To create a neighbourhood relationship, a router running BGP opens a TCP connection with a neighbour and sends an open message.

Update Message

The update message is the heart of the BGP protocol. It is used by a router to withdraw destinations that have been advertised previously, to announce a route to a new destination, or both. Note that BGP can withdraw several destinations that were advertised before, but it can only advertise one new destination (or multiple destinations with the same path attributes) in a single update message.

Keepalive Message

The BGP peers that are running exchange keepalive messages regularly (before their hold time expires) to tell each other that they are alive.

Notification

A notification message is sent by a router whenever an error condition is detected or a router wants to close the session.

Internet Control Message Protocol (ICMPv4)

IPv4 has no mechanism for error reporting or error correcting.

IPv4 protocol also lacks a mechanism for host and management queries.

The Internet Control Message Protocol version 4 (ICMPv4) has been designed to compensate for the above two deficiencies. It is a companion to the IP protocol.

General Format for ICMP messages

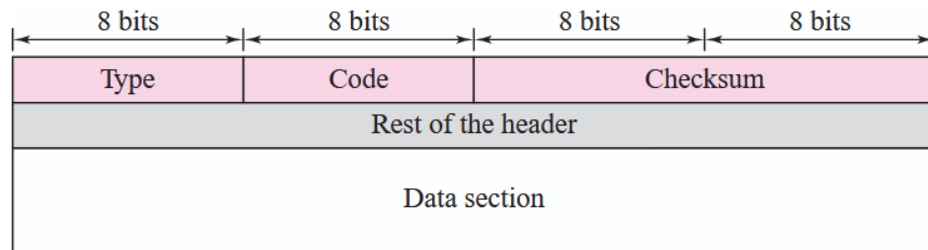


Fig. 2.18 General format for ICMP message

The first field, ICMP type, defines the type of the message. The code field specifies the reason for the particular message type. The last common field is the checksum field. The rest of the header is specific for each message type.

There are two categories of ICMP messages

Error Reporting Message

- Destination Unreachable
- Source Quench
- Time Exceeded
- Parameter problem
- Redirection

Query Message

- Echo Request and Echo Reply
- Timestamp Request and Timestamp Reply
- Router Solicitation and Advertisement
- Address Mask and Reply

Error Reporting Message

Destination Unreachable (type 3)

When a router cannot route a datagram or a host cannot deliver a datagram, the datagram is discarded and the router or the host sends a destination-unreachable message back to the source host that initiated the datagram. Note that destination unreachable messages can be created by either a router or the destination host.

Source Quench

A source-quench message informs the source that a datagram has been discarded due to congestion in a router or the destination host. The source must slow down the sending of datagram's until the congestion is relieved.

Time Exceeded

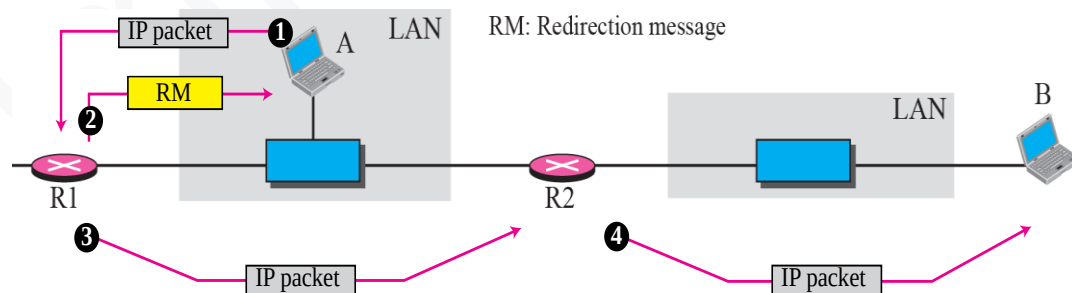
Whenever a router decrements a datagram with a time-to-live value to zero, it discards the datagram and sends a time-exceeded message to the original source.

When the final destination does not receive all of the fragments in a set time, it discards the received fragments and sends a time-exceeded message to the original source.

Parameter Problem

Any ambiguity in the header part of a datagram can create serious problems as the data- gram travels through the Internet. If a router or the destination host discovers an ambiguous or missing value in any field of the datagram, it discards the datagram and sends a parameter-problem message back to the source.

Redirect Message



- R1 will send a Redirection Message to Host A that Host A can send datagram directly to R2.
- A host usually starts with a small routing table that is gradually augmented and updated. One of the tools to accomplish this is the redirection message.

Query Messages (ICMP)

ECHO-REQUEST(type 8) and REPLY(type 0)

The echo-request and echo-reply messages can be used to determine if there is communication at the IP level. Because ICMP messages are encapsulated in IP datagram's, the receipt of an echo-reply message by the machine that sent the echo request is proof that the IP protocols in the sender and receiver are communicating with each other using the IP datagram.

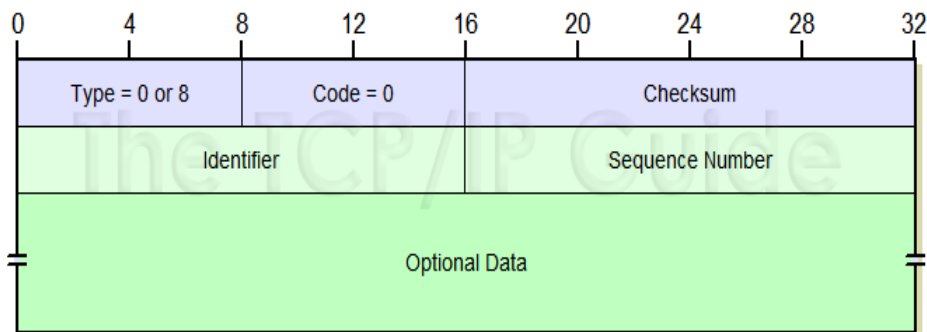


Fig 2.19 ICMP Echo Request/Reply Format

Identifier

An identifier to match Timestamp Replies to the corresponding Timestamp Requests

Sequence Number:

A sequence number to match Timestamp Replies to the corresponding Timestamp Requests

TIMESTAMP REQUEST (type 13) AND REPLY (type 14)

Two machines (hosts or routers) can use the timestamp request and timestamp reply messages to determine the round-trip time needed for an IP datagram to travel between them.

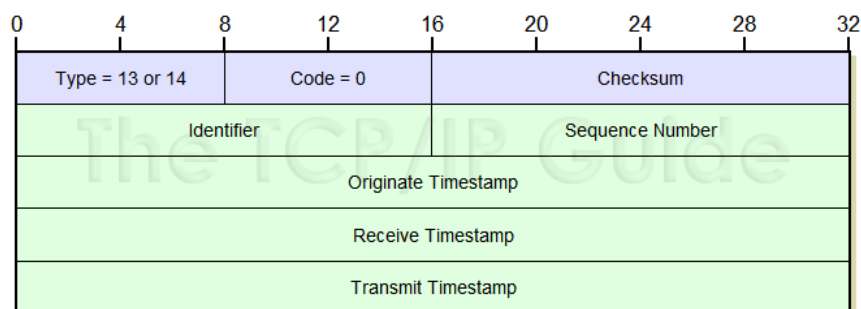


Fig 2.20 ICMP Echo Request/Reply Format

Originate Timestamp

A time value filled in by the originating device just before sending the Timestamp message.

Receive Timestamp

A time value filled in by the responding device just as it receives the Timestamp message.

Transmit Timestamp

A time value filled in by the responding device just before sending back the Timestamp Reply message.

ADDRESS MASK REQUEST (type 17) & ADDRESS MASK REPLY (type 18):

A host may be aware of its own IP address but it is not necessary that it is aware of its own subnet mask. To know your subnet mask sends the address mask request to the host router and the router sends this host's subnet mask address as the mask reply message. If the host is aware of the router's address, then he sends the request directly to the router, otherwise this request is broadcast.

ROUTER SOLICITATION (type 10) AND ADVERTISEMENT (type 9)

As we discussed in the redirection message section, a host that wants to send data to a host on another network needs to know the address of routers connected to its own network. In addition, the host must know if the routers are alive and functioning. The router-solicitation and router-advertisement messages can help in this situation.



Advance Computer Network

IMRAN SHAIKH

Computer Engineering | Networking | 2025

Chapter 3 **Transport Layer Protocols** (14 marks)

3.1 Process to Process Delivery-Client/Server paradigm, Multiplexing and Demultiplexing, Connectionless vs. Connection-Oriented Service.

3.2 User Datagram Protocol (UDP)-Ports-Well known ports for UDP header format, features and applications.

3.3 Transmission Control Protocol(TCP)-TCP services, TCP features, Segment, Three way handshaking, Flow control, Error control, Congestion control-Open loop, Closed loop.

3.4 TLS(Transport Layer Security)-working and applications

3.5 Stream Control Transmission Protocol (SCTP)- services and features.

Transport Layer

The Transport Layer is responsible for process-to-process delivery — ensuring that data is transferred between specific applications (processes) on different hosts.

Process-to-Process Delivery

The Network Layer delivers packets host-to-host (identified by IP addresses), but on each host, many processes (application programs) can run simultaneously — e.g., browser, mail, file transfer, etc.

The Transport Layer ensures that data is delivered to the correct process on the correct host.

Each process is identified by a **port number** (e.g., 80 for HTTP, 25 for SMTP).

Client-Server Paradigm

Most applications follow the Client/Server model.

Server: Provides a specific service (e.g., Web server on port 80).

Client: Requests service from the server.

Example

Client: Web browser

Server: Web server (listening on port 80 or 443)

“Transport layer ensures that client data reaches the right service (port) on the server and the response returns to the correct client process.”

Multiplexing and Demultiplexing

Multiplexing

Occurs at the sender side.

Multiple applications generate data that must be sent over the same network connection.

The Transport Layer adds a header (with source port) to identify each process and combines them onto a single link.

Demultiplexing:

Occurs at the receiver side.

The Transport Layer examines the destination port number in each segment and delivers data to the correct process.

Connection Oriented vs Connection Less Service

Connectionless Service (UDP)

No connection setup before sending data.

Data is sent in independent datagrams.

Each datagram is handled separately (no guarantee of order or delivery).

Low overhead and faster.

Suitable for applications that can tolerate loss.

UDP is a protocol at transport layer supports connection less service.

Connection-Oriented Service (TCP)

A connection is established before communication (handshake).

Data is delivered in sequence and reliably.

Retransmission, acknowledgment, flow and error control are provided.

Suitable for reliable applications (e.g., email, web browsing, file transfer).

Example:

TCP is a protocol at transport layer supports connection oriented service.

Introduction to User Datagram Protocol

The User Datagram Protocol (UDP) is called a connectionless, unreliable transport layer protocol.

It does not add anything to the services of IP except to provide *process-to-process* communication instead of host-to-host communication. In addition, it performs very limited error checking.

Why UDP?

UDP is a very simple protocol using a minimum of overhead. If a process wants to send a small message and does not care much about reliability, it can use UDP. Sending a small message by using UDP takes much less interaction between the sender and receiver than using TCP or SCTP.

USER DATAGRAM

UDP packet is called as User Datagrams. User Datagrams have a header of fixed size that is 8byte. Each field in user datagram header is of 2byte, so total field consist are four.

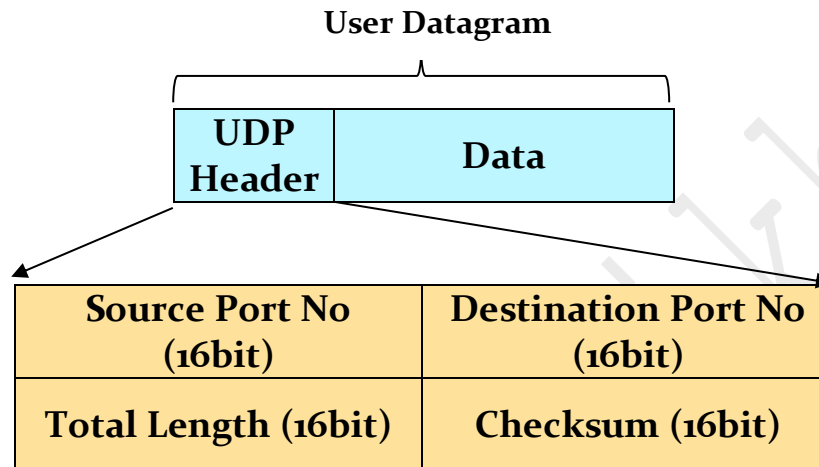


Fig. 4.1 UDP Header Format

Source port number.

- This is the port number used by the process running on the source host. It is 16 bits long, which means that the port number can range from 0 to 65,535.
- If the source host is the client (a client sending a request), the port number, in most cases, is an ephemeral port number requested by the process and chosen by the UDP software running on the source host.
- If the source host is the server (a server sending a response), the port number, in most cases, is a well-known port number.

Destination port Number

- This is the port number used by the process running on the destination host. It is also 16 bits long.
- If the destination host is the server (a client sending a request), the port number, in most cases, is a well-known port number.
- If the destination host is the client (a server sending a response), the port number, in most cases, is an ephemeral port number. In this case, the server copies the ephemeral port number it has received in the request packet.

Total Length

- This is a 16-bit field that defines the total length of the user datagram, header plus data. The 16 bits can define a total length of 0 to 65,535 bytes.
- However, the total length needs to be much less because a UDP user datagram is stored in an IP datagram with a total length of 65,535 bytes.

Checksum

- This field is used to detect errors over the entire user datagram (header plus data).

UDP Services

Process-to-Process Communication

- UDP provides process-to-process communication using socket addresses, a combination of IP addresses and port numbers.

Message-Oriented Protocol

- Data is sent as individual messages (datagrams).
- Each message is independent — boundaries are preserved.

Connectionless Services

- As mentioned previously, UDP provides a connectionless service. This means that each user datagram sent by UDP is an independent datagram. Therefore, in UDP each user datagram can travel on a different path.

Low Overhead

- UDP header size is only 8 bytes, compared to TCP's 20 bytes (minimum).
- Ideal for time-sensitive applications.

Fast Service

- Since there's no connection setup, retransmission, or acknowledgment, UDP is faster than TCP.

No Acknowledgment or Handshaking

- Data is sent directly without any acknowledgment (no handshake like TCP's 3-way).
- This makes it fast and lightweight.

No Flow Control and No Error Control

- UDP is a very simple, unreliable transport protocol. *There is no flow control* and hence no window mechanism. The receiver may overflow with incoming messages.
- There is *no error control mechanism in UDP* except for the checksum.
- This means that the sender does not know if a message has been lost or duplicated. When the receiver detects an error through the checksum, the user datagram is silently discarded.

UDP Applications

- UDP is suitable for a process that requires simple request-response communication with little concern for flow and error control. It is not usually used for a process that needs to send bulk data.
- UDP is suitable for a process with internal flow and error control mechanisms. For example, the Trivial File Transfer Protocol (TFTP) process includes flow and error control. It can easily use UDP.
- UDP is used for management processes such as SNMP.
- UDP is used for some route updating protocols such as Routing Information Protocol (RIP).

Example 1

The following is the content of a UDP header in hexadecimal format.

- a) What is the source port number?
- b) What is the destination port number?
- c) What is the total length of the user datagram?
- d) What is the length of the data?
- e) Is the packet directed from a client to a server or vice versa?

Solution

- a) The source port number is the first four hexadecimal digits (CB84)₁₆, which means that the source port number is 52100.
- b) The destination port number is the second four hexadecimal digits (000D)₁₆, which means that the destination port number is 13.
- c) The third four hexadecimal digits (001C)₁₆ define the length of the whole UDP packet as 28 bytes.
- d) The length of the data is the length of the whole packet minus the length of the header, or $28 - 8 = 20$ bytes.

- e) Since the destination port number is 13 (well-known port), the packet is from the client to the server.

Transmission Control Protocol (TCP)

TCP is a connection oriented reliable protocol. TCP explicitly establishes a connection and then sends data in a form of segment. TCP provides error control, flow control and congestion control features which adds reliability in TCP communication.

TCP Services

Process-to-Process Communication

- Like UDP, TCP provides process-to-process communication using port numbers.

Stream Delivery Service

- In UDP, a process (an application program) sends messages, with predefined boundaries, to UDP for delivery.
- TCP, on the other hand, allows the sending process to deliver data as a stream of bytes and allows the receiving process to obtain data as a stream of bytes.

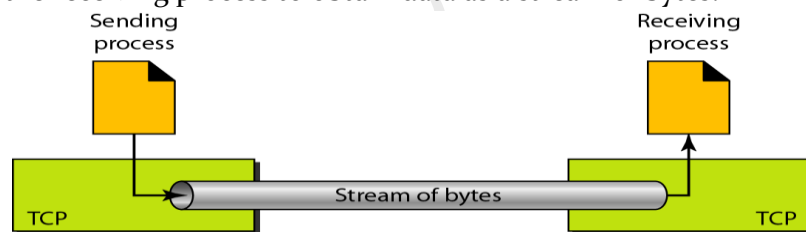


Fig. 4.5 Streams in TCP

Full Duplex Communication

- TCP offers full-duplex service, in which data can flow in both directions at the same time. Each TCP then has a sending and receiving buffer, and segments move in both directions

Connection Oriented Service

- TCP, unlike UDP, is a connection-oriented protocol. When a process at site A wants to send and receive data from another process at site B, the following occurs:
 - i. The two TCPs establish a connection between them.
 - ii. Data are exchanged in both directions.
 - iii. The connection is terminated.

Reliable Service

- TCP is a reliable transport protocol. It uses an acknowledgment mechanism to check the safe and sound arrival of data. We will discuss this feature further in the section on error control.

TCP Features

Numbering System

- The bytes of data being transferred in each connection are numbered by TCP. The numbering starts with a randomly generated number.

Sequence Number

- After the bytes have been numbered, TCP assigns a sequence number to each segment that is being sent. The sequence number for each segment is the number of the first byte carried in that segment.

Flow Control

- TCP, unlike UDP, provides flow control. The receiver of the data, controls the amount of data that are to be sent by the sender. This is done to prevent the receiver from being over-whelmed with data.

Error Control

- To provide reliable service, TCP implements an error control mechanism.

Congestion Control

- TCP, unlike UDP, takes into account congestion in the network. The amount of data sent by a sender is not only controlled by the receiver (flow control), but is also determined by the level of congestion in the network.

TCP Segment

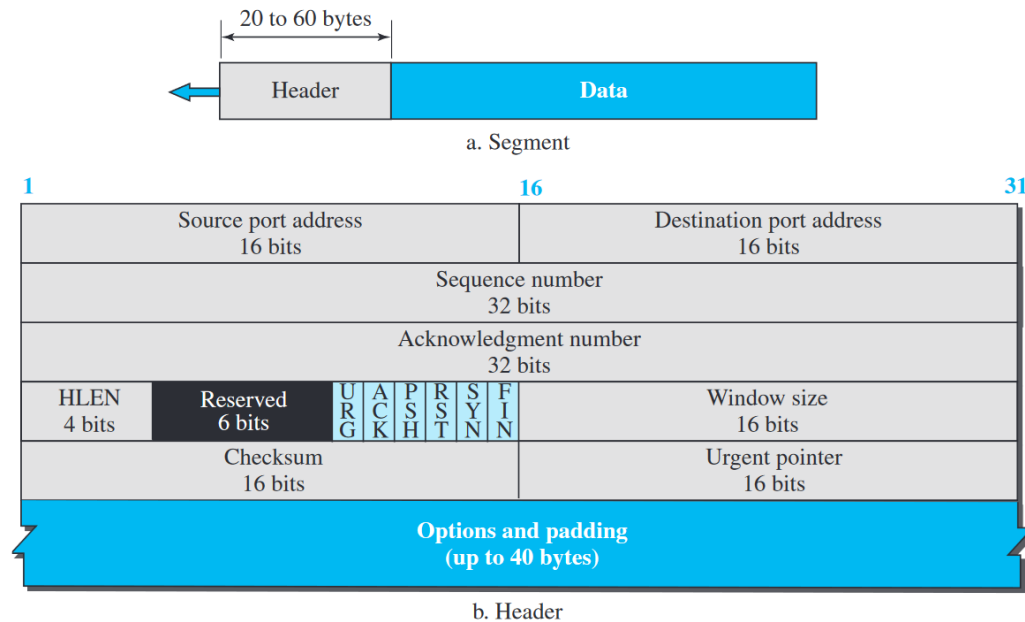


Fig. 4.6a TCP Segment

Fig. 4.6b TCP Segment Header

Source Port Number

This is a 16-bit field that defines the port number of the application program in the host that is sending the segment.

Destination Port Number

This is a 16-bit field that defines the port number of the application program in the host that is receiving the segment.

Sequence Number

This 32-bit field defines the number assigned to the first byte of data contained in this segment.

Acknowledgment Number

This 32-bit field defines the byte number that the receiver of the segment is expecting to receive from the sender.

Header Length

This 4-bit field indicates the number of 4-byte words in the TCP header. The length of the header can be between 20 and 60 bytes.

Therefore, the value of this field can be between 5 ($5 \times 4 = 20$) and 15 ($15 \times 4 = 60$).

if HLEN = 0101 = 5 in decimal

$$5 * 4B = 20B \text{ (Minimum length of TCP Header)}$$

if HLEN = 1111 = 15 in decimal

$$15 * 4B = 60B \text{ (Maximum length of TCP Header)}$$

Reserved

This is a 6-bit field reserved for future use.

Window Size

This 16 bit field defines the size of the window, in bytes, that the receiver must maintain.

maximum size of the window is 65,535 bytes (2^{16}).

This value is normally referred to as the receiving window (rwnd) and is determined by the receiver.

The sender can send number of bytes equal to window size, to receiver without worrying acknowledgement because receiver have that size (window size) buffer to save the data bytes.

Checksum

This 16-bit field contains the checksum. The calculation of the checksum for TCP follows the same procedure as the one described for UDP (refer slide 6 and 7).

The inclusion of the checksum for TCP is mandatory.

The same pseudoheader (as UDP), serving the same purpose, is added to the segment. For the TCP pseudoheader, the value for the protocol field is 6.

Urgent Pointer (16bit)

This 16-bit field, which is valid only if the urgent flag is set. It is used when the segment contains urgent data. It defines the number that must be added to the sequence number to obtain the number of the last urgent byte.

Options

There can be up to 40 bytes of optional information in the TCP header.

CONTROL FLAGS in TCP

URG flag

The urgent pointer flag is used to identify incoming data as urgent.

ACK flag

It is used to acknowledge packets which are successfully received by the host.

SYN flag

It is used in the first step of connection establishment phase or 3-way handshake process between the two hosts. Only the first packet from sender as well as receiver should have this flag set.

FIN flag

It is used to request for connection termination i.e. when there is no more data from the sender, it requests for connection termination.

RST flag

Abruptly tells the other side to stop communication, because sender/receiver feels something is wrong with the TCP communication.

PSH flag

It tells the receiver to process these packets as they are received instead of buffering them.

TCP Connection

There are three phases during TCP connection

- Connection Establishment (Three Way Handshaking)
- Data Transfer
- Connection Termination

TCP Connection Establishment (Three Way Handshaking)

Three-Way Handshaking The connection establishment in TCP is called three-way handshaking.

The three steps in this phase are as follows.

SYN

The client sends the first segment, a SYN segment, in which only the SYN flag is set. This segment is for synchronization of sequence numbers.

SYN + ACK

The server sends the second segment, a SYN +ACK segment, with 2 flag bits set.

ACK

The client sends the third segment. This is just an ACK segment. It guarantees the completion of three way handshaking.

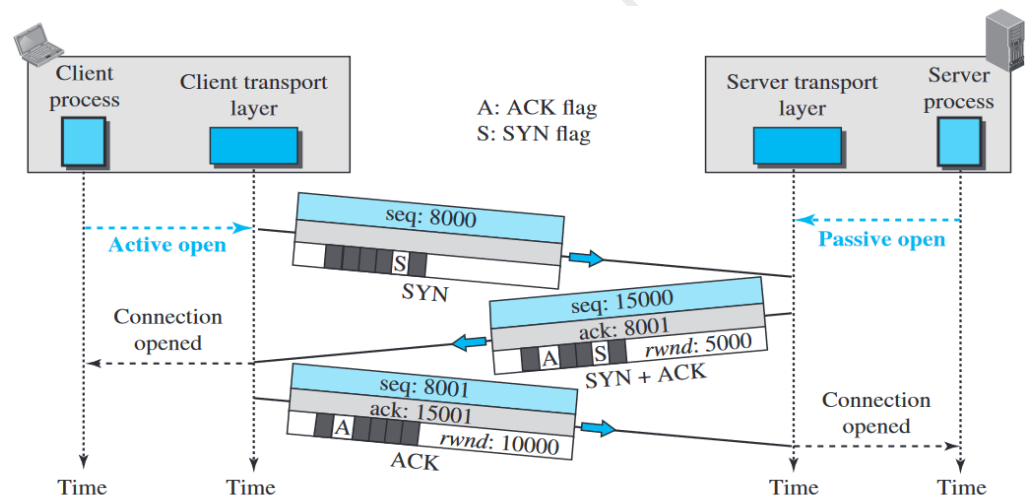


Fig. 4.7 TCP Connection Establishment

Error Control in TCP

TCP provides reliability using error control.

- Error control includes mechanisms for detecting corrupted segments, lost segments, out-of-order segments, and duplicated segments.
- Error control also includes a mechanism for correcting errors after they are detected.

Error detection and correction in TCP is achieved through the use of three simple tools.

checksum

Each segment includes a checksum field which is used to check for a corrupted segment. If the segment is corrupted, it is discarded by the destination TCP and is considered as lost. TCP uses a 16-bit checksum that is mandatory in every segment.

Acknowledgment

TCP uses acknowledgments to confirm the receipt of data segments. Control segments that carry no data but consume a sequence number are also acknowledged. ACK segments are never acknowledged.

Retransmission

In modern implementations, a retransmission occurs

- If the retransmission timer expires **OR**
- Three duplicate ACK segments have arrived.

Timers in TCP (consider fig. 4.12)

The sender sends segments 1 and 2, which are acknowledged immediately by an ACK. Segment 3, however, is lost. The receiver receives segment 4, which is out of order. The receiver stores the data in the segment in its buffer but leaves a gap to indicate that there is no continuity in the data. The receiver immediately sends an acknowledgment to the sender, displaying the next byte it expects. Note that the receiver stores bytes 801 to 900, but never delivers these bytes to the application until the gap is filled.

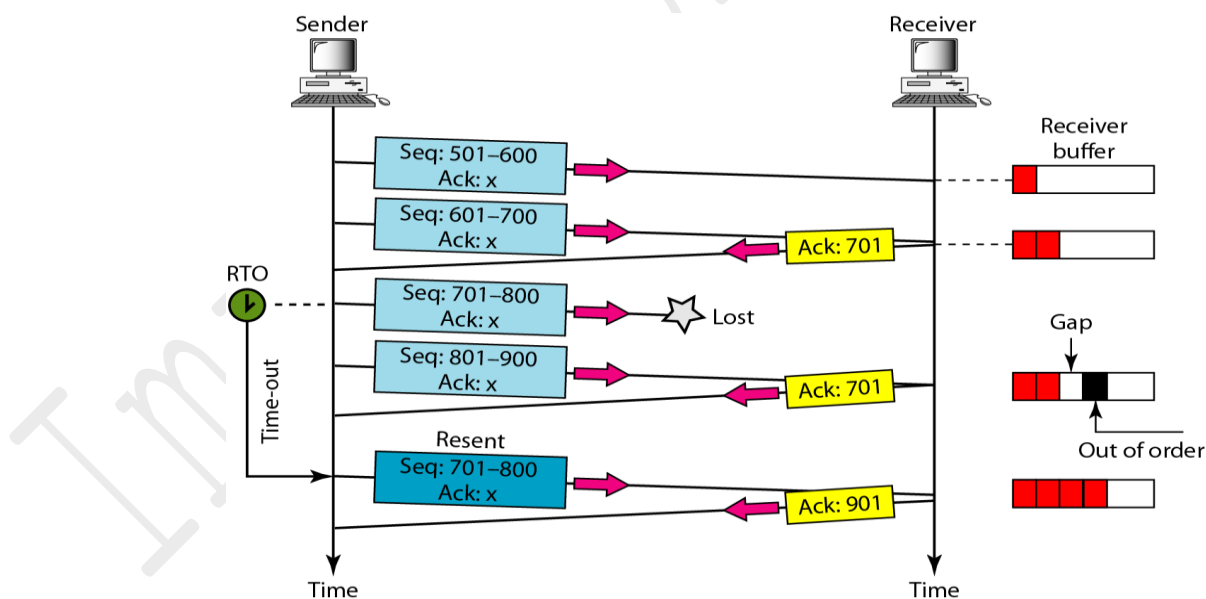


Fig. 4.12 Retransmission based on Timer

We have shown the timer for the earliest outstanding segment. The timer for this definitely runs out because the receiver never sends an acknowledgment for lost or out-of-order segments. When the timer matures, the sending TCP resends segment 3, which arrives this time and is acknowledged properly. Note that the value in the second and third acknowledgments differs according to the corresponding rule.

Fast Retransmission

In this scenario, we want to show the idea of fast retransmission. Our scenario is the same as the second except that the RTO has a higher value. In fast transmission even the timer not matured (expired) still after receiving three ACK segments for segment 3, sender will resend segment 3.

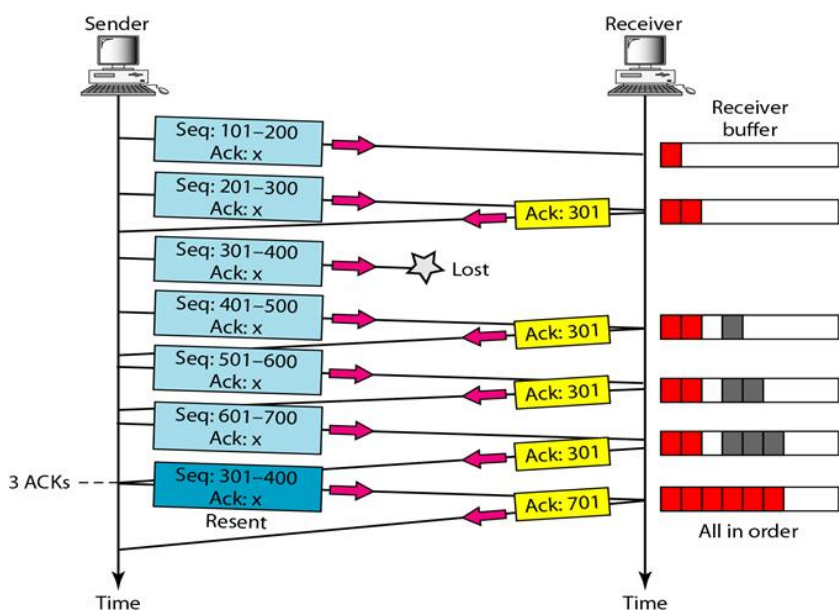


Fig. 4.11 Fast Retransmission

Flow Control in TCP

Flow control in TCP means sender must send data with an amount that can be handled by receiver without overwhelming.

Receive Window

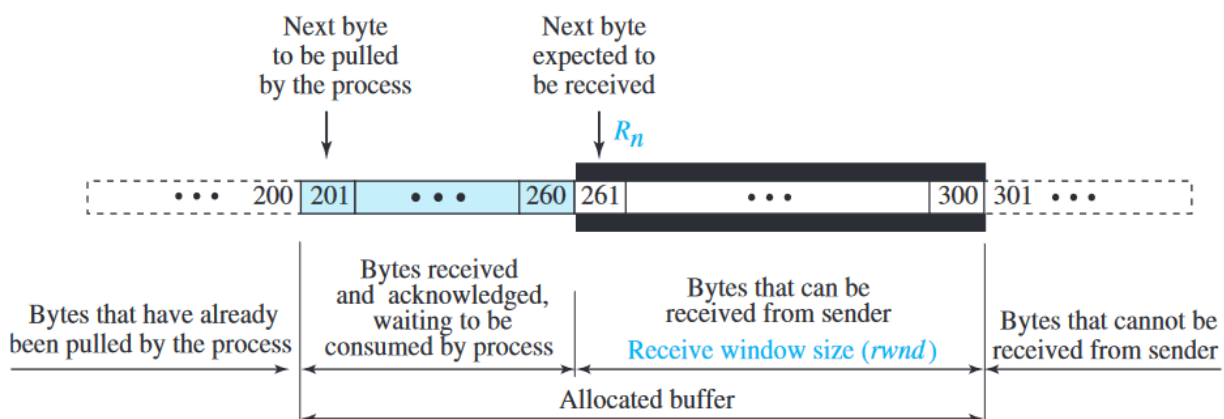


Fig. 4.12 Receive Window

The receiver advertises a window size i.e. **rwnd** (the amount of data receiver is ready to accept) to the sender. This size represents the free space in the receiver's buffer (*can be calculated by subtracting bytes not consumed the process from allocated buffer as shown in fig. 4.12*) and the sender must stay within this limit when transmitting data.

Send window

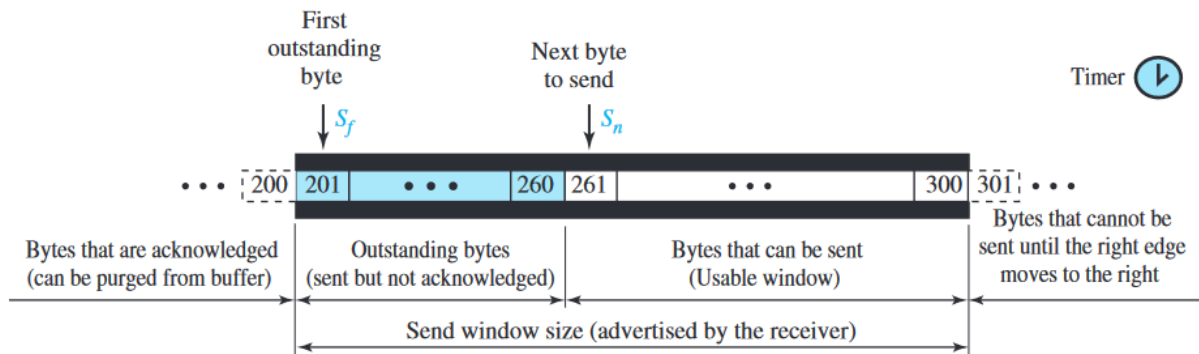


Fig. 4.13 Send Window

The sender adjusts its transmission rate based on the **rwnd** value received from the receiver.

If the **rwnd** is large, the sender can send more data at once. If it is small, the sender must slow down or stop until more space is available.

The sender keeps track of the amount of data that has been sent but not yet acknowledged (*as shown in fig. 4.13*) and ensures that this amount stays within the receiver's advertised window.

Congestion Control in TCP

Congestion Window (cwnd)

- Use of **rwnd** guarantees that, there is no congestion at the other end, but there may be *congestion in the middle*. An intermediate router may receive data from more than one sender. No matter how large the buffers of a router may be, it may be overwhelmed with data, results congestion at middle.
- To control the number of segments to transmit, TCP uses another variable called a congestion window, **cwnd**, whose size is controlled by the congestion situation in the network (as we will explain shortly).
- The **cwnd** variable and the **rwnd** variable together define the size of the send window in TCP. The first is related to the congestion in the middle (network); the second is

related to the congestion at the end. The actual size of the window is the minimum of these two.

Actual window size = minimum (rwnd, cwnd)

Congestion Detection

A very interesting point in TCP congestion is that the TCP sender uses only one feedback from the other end to detect congestion: **ACKs**.

The lack of regular, timely receipt of ACKs, which results in a *time-out*, is the sign of a strong congestion; the receiving of *three duplicate ACKs* is the sign of a weak congestion in the network.

Open Loop Congestion Control (Prevention)

Goal: Prevent congestion before it happens.

Closed Loop Congestion Control (Removal)

Goal: Detect and remove congestion after it occurs.

Slow Start (exponential increase) and Congestion Avoidance (additive increase) belong to the Closed Loop Congestion Control mechanism of TCP.

Closed Loop Congestion Policies

1. Slow Start – Exponential Increase

The Slow Start – Exponential Increase algorithm is based on the idea that the size of the congestion window (cwnd) starts with one maximum segment size (MSS), but it increases exponentially as follows.

Slow start: $\rightarrow \text{cwnd} = 2^0 = 1$

After Receiving 1 RTT $\rightarrow \text{cwnd} = 2^1 = 2$

After Receiving 2 RTT $\rightarrow \text{cwnd} = 2^2 = 4$

After Receiving 3 RTT $\rightarrow \text{cwnd} = 2^3 = 8$

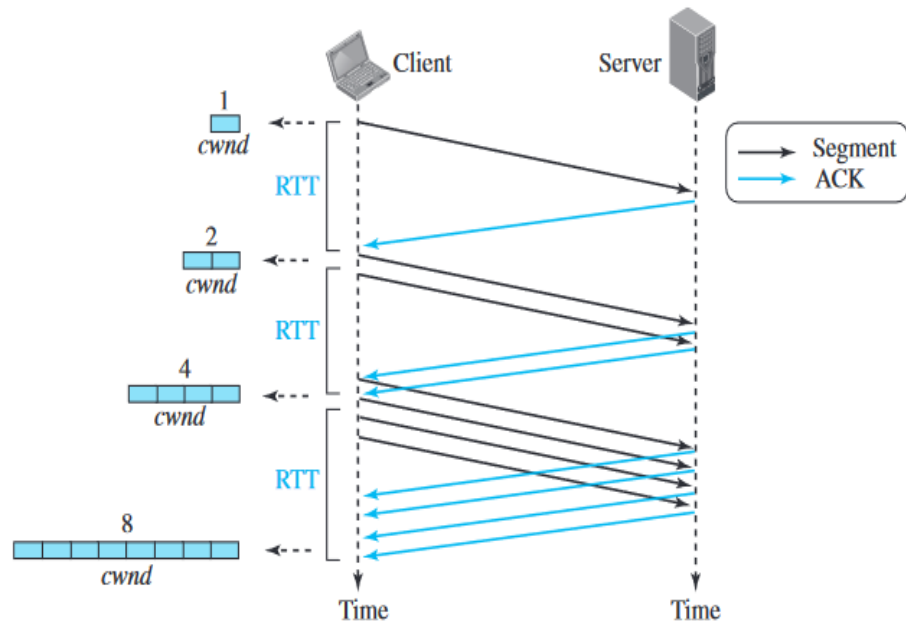


Fig. 4.14 Slow start – Exponential increase

2. Congestion Avoidance – additive increase

TCP defines another algorithm called congestion avoidance, which increases the *cwnd* additively instead of exponentially.

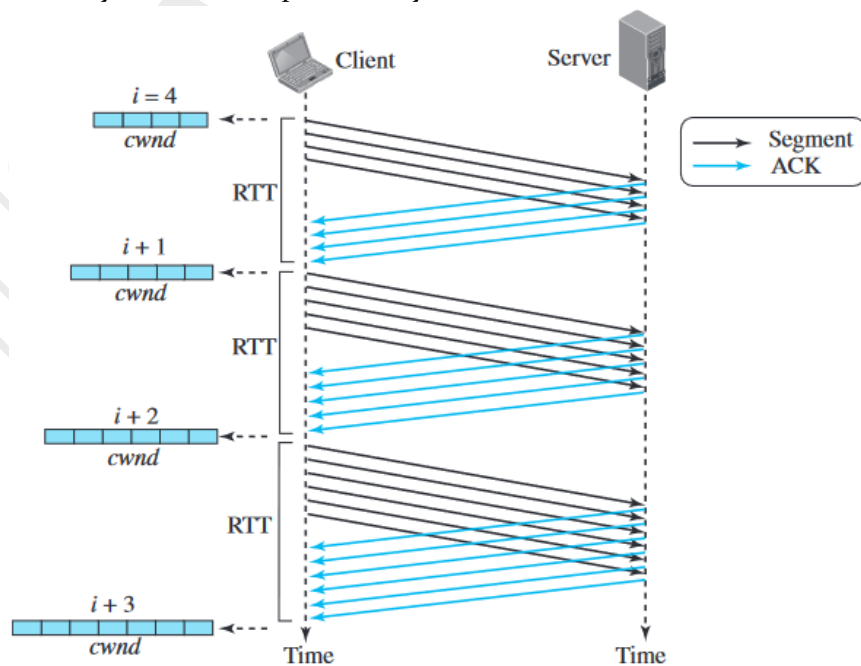


Fig. 4.15 Congestion avoidance – additive increase

SCTP – Stream Control Transmission Protocol

- Stream Control Transmission Protocol (SCTP) is a new reliable, message-oriented transport layer protocol.
- SCTP combines the best features of UDP and TCP. SCTP is a reliable message-oriented protocol. It preserves the message boundaries and at the same time detects lost data, duplicate data, and out-of-order data. It also has congestion control and flow control mechanisms.

SCTP Services

- Process-to-Process Communication
- Multiple Streams :
- Multihoming.
- Full-Duplex Communication
- Connection-Oriented Service
- Reliable Service

Multiple Stream

SCTP allows multi stream service in each connection, which is called association in SCTP terminology. If one of the streams is blocked, the other streams can still deliver their data.

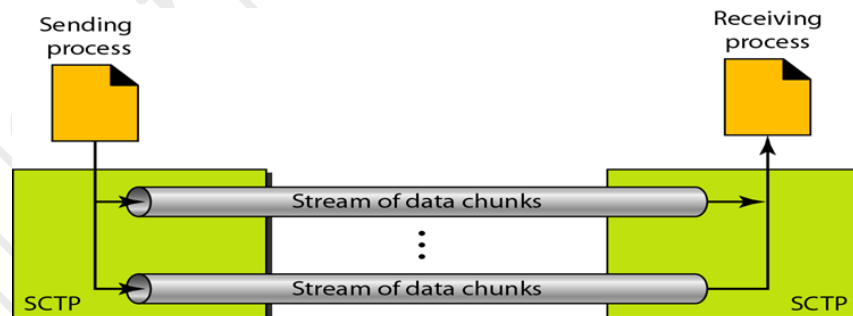


Fig. 4.16 Multiple streams in SCTP association

Multihoming concept in SCTP

An SCTP association supports multihoming service.

The sending and receiving host can define multiple IP addresses in each end for an association. In this fault-tolerant approach, when one path fails, another interface can be used for data delivery without interruption.

SCTP Features

TSN

The Transmission Sequence Number (TSN) is a *globally unique sequence number* assigned to every SCTP data chunk transmitted *within an SCTP association*.

SSN

The Stream Sequence Number (SSN) is a sequence number assigned to every SCTP data chunk transmitted within a particular stream in SCTP association. Its an identifier for data chunk over a stream.

SI

The Stream Identifier (SI) is a unique number used to identify a specific stream within an SCTP association.

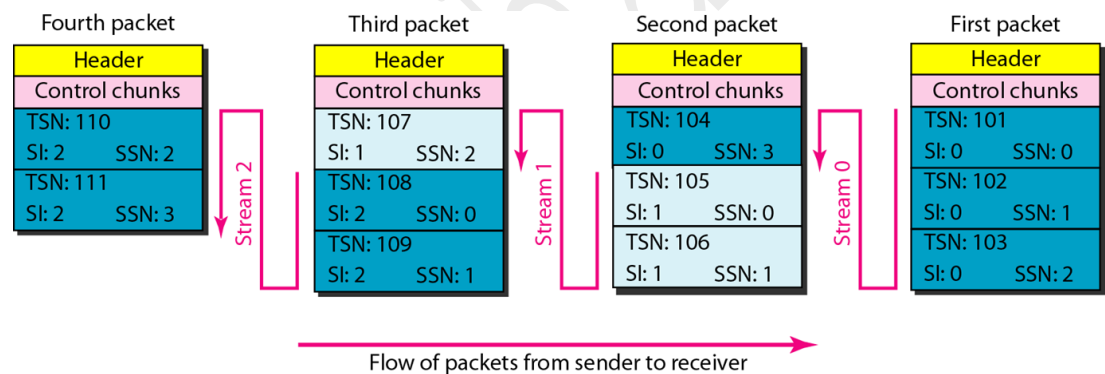


Fig. 4.17 Packets, data chunks, and streams

Packets

In SCTP packet is collection of GENERAL HEADER, CONTROL CHUNKS, DATA CHUNKS.

Acknowledgment Number

In SCTP, acknowledgment numbers are used to acknowledge only data chunks; control chunks are acknowledged by other control chunks if necessary.

Flow Control

Like TCP, SCTP implements flow control to avoid overwhelming the receiver.

Error Control

Like TCP, SCTP implements error control to provide reliability. TSN numbers and acknowledgment numbers are used for error control. We will discuss error control later in the chapter.

Congestion Control

Like TCP, SCTP implements congestion control to determine how many data chunks can be injected into the network.

SCTP Packet Format

- An SCTP packet has a mandatory general header and a set of blocks called chunks.
- There are two types of chunks: control chunks and data chunks.
- A control chunk controls and maintains the association, a data chunk carries user data.

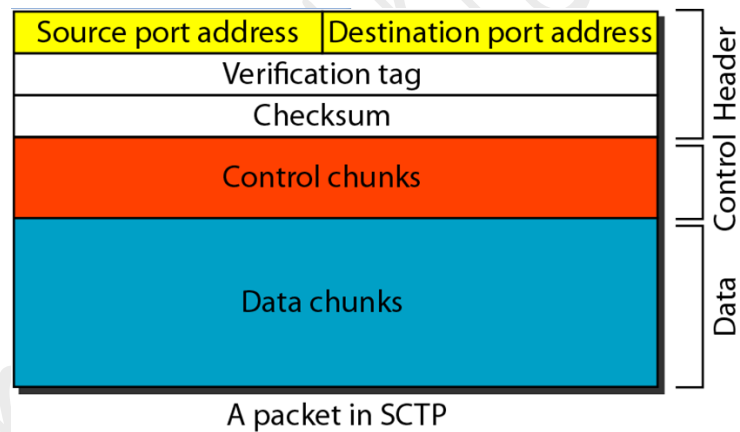


Fig. 4.18 SCTP Packet Format

GENERAL HEADER

Source port address.

This is a 16-bit field that defines the port number of the process sending the packet.

Destination port address.

This is a 16-bit field that defines the port number of the process receiving the packet.

Verification tag.

This is a number that matches a packet to an association. This prevents a packet from a previous association from being mistaken as a packet in this association. It

serves as an identifier for the association; it is repeated in every packet during the association. There is a separate verification used for each direction in the association.

Checksum

This 32-bit field contains a CRC-32 checksum. Note that the size of the checksum is increased from 16 (in UDP, TCP, and IP) to 32 bits to allow the use of the CRC-32 checksum.

CHUNKS

Control information or user data are carried in chunks. Chunks have a common layout, as shown in Figure 4.19. The first three fields are common to all chunks; the information field depends on the type of chunk.

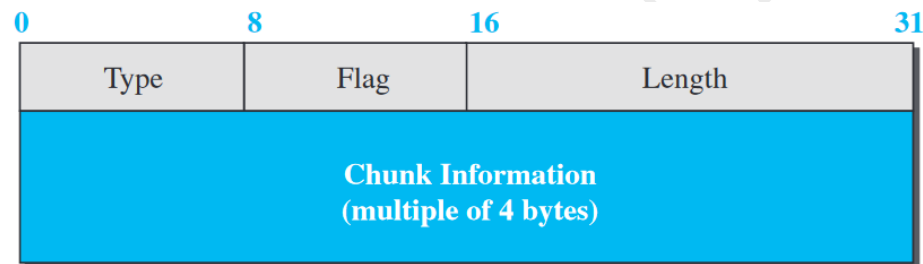


Fig. 4.18 common layout of a SCTP chunk

Types of Chunks

Type	Chunk	Description
0	DATA	User data
1	INIT	Sets up an association
2	INIT ACK	Acknowledges INIT chunk
3	SACK	Selective acknowledgment
4	HEARTBEAT	Probes the peer for liveliness
5	HEARTBEAT ACK	Acknowledges HEARTBEAT chunk
6	ABORT	Aborts an association
7	SHUTDOWN	Terminates an association
8	SHUTDOWN ACK	Acknowledges SHUTDOWN chunk
9	ERROR	Reports errors without shutting down
10	COOKIE ECHO	Third packet in association establishment
11	COOKIE ACK	Acknowledges COOKIE ECHO chunk
14	SHUTDOWN COMPLETE	Third packet in association termination
192	FORWARD TSN	For adjusting cumulating TSN

Imran Shaiikh



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Chapter 4 **Application Layer Protocols** (14 marks)

4.1 Domain Name System (DNS) architecture, Domain types, DNS name space, Domain name resolution & mapping to physical addresses

4.2 Electronic mail

i) Message Transfer Agent - Simple Mail Transfer Protocol (SMTP)

Components, Working

ii) Message Access Agent - Post Office Protocol (POP) and Internet Message Access Protocol (IMAP)

4.3 File Transfer Protocol (FTP), Anonymous FTP

4.4 Remote logging: Telnet, Remote Desktop

4.5 World Wide Web (WWW) and Hyper Text Transfer Protocol (HTTP)- Architecture, Types of web documents, HTTP transaction

4.6 Pretty Good Privacy (PGP)-Security Parameters, Services, A Scenario or Overview of -PGP algorithms, Key rings, PGP certificates

Domain Name System – Architecture

DNS architecture is a hierarchical, distributed system of servers that translates human-readable domain names into IP addresses.

It consists of Root, Top-Level Domain (TLD), Authoritative name servers and Resolvers organized in an inverted tree structure with the root domain at the top.

Resolvers

The resolvers are the computers that actually query the DNS servers to resolve domain names to IP addresses. Resolvers can be web browsers , email clients, or any other application that needs to access resources on the Internet.

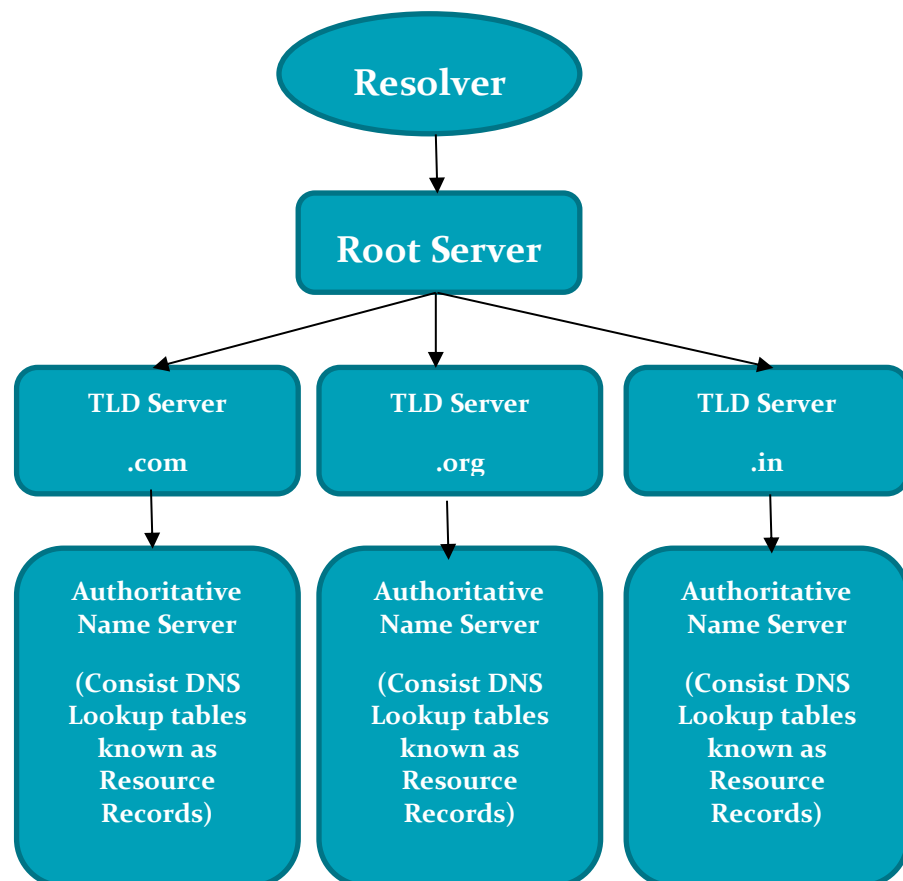


Fig. 4.1 Domain Name System – Architecture

Root DNS Server

The top-level of the DNS hierarchy.

Directs queries to appropriate Top-Level Domain (TLD) servers (like .com, .org, .in).

TLD Servers

The TLD servers are responsible for the mapping to their respective authoritative name servers. There can be more than one authoritative name server for one DNS, so TLD servers return IP address mapping to them so that resolver can go to authoritative name server.

Authoritative name servers

The authoritative name servers are responsible for storing the mapping of domain names to IP addresses for a particular TLD or domain.

Domain Types

Domains are categorized based on their position in the DNS hierarchy.

- **Top-Level Domain (TLD):**
 - The highest level in the DNS hierarchy.
 - Examples:
 - Generic TLDs (gTLDs): .com, .org, .net, .edu
 - Country Code TLDs (ccTLDs): .in, .uk, .us
- **Second-Level Domain (SLD):**
 - Directly below the TLD in the hierarchy.
 - Example: In google.com, google is the second-level domain.
- **Subdomain:**
 - A part of a larger domain.
 - Example: mail.google.com or docs.google.com are subdomains of google.com.

- **Root Domain:**

- The topmost part of the hierarchy, represented by a dot (.) at the end of a domain name (usually hidden in browsers).

DNS Namespace

The DNS namespace is the hierarchical tree structure that organizes all domain names on the Internet.

- The tree starts from the root (.) at the top.
- Below the root are TLDs, then second-level domains, and then subdomains.

Label

- A label is a single part (component) of a domain name, separated by dots (.).
- Each level in the DNS hierarchy is represented by one label.
- A label can have up to 63 characters. Labels are written from left to right, with the most specific part on the left and the least specific (root) on the right.

In the domain name **www.example.com**:

www → Label for a host

example → Label for the organization or domain

com → Label for the top-level domain (TLD)

So, this domain has three labels: **www**, **example**, and **com**.

Domain Name

A full domain name is a sequence of labels separated by dots (.).

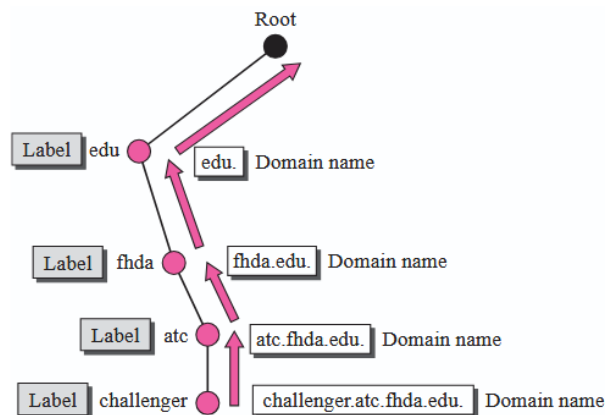


Fig. 4.2 DNS Space – Labels and Domain Names

Domain Name Resolution

The following six steps map the host name to an IP address.

1. The user passes the host name to the file transfer client.
2. The file transfer client passes the host name to the DNS client.
3. We know from Chapter 18 that each computer, after being booted, knows the address of one DNS server. The DNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
4. The DNS server responds with the IP address of the desired file transfer server.
5. The DNS client passes the IP address to the file transfer server.
6. The file transfer client now uses the received IP address to access the file transfer server.

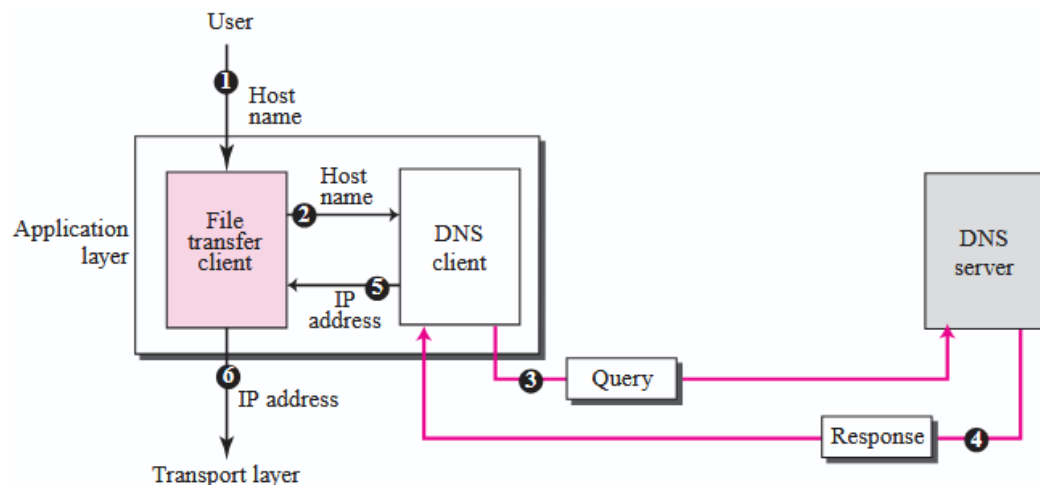


Fig. 4.3 Domain Name Resolution

Electronic Mail

Email Architecture

When both sender and receiver are connected to the mail server via a LAN or a WAN, we need two UAs, two pairs of MTAs (client and server), and a pair of MAAs (client and server).

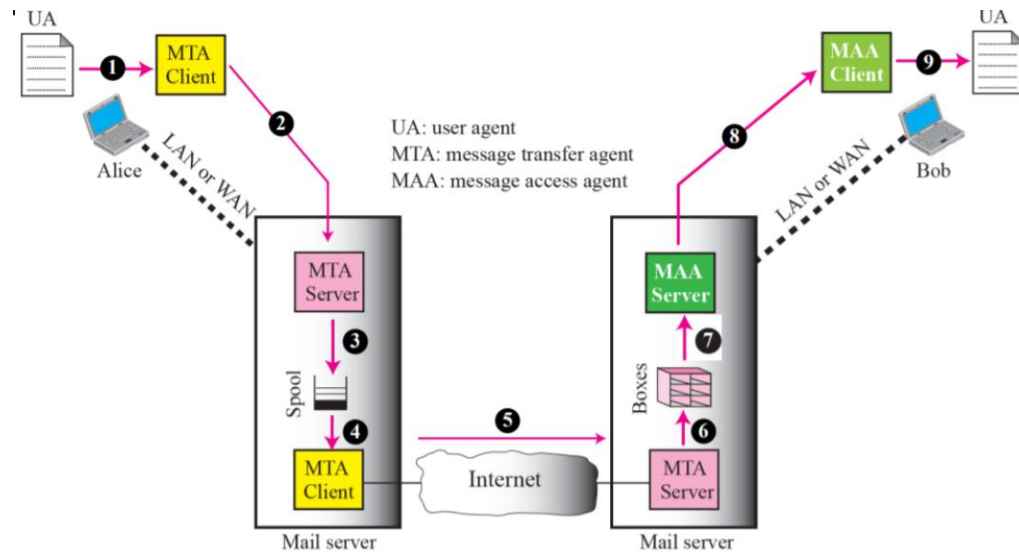


Fig. 4.4 Email Architecture (Common Scenario)

1. Sender runs a UA program to prepare the message and send it to its own mail server(step 1,2); message is stored in pool at sender's mail server(step 3). (Communication between MTA Client at Senders PC and MTA server at senders mail server)
2. Sender mail server now sent the message to receivers mail server. (step 4,5).(Communication between MTA Client at Senders mail server and MTA server at receivers mail server).
3. MTA server at Receivers mail server stores messages in to mail boxes (step 6).
4. When receiver PC request for any message from mail box, it act as a MAA client to request message and receivers mail server as a MAA server to serve that message to receivers PC (MAA client).

Email Components

- **User Agent**
- **Mail Transfer Agent**
- **Mail Access Agent**

User Agent (UA)

- The first component of an electronic mail system is the user agent (UA).
- It provides service to the user to make the process of sending and receiving a message easier.
- Some examples of GUI-based user agents are Eudora, Outlook, And Netscape.
- UA provide following services as shown in figure.

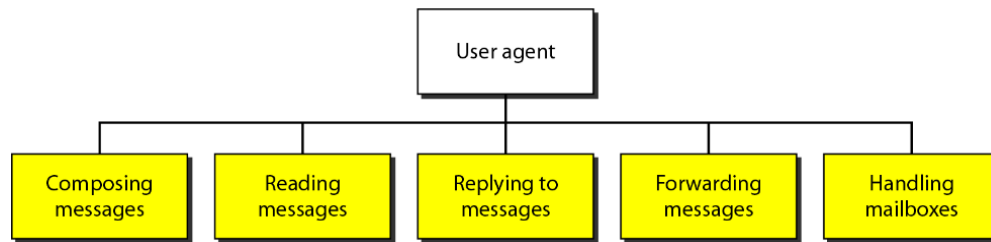


Fig. 4.5 User Agent Services

Message Transfer Agent (MTA)

- The actual mail transfer is done through message transfer agents (MTAs).
- To send mail, a system must have the client MTA, and to receive mail, a system must have a server MTA.
- The formal protocol that defines the MTA client and server in the Internet is called Simple Mail Transfer Protocol (SMTP).

Message Access Agent (MAA)

- It is responsible for retrieving messages from the receivers mail server and making them available for reading by the user.
- The MAA does not send emails; it only retrieves and manages the emails stored on the server.

Simple Mail Transfer Protocol (SMTP)

- SMTP work as a Message Transfer Agent (MTA) in Email Technology.
- SMTP works on port no 25.
- SMTP can transfer only 7 bit ASCII data.
- SMTP transfer the mails from Senders PC to Senders mail server and from Senders mail server to Receivers mail server as shown in figure.

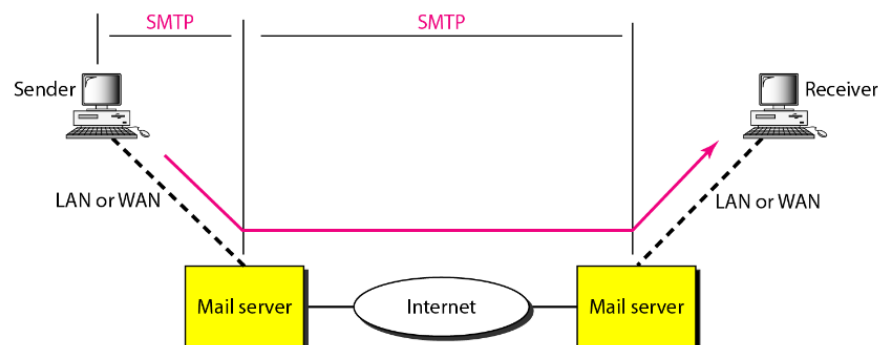


Fig. 4.6 SMTP

SMTP Working

SMTP works in three steps

- Connection establishment
- Message data transfer
- Connection Termination

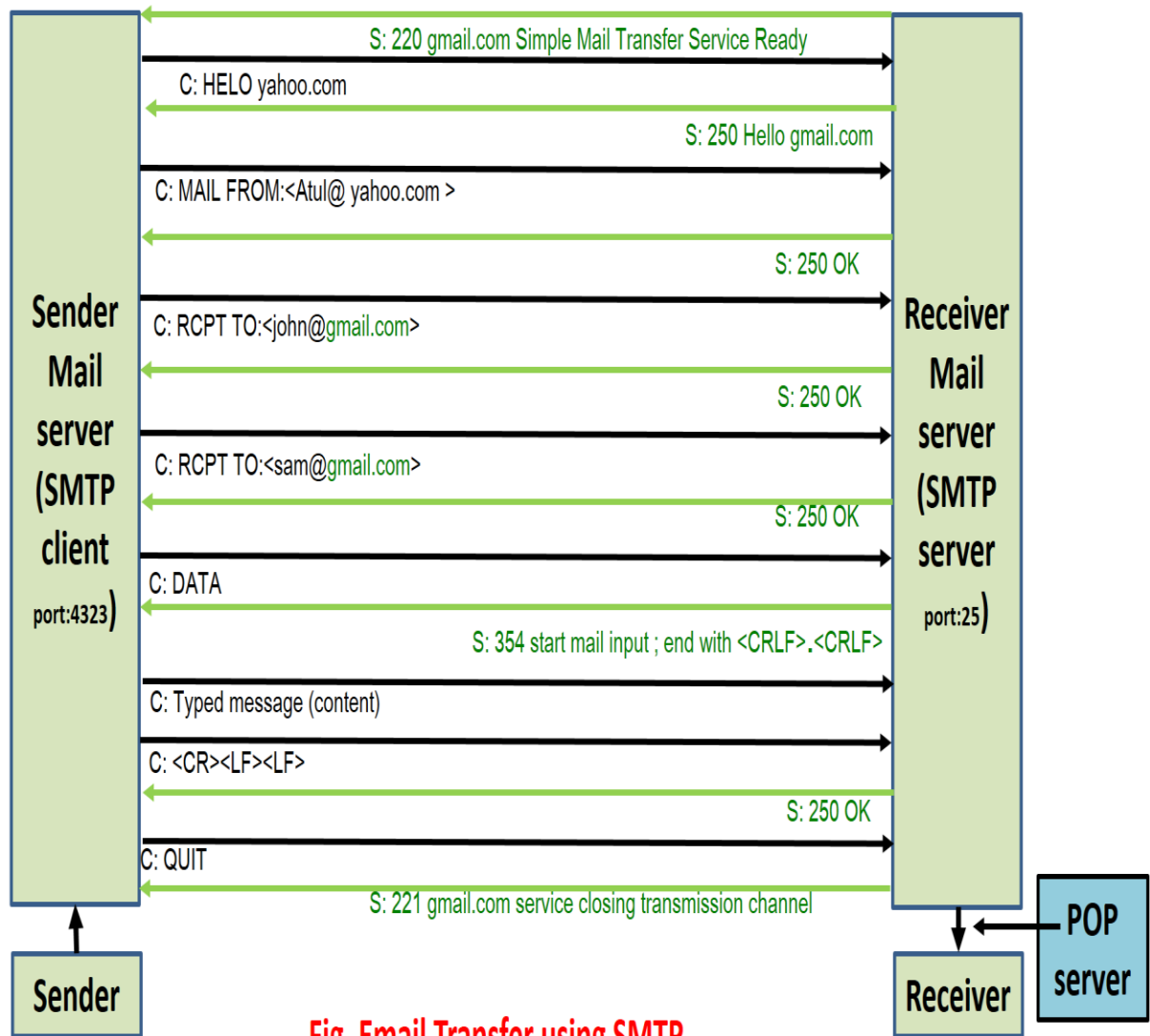


Fig. Email Transfer using SMTP

Fig. 4.7 SMTP working

SMTP Commands

HELO [domain] or EHLO [domain]:

Initiates the conversation with the SMTP server. EHLO is the extended version and is preferred for modern servers.

Example: EHLO example.com

MAIL FROM:<sender@example.com>:

Specifies the sender's email address.

Example: MAIL FROM:<alice@example.com>

RCPT TO:<recipient@example.com>:

Specifies the recipient's email address. You can issue multiple RCPT TO commands if there are multiple recipients.

Example: RCPT TO:bob@example.com

DATA:

Signals the server that message content is about to be sent. After issuing this command, you type the message body and end with a single period . on a line by itself.

QUIT:

Ends the SMTP session and disconnects from the server.

Post Office Protocol 3 (POP3)

POP3 is a standard protocol used for retrieving emails from a remote server to a local client. It is designed to enable users to download their emails from a mail server so they can be read locally on a device.

POP3 working

- Connection Establishment
- Retrieving Emails
- Ending the Session

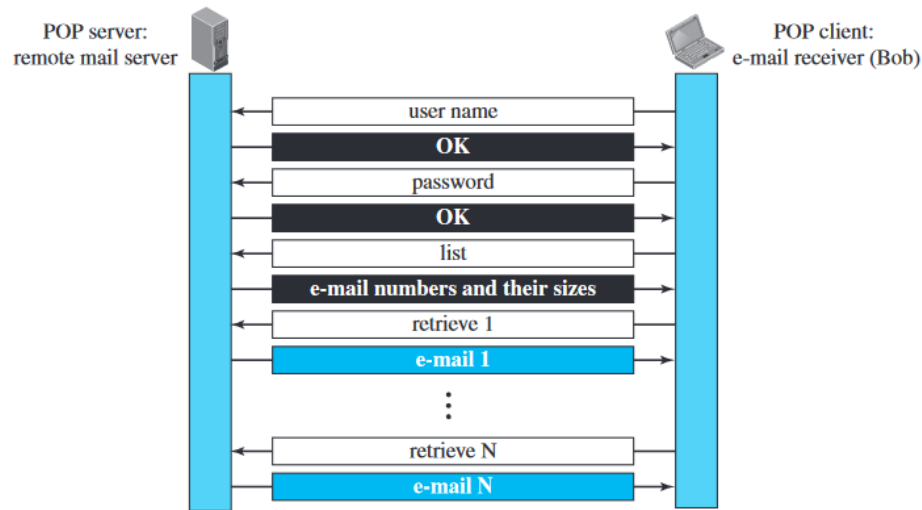


Fig. 4.8 POP₃ working

Internet Message Access Protocol (IMAP)

IMAP is a protocol used for retrieving emails from a mail server to a local client, such as a computer or mobile device.

It allows users to access and manage their emails directly on the server, making it particularly useful for accessing email from multiple devices.

IMAP operates at the application layer of the TCP/IP model.

It uses TCP for reliable communication, typically over port 143 for unencrypted communication and port 993 for IMAPS (IMAP over SSL/TLS).

IMAP₄ provides the following extra functions as compare to POP

- A user can check the e-mail header prior to downloading.
- A user can search the contents of the e-mail for a specific string of characters prior to downloading.
- A user can partially download e-mail. This is especially useful if bandwidth is limited and the email contains multimedia with high bandwidth requirements.
- A user can create, delete, or rename mailboxes on the mail server.
- A user can create a hierarchy of mailboxes in a folder for e-mail storage.

Difference between POP₃ and IMAP

Parameter	POP ₃	IMAP
Functionality	POP ₃ client downloads messages from server and then deleted them from server.	IMAP client Synchronize with server, allows users to access and manage their emails directly on the server
Storage	Less storage required at server.	Server need high storage capacity.
Offline Access	Yes	No
Email Management	No cannot manage mails on server.	Yes, emails can be stored in folders.
Uses	Suitable for Single device use.	Suitable for multiple device use.
Port No	110	143

Difference between SMTP and POP₃

Parameter	SMTP	POP ₃
Purpose	Sending emails	Receiving emails
Direction	Outgoing	Incoming
Connection	Persistent	Temporary
Port	25	110
Email Storage	Does not store emails	Download to client.

File Transfer Protocol

- FTP uses the services of TCP. It needs two TCP connections.
- The well-known port **21** is used for the control connection and the well-known port **20** for the data connection.
- The control connection remains connected during the entire interactive FTP session.
- The data connection is opened and then closed for each file transferred.
- Control connection is there between client & server Control processes.
- Data connection is there between client & server Data transfer processes.

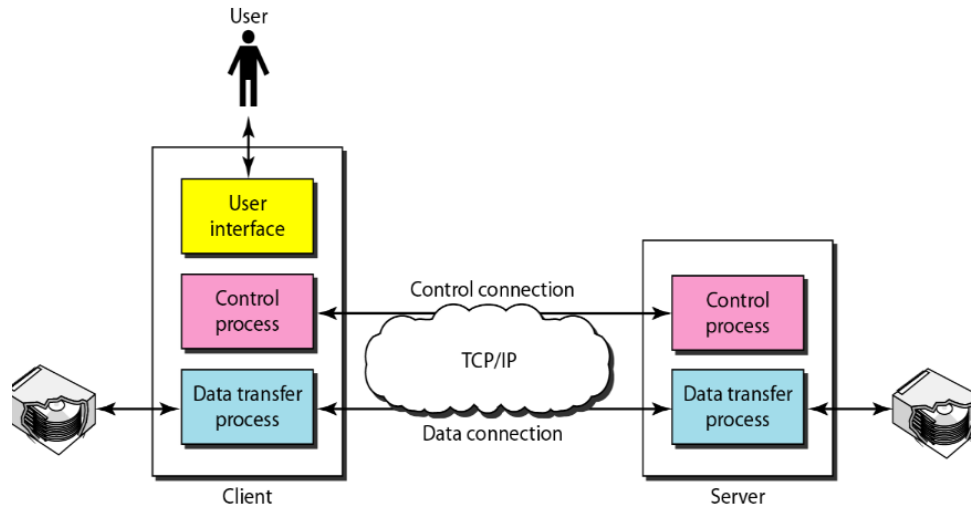


Fig. 4.9 File Transfer Protocol (FTP)

Communication over FTP Control Connection

- It uses the 7-bit ASCII character set (see Figure).
- Communication is achieved through commands and responses.
- Each command or response is only one short line, so we need not worry about file format or file structure.

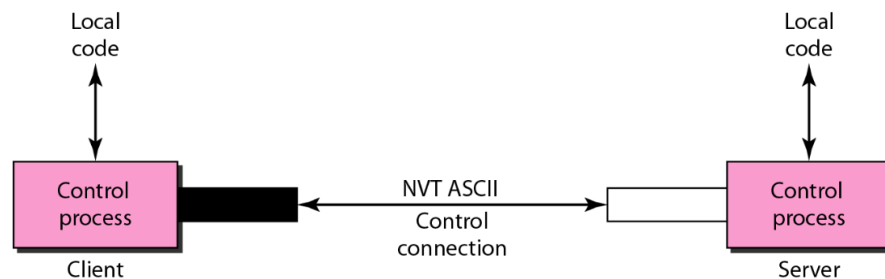


Fig. 4.10 FTP Control Connection

Communication over FTP Data Connection

- The purpose of the data connection is different from that of the control connection. It is used to transfer files.
- One of problem is that file structure at client and server may be differ, known as heterogeneity problem.
- The heterogeneity problem is resolved by defining three attributes of communication: **file type**, **data structure**, and **transmission mode**.

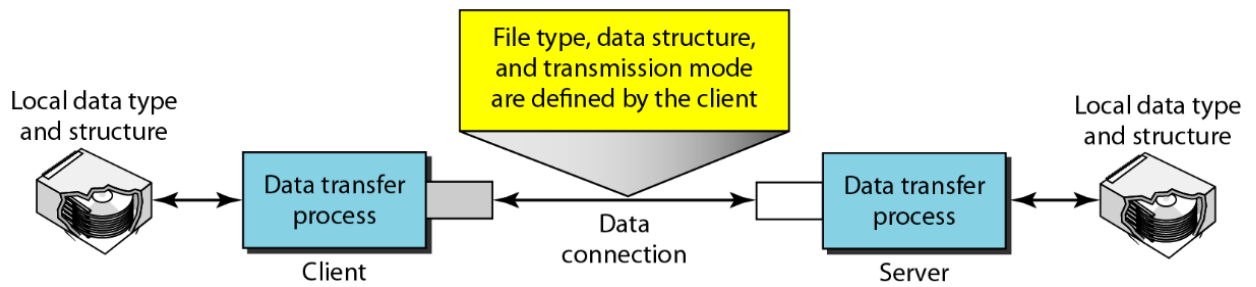


Fig. 4.11 FTP Data Connection

FTP Commands

get [filename]: Downloads a file from the FTP server to your local directory.

mget [file1 file2 ...]: Downloads multiple files from the server.

put [filename]: Uploads a file from your local machine to the FTP server.

mput [file1 file2 ...]: Uploads multiple files to the FTP server.

delete [filename]: Deletes a file from the FTP server.

rename [oldname] [newname]: Renames a file on the server.

mkdir [directory]: Creates a new directory on the FTP server.

rmdir [directory]: Removes an empty directory on the FTP server.

Main differences between FTP and TFTP are

1. FTP uses TCP port numbers 20 and 21, TFTP uses UDP port number 69.
2. FTP is a complete, session-oriented, general purpose file transfer protocol. TFTP is used as a special purpose file transfer protocol.
3. FTP can be used interactively. TFTP allows only unidirectional transfer of files.
4. FTP depends on TCP, is connection oriented, and provides reliable control. TFTP depends on UDP, requires less overhead, and provides virtually no control.
5. FTP provides user authentication. TFTP does not.

Anonymous FTP

Anonymous FTP is a method that allows users to access files on an FTP server without needing a personal username and password.

It is commonly used to provide public access to files such as software updates, documents, or datasets.

Remote Login - TELNET

- TELNET is an abbreviation for TErminaL NETwork. It is the standard TCP/IP protocol for virtual terminal service, which runs at port no 23.
- TELNET enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system.
- There are two parties involved TELNET Client and TELNET server.
- TELNET client, which transforms the characters to a universal character set called network virtual terminal (NVT) characters and delivers them to the local TCP/IP protocol stack.

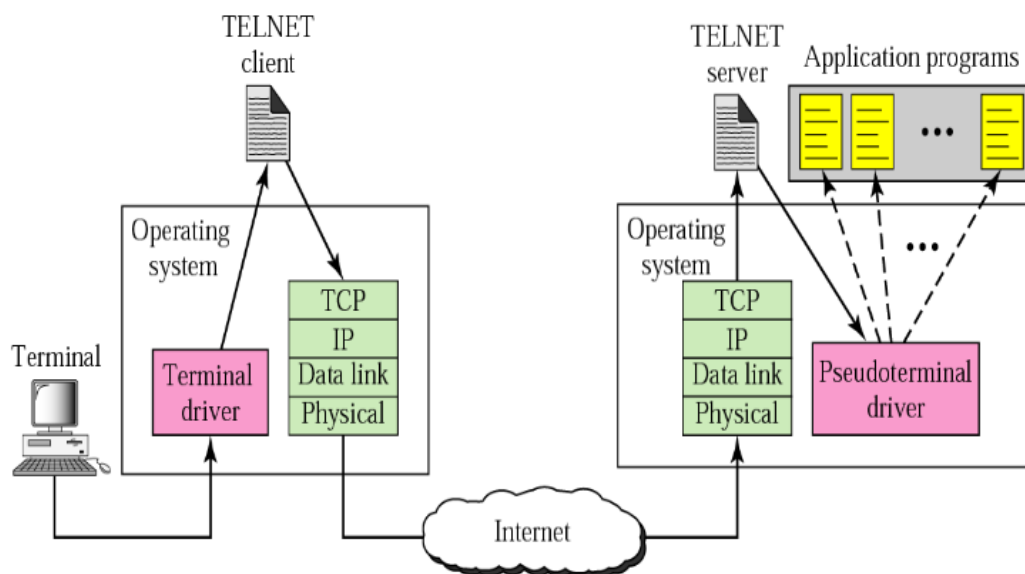


Fig. 4.12 TELNET Working

- The commands or text, in NVT form, travel through the Internet and arrive at the TCP/IP stack at the remote machine. Here the characters are delivered to the operating system and passed to the TELNET server, which changes the characters to the corresponding characters understandable by the remote computer, which is understandable by OS of remote machine.

World Wide Web (WWW)

- WWW is a repository of information linked together from points all over the world.
- The WWW has a unique combination of flexibility, portability, and user-friendly features that distinguish it from other services provided by the Internet.

WWW Architecture

The WWW today is a distributed client/server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called sites.

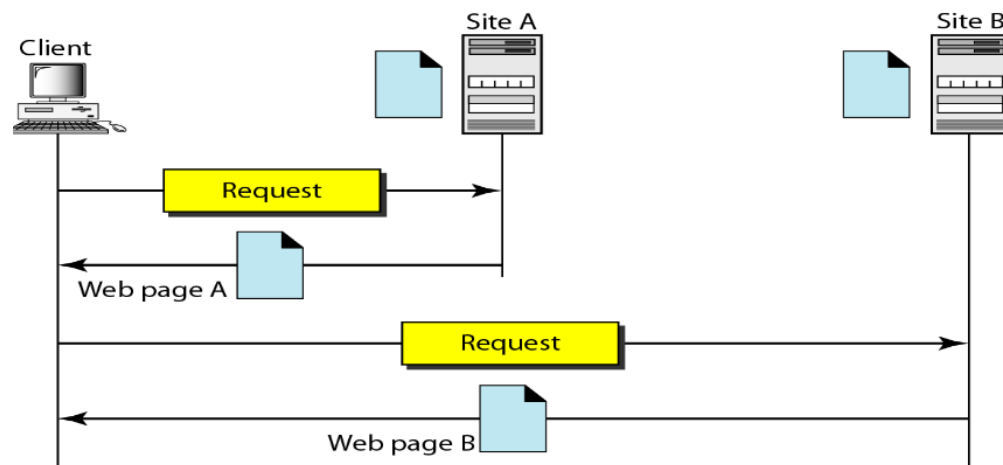


Fig. 4.13 Architecture of WWW

Hyper Text Transfer Protocol (HTTP)

- The Hypertext Transfer Protocol (HTTP) is a Application layer protocol used mainly to access data on the World Wide Web.
- HTTP uses the services of TCP on well-known port 80.

Types of web documents

Web documents are the different kinds of files or content that can be accessed through the World Wide Web (WWW) using a browser.

They are usually identified by a URL (Uniform Resource Locator) and can contain text, images, multimedia, or scripts.

Static Web Documents

These are fixed content pages created using HTML.

The content does not change automatically — the same page is shown to every user.

Any change requires the developer to manually edit the HTML file on the server.

Example:

A company's "About Us" page or contact information page.

Dynamic Web Documents

Content is generated on the server side each time a user requests the page.

Uses server-side scripting languages like PHP, Python, ASP, JSP, or Node.js.

Data is often fetched from a database and displayed dynamically.

Example:

News websites, e-commerce pages, or social media feeds.

Active Web Documents

These are interactive pages that can change or update after loading, without reloading the whole page.

Uses client-side scripting languages such as JavaScript, AJAX, or HTML5 APIs.

Example:

Live chat windows, instant search suggestions, online games, or maps.

HTTP Transaction

HTTP Transaction Figure 5.5 illustrates the HTTP transaction between the client and server. The client initializes the transaction by sending a request message. The server replies by sending a response.

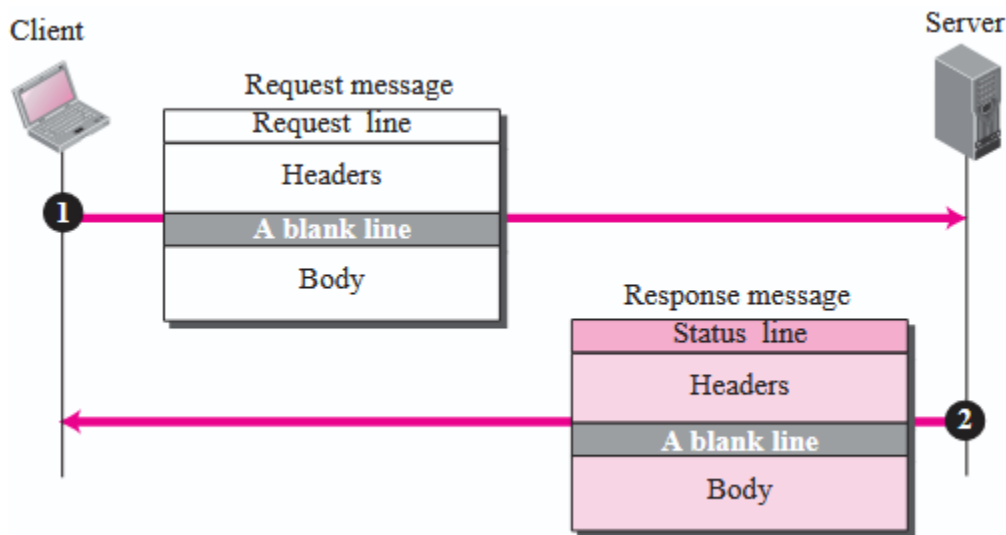


Fig. 4.14 HTTP transaction

HTTP Request Message

Request Line

The Request-Line begins with a Request type (method), followed by the Request-URL and the protocol version.

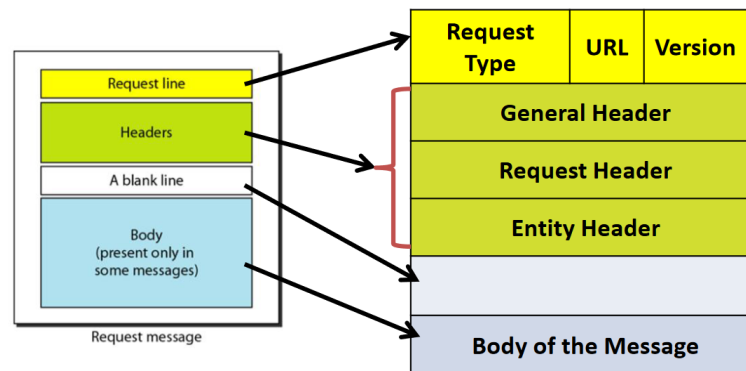


Fig. 4.15 HTTP Request Message

Headers

- General Header
- Request Header
- Entity Header

General Header

The general header gives general information about the message and can be present in both a request and a response.

E.g.

Cache-control: to specify information about caching.

Date: to specify current date.

Request Header

It specifies the client's configuration and the client's preferred document format.

E.g.

Accept: to specify format the client can accept.

Accept-charset: to specify character set the client can handle.

Entity Header

The entity header gives information about the body of the document.

E.g.

Accept: to specify format the client can accept.

Accept-charset: to specify character set the client can handle.

Body of the document

It contains the data that can be sent.

HTTP Response Message

Status Line

The status-Line begins with a protocol version, then status code and status phrase.

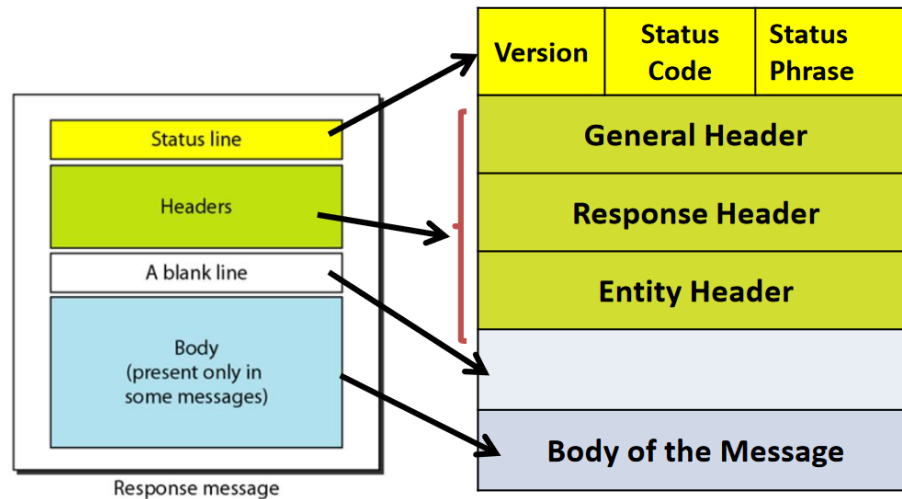


Fig. 4.16 HTTP Response Message

Headers

- General Header
- Response Header
- Entity Header

General Header

The general header gives general information about the message and can be present in both a request and a response.

E.g.

Cache-control: to specify information about caching.

Date: to specify current date.

Response Header

It specifies the server's configuration and special information about the request.

E.g.

Server: to specify server name and version number.

Entity Header

The entity header gives information about the body of the document.

E.g.

Accept: to specify format the client can accept.

Accept-charset: to specify character set the client can handle.

Body of the document

It contains the data that can be sent.

Application Layer Security

Email Security

Two protocols providing security services for e-mails: Pretty Good Privacy (PGP) and Secure/Multipurpose Internet Mail Extension (S/MIME).

Pretty Good Privacy

Security Parameters of PGP

PGP provides several security parameters (objectives) to ensure safe communication

Confidentiality

Ensures that only the intended recipient can read the message (using encryption).

Authentication

Confirms the sender's identity (using digital signatures).

Integrity

Ensures that the message has not been altered during transmission (via hashing).

Non-repudiation

Prevents the sender from denying having sent the message (digital signatures provide proof).

Compression

Reduces message size to improve transmission efficiency and security.

Services Provided by PGP

PGP provides the following security services:

Message Encryption (Confidentiality)

- Uses a session key (symmetric encryption) to encrypt the message.
- The session key itself is encrypted using the receiver's public key (asymmetric encryption).

Digital Signature (Authentication & Integrity)

- The sender computes a hash of the message (using SHA or MD5).
- The hash is encrypted using the sender's private key, forming a digital signature.

Compression

- The message is compressed before encryption to save bandwidth and add an extra layer of security.

Email Compatibility

- Converts binary data into ASCII format for easy email transmission (via Radix-64 encoding).

PGP Working

Sender Site (Encryption Process):

- The sender generates a random symmetric key (also called a session key) to encrypt the email using a fast symmetric encryption algorithm (e.g., AES).
- This session key is then encrypted with the recipient's public key using asymmetric encryption (e.g., RSA).
- Email is hashed and a digest is created, then email digest is encrypted using private key of sender.
- The encrypted session key, Email digest and the encrypted Email are sent to the recipient.

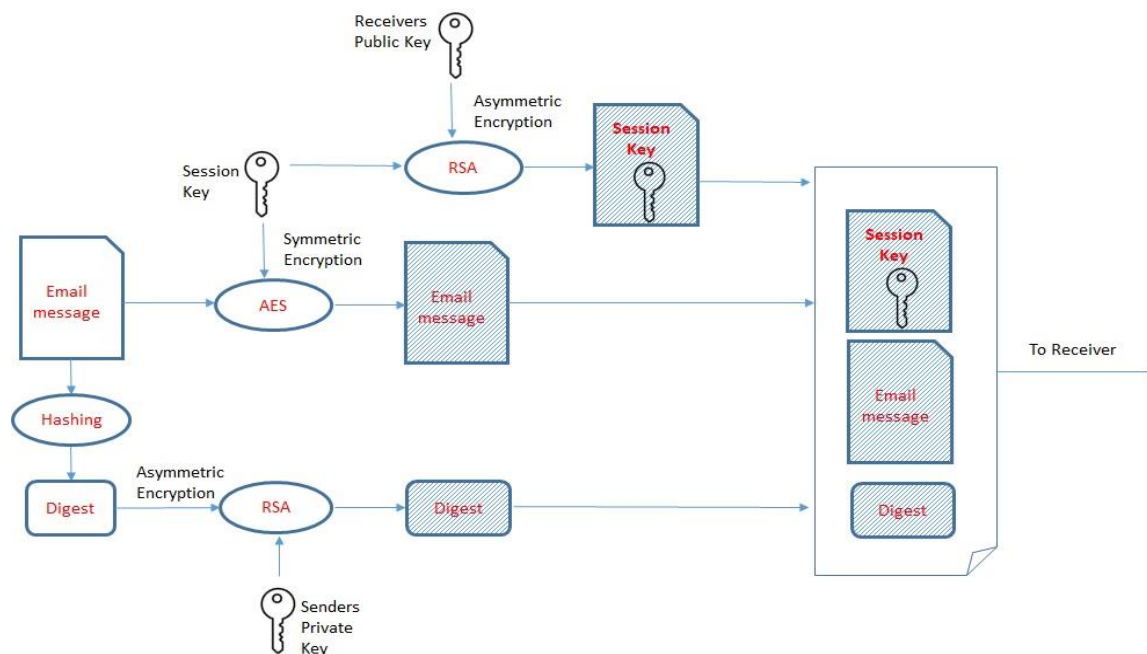


Fig. 4.17 PGP Working

Receiver Site (Decryption Process)

- The recipient first uses their private key to decrypt the session key.
- The decrypted session key is then used to decrypt the actual email message. Recipient hash the email message to create email digest at receiver side.
- The recipient decrypt email digest received from sender using public key of sender, then compare email digest from sender with email digest generated at its own site, if match is found email is accepted otherwise email is discarded.

PGP Algorithms

PGP defines a set of asymmetric-key and symmetric-key algorithms, cryptography hash functions, and compression methods.

Function	Algorithm Used	Purpose
Message encryption	AES	Symmetric encryption for fast data encryption
Public key encryption	RSA	To encrypt session keys and verify digital signatures
Hashing	MD5	To ensure message integrity
Compression	ZIP	To compress message before encryption
Radix-64 Conversion	Base64 encoding	To convert binary data to ASCII for email transmission

Key Rings

In all previous scenarios, we assumed that Sender needs to send a message only to Receiver. That is not always the case. Sender may need to send messages to many people; he/she needs key rings.

PGP maintains two key storage areas called key rings:

Public Key Ring

Stores the public keys of other users (used to encrypt messages to them or verify their signatures).

Private Key Ring

Stores the user's own private/public key pairs (used to sign or decrypt messages).

PGP Certificates

- A **PGP certificate** binds a **user's identity (name, email)** to their **public key**.
- Unlike traditional digital certificates (which rely on Certificate Authorities), PGP uses a **Web of Trust** model.

Web of Trust Concept:

- Users personally sign each other's public keys to verify authenticity.
- Trust builds through multiple verifications among users instead of a central authority.

Example:

If Alice trusts Bob, and Bob trusts Carol, then Alice can trust Carol's public key because Bob vouched for it.

Unit - V

Wireless Network Technologies

(Marks = 14)

5.1 Wireless Network Communication- 3G, 4G, 5G

5.2 SDN (Software Defined Network)- Architecture, Working, Applications

5.3 Network Functions Virtualization (NFV)-Architecture, Benefits, Applications

5.4 Edge Computing and Edge Networking-Definition, Components, Challenges, Applications

5.5 Multimedia Wireless Networks – Streaming Audio and Video, Voice Over Internet Protocol (VoIP), Protocols – Real time Transport Protocol(RTP), Real-Time Streaming Protocol (RTSP)

Wireless Network Communication- 3G, 4G, 5G

3G - Third Generation

Introduced in Early 2000s.

3G uses WCDMA/UMTS to send multiple signals over a single channel by assigning each user a unique code.

3G mainly uses microwave frequencies in the lower GHz range, which provide a balance between coverage and data speed. Suitable for voice, video, and mobile internet.

4G – Fourth Generation

Introduced in Around 2010.

4G uses LTE and WiMAX technologies based on IP packet switching, LTE provides faster data transfer and lower latency.

4G uses higher microwave frequencies (especially 2.3 GHz, 2.6 GHz, and 3.5 GHz) for faster data transfer.

These waves carry more data but have slightly shorter range.

5G – Fifth Generation

Introduced in around 2019–2020.

5G uses millimeter waves (24 – 100 GHz) for higher bandwidth, 6 GHz bands are used for broader area coverage.

5G uses millimeter waves that provide ultra-high speed and low latency, but shorter coverage range.

Feature	3G	4G	5G
Introduction Year	2001	2010	2019
Technology	WCDMA / HSPA	LTE / WiMAX	mmWave / Massive MIMO
Frequency Band used	1.8 GHz – 2.5 GHz	2 GHz – 8 GHz	Low Band: 600 MHz – 1 GHz Mid Band: 2.5 – 6 GHz High Band (mmWave): 24 – 100 GHz
Max Speed	Up to 2 Mbps	Up to 1 Gbps	Up to 10 Gbps
Latency	~100 ms	~50 ms	~1 ms
Applications	Voice, video calls, basic internet	HD streaming, gaming	IoT, AI, automation
Spectrum Efficiency	Moderate	High	Very High
Device Connection Density	Limited	Moderate	Massive (IoT)

SDN (Software Defined Network)

SDN (Software-Defined Networking) is a modern networking approach that separates the control plane (decision-making) from the data plane (packet forwarding).

This allows network behavior to be programmed and managed centrally through software not hardware.

SDN applications are the software programs that run on top of the SDN controller to manage, optimize, and secure the network.

SDN Architecture

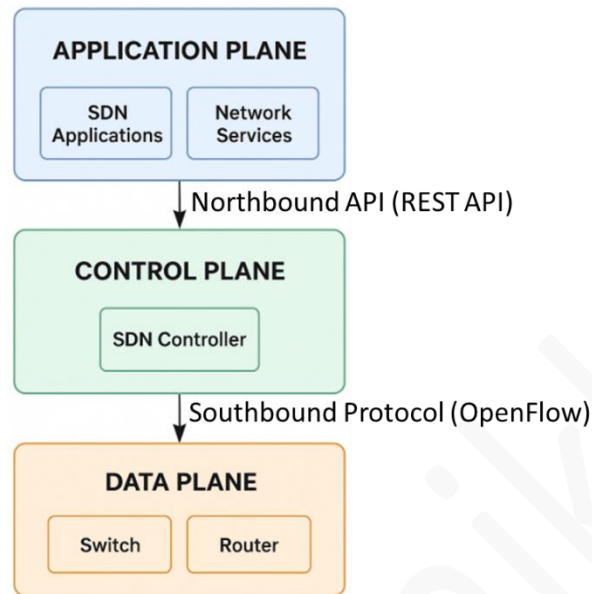


Fig. SDN Architecture

Applications tell the controller what they want, and the controller tells the switches, routers how to forward traffic, to complete requirements of Application plane

SDN architecture can be divided into three planes.

- Application Plane
- Control Plane
- Data Plane

Application plane

It doesn't directly control hardware — instead, it communicates with the SDN Controller (in the Control Plane) through northbound APIs.

The Application Plane talks to the Control Plane using Northbound APIs (usually REST APIs). These APIs are abstract, so the app developer doesn't need to know hardware details.

Example:

An app might just say:

“Ensure 10 Mbps guaranteed for Host A.”

The controller translates this into OpenFlow rules for the switches.

Control Plane

The control plane is responsible for making decisions about where and how packets should be forwarded.

In SDN, this logic is centralized in a controller (software), rather than being embedded in each network device.

Control plane installs flow rules into switches, routers and decides routing paths.

SDN Controller

A software platform (like ONOS, OpenDaylight, Ryu) that acts as the central decision-maker.

Example

Imagine a new device connects to the network. The switch doesn't know how to handle its packets, so it asks the SDN controller.

The controller decides, "Send these packets to Server A" and sends a flow rule to the switch.

Data Plane

The data plane (also called the forwarding plane) is responsible for moving packets based on rules given by the control plane.

Forwards, drops, or modifies packets according to flow table entries.

Example

Once the controller installs a rule like

"Packets from Host X to Host Y → output port 3,"

the switch forwards those packets directly without asking the controller flag

SDN Applications**Network Security**

Centrally monitors and enforces security policies across the network.

Traffic Engineering

Optimizes the flow of data through the network to avoid congestion and improve performance.

WAN Optimization (SD-WAN)

Improves performance of wide area networks (WANs) using centralized control and dynamic routing.

Network Virtualization

Combines multiple physical networks into one virtual network or divides one physical network into multiple virtual networks.

Network Function Virtualization (NFV) is a modern approach to networking where traditional network services (like firewalls, load balancers, intrusion detection systems, etc.) that used to run on dedicated hardware devices are now implemented as software functions running on standard servers, storage, and switches.

Instead of buying special hardware for each function, NFV lets operators use virtualized network functions (VNFs) that run in a virtual environment.

NFV (Network Function Virtualization)

NFV (Network Function Virtualization) is a modern networking technology that replaces traditional hardware-based network devices (like routers, firewalls, and load balancers) with software-based virtual functions that run on standard servers.

NFV Architecture

The NFV architecture consists of three main components:

i. NFV Infrastructure (NFVI):

This is the foundation layer that provides the hardware and virtualization environment for VNFs.

It includes:

- Compute: Standard servers (CPU, RAM).
- Storage: Hard disks, SSDs.
- Network: Physical network to interconnect servers.

Virtualization Layer: Hypervisors (like VMware, KVM, Xen) or container runtimes (like Docker).

Basically, NFVI supplies the resources that VNFs need to run.

ii. Virtual Network Functions (VNFs):

- These are software-based implementations of traditional network functions.
- Each VNF performs a specific task such as firewalling, routing, NAT, or intrusion detection.
- Example: Virtual Firewall, Virtual Router, Virtual Load Balancer.

iii. NFV Management and Orchestration (MANO):

This is the "brain" of the architecture.

Responsible for:

- Deploying VNFs.
- Scaling VNFs up/down when traffic changes.
- Monitoring performance and failures.
- Managing resources in NFVI.

NFV Management and Orchestration			
Orchestrator (NFVO)	VNF Manager (VNFM)	Infrastructure Manager (VIM)	
NFV Infrastructure (NFVI)			
Compute	Storage	Network	Virtualization Layer
Virtualized Network Functions (VNFs)			
<u>vRouter</u>	<u>vFirewall</u>	<u>vLoad Balancer</u>	<u>vIDS</u>

Fig. NFV Architecture

NFV Benefits

Cost Reduction

Reduces the need for expensive dedicated hardware devices; uses general-purpose servers instead.

Faster Deployment

Network functions can be installed as software, reducing time compared to hardware setup.

Vendor Independence

Works on any hardware platform, reducing dependency on specific hardware vendors.

Energy Efficiency

Using fewer hardware devices reduces power and cooling costs.

NFV Applications

Virtual Firewalls (vFW)

Provides software-based network protection that can be easily updated and scaled.

Virtual Routers (vRouter)

Performs routing tasks using software on virtual machines or containers.

Virtual Load Balancers

Distributes network traffic among multiple servers to ensure high availability.

IoT Networks

Helps manage and secure large-scale IoT device communication virtually.

Edge Computing

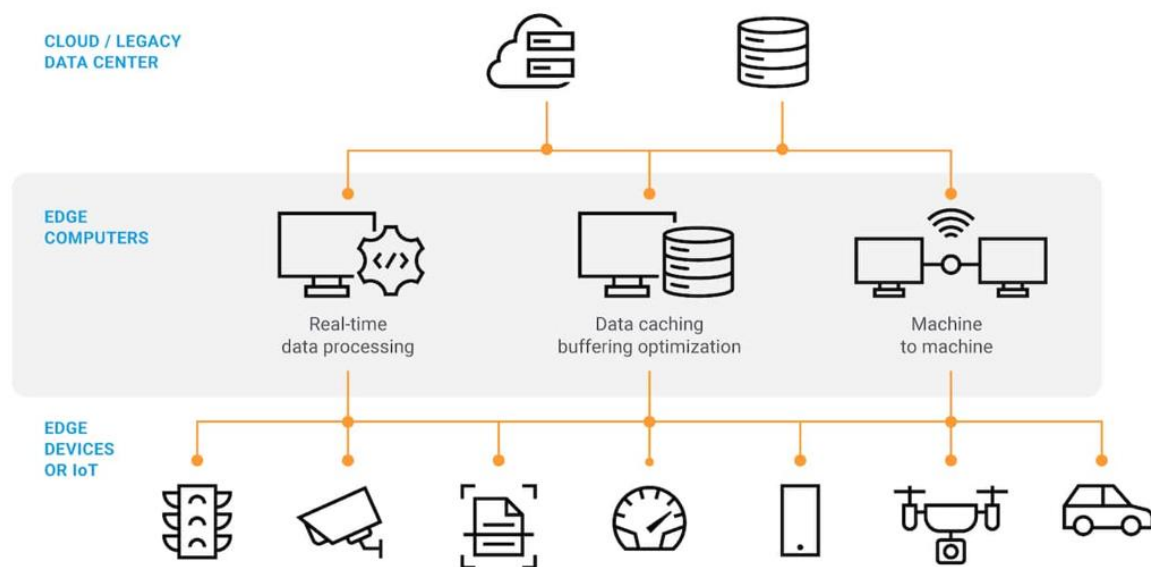
Edge Computing is a distributed computing model where data processing and computation occur near the data source or at the edge of the network, instead of relying entirely on a centralized cloud data center.

The “edge” means devices like routers, gateways, sensors, or local servers located close to the users or data generators.

The goal is to reduce latency, save bandwidth, and enable real-time processing.

Example:

Processing video data locally in a surveillance camera before sending only relevant alerts to the cloud.



Components

Edge Devices

End devices that collect or generate data.

Examples: IoT sensors, smartphones, CCTV cameras, smart meters.

Edge Nodes / Edge Servers

Local computing units that process and store data near the edge.

Examples: Gateway devices, micro data centers.

Edge Gateway

Connects edge devices to the cloud or central data centers.

Handles data filtering, protocol translation, and security.

Edge Network Infrastructure

Includes routers, switches, 5G base stations, and communication links that enable fast and reliable connectivity.

Cloud or Central Data Center

Used for long-term storage, large-scale analytics, and machine learning model updates.

Edge Management Platform

Manages and monitors edge resources, updates, and security policies.

Applications

- **IoT**
- **Healthcare**
- **Autonomous Vehicles**
- **Smart Cities**

Multimedia Wireless Networks**Streaming Audio and Video**

Streaming means playing audio or video content directly from the internet without downloading the entire file.

The data is sent in small packets and played immediately as it arrives.

Types of Streaming:

- Live Streaming – Real-time broadcast (e.g., live classes, sports).
- On-demand Streaming – Pre-recorded content (e.g., Netflix, YouTube).

Voice Over Internet Protocol (VoIP)

VoIP is a technology that allows voice communication over IP networks (like the Internet) instead of traditional telephone lines.

It converts voice signals into digital packets, sends them through IP, and converts them back into sound at the destination.

Two protocols have been designed to handle this type of communication: **SIP** and **H.323**.

SIP

SIP is an application layer protocol that establishes, manages, and terminates a multimedia session (call).

SIP is a text-based protocol like HTTP. SIP, like HTTP, uses messages. Six messages are

Message	Message Name	Direction
INVITE	From Caller → Callee	Starts a session. It asks the receiver to join a call or conference. Contains details like codec, IP, and port.
ACK	Caller → Callee	Confirms that the callee has received the final response (200 OK). It completes the call setup.
BYE	Either direction	Ends an existing call or session.
CANCEL	Caller → Callee	Cancels a pending call before it's answered.
REGISTER	Client → SIP Server	Registers the user's current location (IP address and username) with the SIP server so others can find them.
OPTIONS	Either direction	Used to check the availability or capabilities of another SIP device (like pinging).

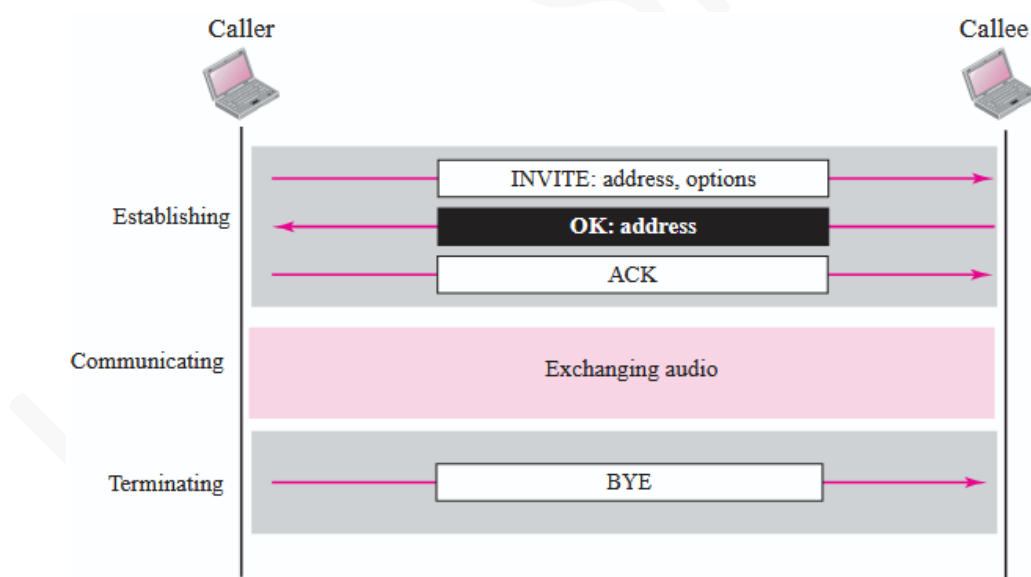


Fig. SIP simple session

H.323

H.323 is a standard designed by ITU to allow telephones on the public telephone network to talk to computers (called terminals in H.323) connected to the Internet.

Protocols Used in Multimedia and VoIP

Real time Transport Protocol(RTP)

- RTP is designed to handle real time traffic on Internet e.g. Audio and Video.
- RTP doesn't have delivery mechanism (port no, multicasting), for which it is used with UDP.
- RTP stands in between Application Program and UDP.

RTP services are

- **Timestamping** –
 - To prevent *jitter*, Jitter means packets don't arrive at the receiver at regular intervals — some come early, some come late.
 - In video calls, it synchronizes **audio with video frames** so sound matches mouth movement.
- **Sequencing** - A sequence number on each packet is required for real-time traffic, to detect packet loss so that ordering can be corrected.
- **Mixing** - Mixing means combining multiple RTP streams (for example, several voices in a conference call) into a single stream before sending it to all participants.

Real-Time Streaming Protocol (RTSP)

- RTSP used for controlling multimedia streaming sessions between a client and a media server.
- It does not transmit the actual data, but controls how it is played.

Functions

- **PLAY**: Start media playback
- **PAUSE**: Temporarily stop playback
- **TEARDOWN**: End the session
- **RECORD**: Save or capture a stream

QUESTION BANK

CHAPTER 1: Internet Architecture and Network Layer

2marks

- What is the difference between the Internet and an Intranet?
- Define the role of an Internet Service Provider (ISP).
- What does ICANN stand for? List out the role of ICANN.
- Describe the role of ICANN in domain name management.
- List any four fields of the IPv4 header.
- Define Hop Limit field purpose in IPv6.
- What is the purpose of ARP in networking?

4marks

- Explain how the structure of the Internet is organized.
- Explain the purpose of the “Time to Live (TTL)” and “Service Type” fields in the IPv4 header.
- Explain the purpose of the “Next Header” and “flow label” fields in the IPv6 header.
- Compare IPv4 and IPv6 header formats with respect to Header Size, Security, Fragmentation and Packet Processing Time.
- An IPv4 packet has a TTL value of 5. Explain what will happen after the packet passes through 6 routers.
- Explain how ARP maps a logical address to a physical address.

6marks

- Draw IPv4 datagram header format and explain its all fields.
- Draw IPv6 packet format and explain its all fields.
- Draw ARP message format and explain its all fields.
- A diskless workstation knows only its MAC address. Explain how RARP helps it to get its IP address.

TS1 - QUESTION BANK

ACN (315321)

2marks

1. What is the difference between the Internet and an Intranet?
2. Define the role of an Internet Service Provider (ISP).
3. What does ICANN stand for? List out the role of ICANN.
4. Describe the role of ICANN in domain name management.
5. List any four fields of the IPv4 header.
6. What is the purpose of RARP in networking?
7. What is the purpose behind dividing an autonomous system into areas in OSPF?
8. List any two intra-domain routing algorithms with example protocols.
9. List out any two types of ICMP error reporting messages.
10. What is Autonomous system?
11. Give an example of an administrative authority for a real-time autonomous system.
12. List out types of links in OSPF.

4marks

1. Explain the purpose of the “Time to Live (TTL)” and “Service Type” fields in the IPv4 header.
2. Explain the purpose of the “Next Header” and “flow label” fields in the IPv6 header.
3. Compare IPv4 and IPv6 header formats with respect to Header Size, Security, Fragmentation and Packet Processing Time.
4. Explain how ARP maps a logical address to a physical address.
5. Explain the message format of ICMP.
6. Describe four types of BGP messages.
7. Explain any four ICMP error reporting messages with examples.

6marks

1. Draw IPv4 datagram header format and explain its all fields.
2. Draw IPv6 packet format and explain its all fields.
3. Describe the process of distance vector routing in detail with suitable examples.
4. Explain the RIP protocol and draw the RIPv2 message format. Describe all the fields present in the message format.
5. Discuss link state routing in depth and explain how OSPF builds the network topology.

TS2 - QUESTION BANK (With Solution)

ACN (315321)

2marks

1. List out features of TCP.
2. List out services of UDP.
3. List out port numbers for HTTP, DNS, SMTP and IMAP.
4. Define World Wide Web.
5. Define Anonymous FTP.
6. Define Static and Dynamic Web documents.
7. What is the Application are for RTP (Real Time Transfer Protocol).
8. What is the main idea behind Software Defined Networking (SDN)?
9. What is Edge computing?

4marks

1. Explain the TCP Three-Way Handshake process used to establish a connection.
2. Explain working of Transport Layer Security (TLS) protocol.
3. Explain Error Control in TCP.
4. Differentiate between POP3 and IMAP.
5. Explain working of TELNET with help of diagram.
6. Explain working of PGP with help of diagram.
7. Explain SDN architecture.

6marks

1. Explain in detail TCP Flow Control with the help of Receiver Window and Sender Window diagrams.
2. Define **cwnd**. Explain congestion control policies in TCP.
3. Describe FTP. Explain communication over control connection and data connection in FTP.
4. List out SMTP commands. Use an example to show how a mail is sent using SMTP commands.
5. Explain NFV architecture with its Applications.

TS2 - QUESTION BANK SOLUTION

ACN (315321)

2marks

1. List out features of TCP.

- Numbering System
- Sequence Number
- Flow Control
- Error Control
- Congestion Control

2. List out services of UDP.

- Process-to-Process Communication
- Message-Oriented Protocol
- Connectionless Services
- Low Overhead
- Fast Service
- No Acknowledgment or Handshaking
- No Flow Control and No Error Control

3. List out port numbers for HTTP, DNS, SMTP and IMAP.

Protocol	Port No
HTTP	80
DNS	53
SMTP	25
IMAP	143

4. Define World Wide Web.

WWW is a repository of information linked together from points all over the world. The WWW has a unique combination of flexibility, portability, and user-friendly features that distinguish it from other services provided by the Internet.

5. Define Anonymous FTP.

Anonymous FTP (File Transfer Protocol) is a method that allows users to access and download files from an FTP server without needing a personal username or password.

Instead of requiring an account, users log in using the username “anonymous” and usually enter their email address (or nothing) as the password.

6. Define Static and Dynamic Web documents.

Static Web Documents

Static web documents are fixed content pages that are pre-stored on the web server and sent to the client exactly as they are.

They do not change unless the developer manually updates them.

Dynamic Web Documents

Dynamic web documents are generated by the web server at runtime based on user requests or data.

They can change depending on inputs, database content, or session details.

7. What is the Application are for RTP (Real Time Transfer Protocol).

- Video Conferencing
- Voice over IP (VoIP)
- Live Streaming
- Online Gaming
- Internet Radio

8. What is the main idea behind Software Defined Networking (SDN)?

The main idea of SDN is to separate the control plane from the data plane in networking devices.

This means that the control logic (controller) is centralized, while switches and routers only forward packets based on the controller's instructions.

9. What is Edge computing?

Edge computing is a distributed computing model that brings computation and data storage closer to the sources of data.

4marks

1. Explain the TCP Three-Way Handshake process used to establish a connection.

TCP Connection Establishment (Three Way Handshaking)

Three-Way Handshaking The connection establishment in TCP is called three- way handshaking.

The three steps in this phase are as follows.

SYN

The client sends the first segment, a SYN segment, in which only the SYN flag is set. This segment is for synchronization of sequence numbers.

SYN + ACK

The server sends the second segment, a SYN +ACK segment, with 2 flag bits set.

ACK

The client sends the third segment. This is just an ACK segment. It guarantees the completion of three way handshaking.

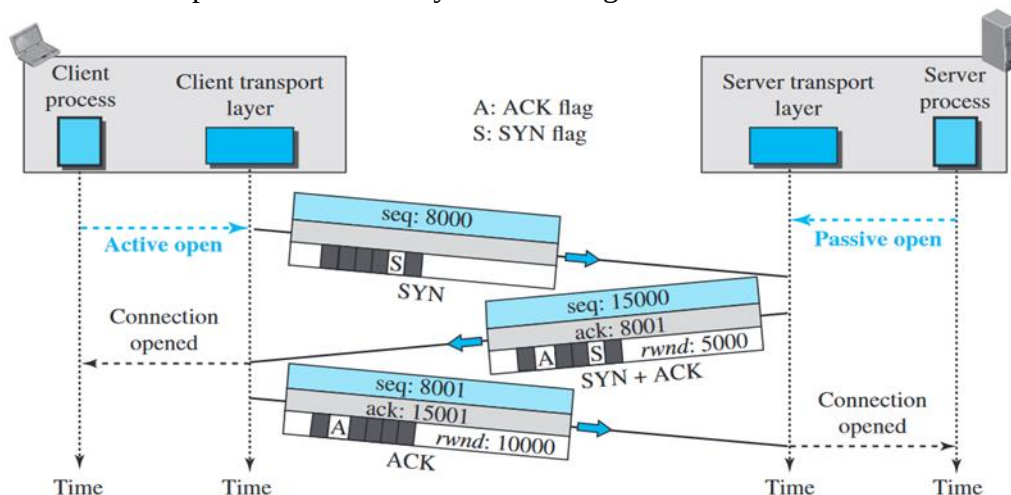


Fig. 4.7 TCP Connection Establishment

2. Explain working of Transport Layer Security (TLS) handshake protocol.

Client Hello:

The client sends a “Hello” message to the server containing supported TLS versions, cipher suites, and a random value.

Server Hello:

The server responds with its chosen TLS version, cipher suite, and its own random value.

Certificate Exchange:

The server sends its digital certificate (issued by a trusted CA) to prove its identity. It contains public key of the server.

Key or Pre Master Secret Exchange:

Client uses Servers public key to Encrypt pre master secret which is used to generate Session Key at both side. We can say that client and server exchanged session key logically.

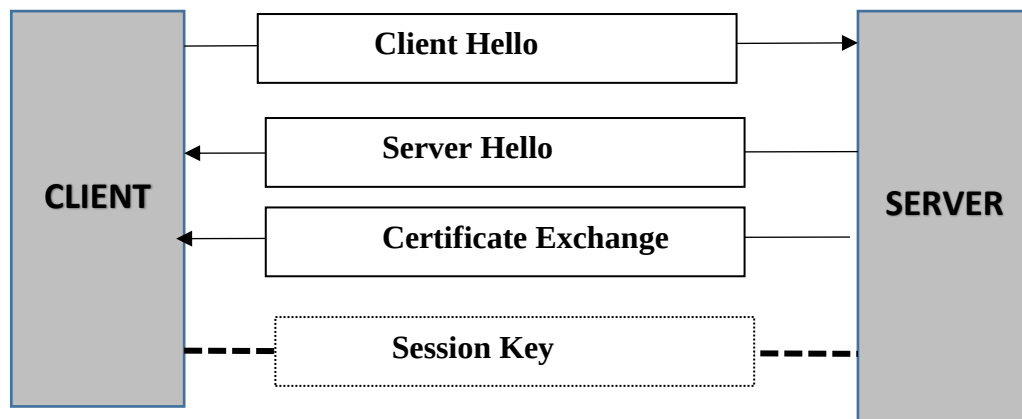


Fig. TLS Handshake Working

3. Explain Error Control in TCP.

Error Control in TCP

TCP provides reliability using error control.

- Error control includes mechanisms for detecting corrupted segments, lost segments, out-of-order segments, and duplicated segments.
- Error control also includes a mechanism for correcting errors after they are detected.

Error detection and correction in TCP is achieved through the use of three simple tools.

checksum

Each segment includes a checksum field which is used to check for a corrupted segment. If the segment is corrupted, it is discarded by the destination TCP and is considered as lost. TCP uses a 16-bit checksum that is mandatory in every segment.

Acknowledgment

TCP uses acknowledgments to confirm the receipt of data segments. Control segments that carry no data but consume a sequence number are also acknowledged. ACK segments are never acknowledged.

Retransmission

In modern implementations, a retransmission occurs

- If the retransmission timer expires **OR**
- Three duplicate ACK segments have arrived.

Retransmission based on Timers

The sender sends segments 1 and 2, which are acknowledged immediately by an ACK. Segment 3, however, is lost. The receiver receives segment 4, which is out of order. The receiver stores the data in the segment in its buffer but leaves a gap to indicate that there is no continuity in the data. The receiver immediately sends an acknowledgment to the sender, displaying the next byte it expects. Note that the receiver stores bytes 801 to 900, but never delivers these bytes to the application until the gap is filled.

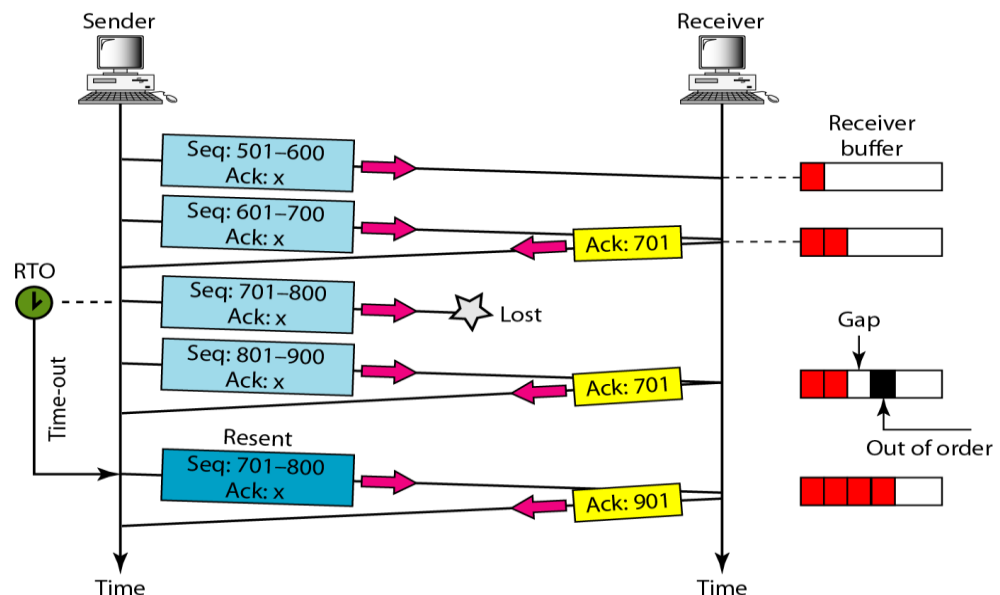


Fig. 4.12 Retransmission based on Timer

We have shown the timer for the earliest outstanding segment. The timer for this definitely runs out because the receiver never sends an acknowledgment for lost or out of-order segments. When the timer matures, the sending TCP resends segment 3, which arrives this time and is acknowledged properly. Note that the value in the second and third acknowledgments differs according to the corresponding rule.

Fast Retransmission based on 3-Duplicate ACK

In this scenario, we want to show the idea of fast retransmission. Our scenario is the same as the second except that the RTO has a higher value. In fast transmission even the timer not matured (expired) still after receiving three ACK segments for segment 3, sender will resend segment 3.

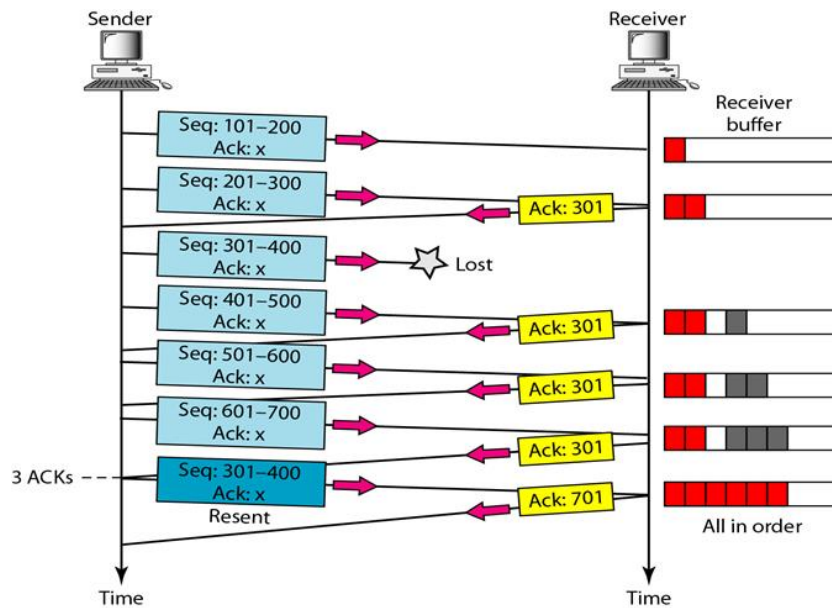


Fig. 4.11 Fast Retransmission

4. Differentiate between POP3 and IMAP.

Parameter	POP3	IMAP
Functionality	POP3 client downloads messages from server and then deleted them from server.	IMAP client Synchronize with server, allows users to access and manage their emails directly on the server
Storage	Less storage required at server.	Server need high storage capacity.
Offline Access	Yes	No
Email Management	No cannot manage mails on server.	Yes, emails can be stored in folders.
Uses	Suitable for Single device use.	Suitable for multiple device use.
Port No	110	143

5. Explain working of TELNET with help of diagram.

- TELNET is an abbreviation for TERminal NETwork. It is the standard TCP/IP protocol for virtual terminal service, which runs at port no 23.
- TELNET enables the establishment of a connection to a remote system in such a way that the local terminal appears to be a terminal at the remote system.
- There are two parties involved TELNET Client and TELNET server.
- TELNET client, which transforms the characters to a universal character set called network virtual terminal (NVT) characters and delivers them to the local TCP/IP protocol stack.

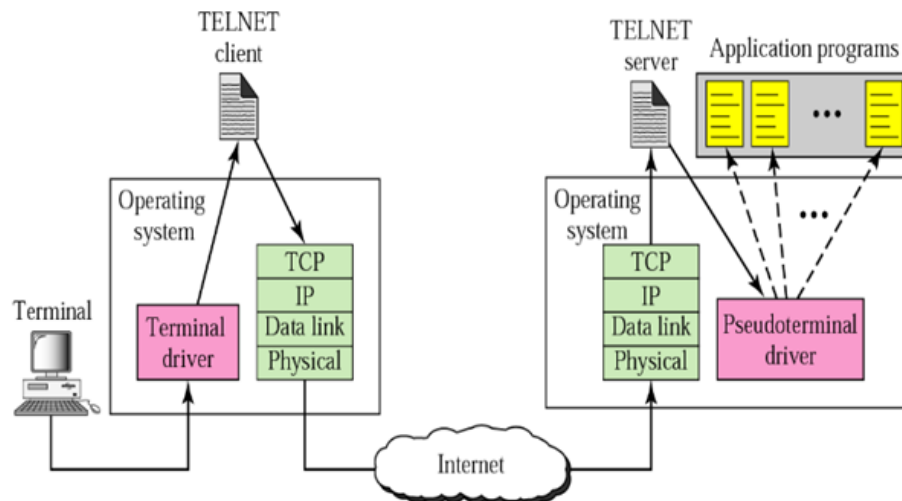


Fig. 5.20 TELNET Working

- The commands or text, in NVT form, travel through the Internet and arrive at the TCP/IP stack at the remote machine. Here the characters are delivered to the operating system and passed to the TELNET server, which changes the characters to the corresponding characters understandable by the remote computer, which is understandable by OS of remote machine.

6. Explain working of PGP with help of diagram.

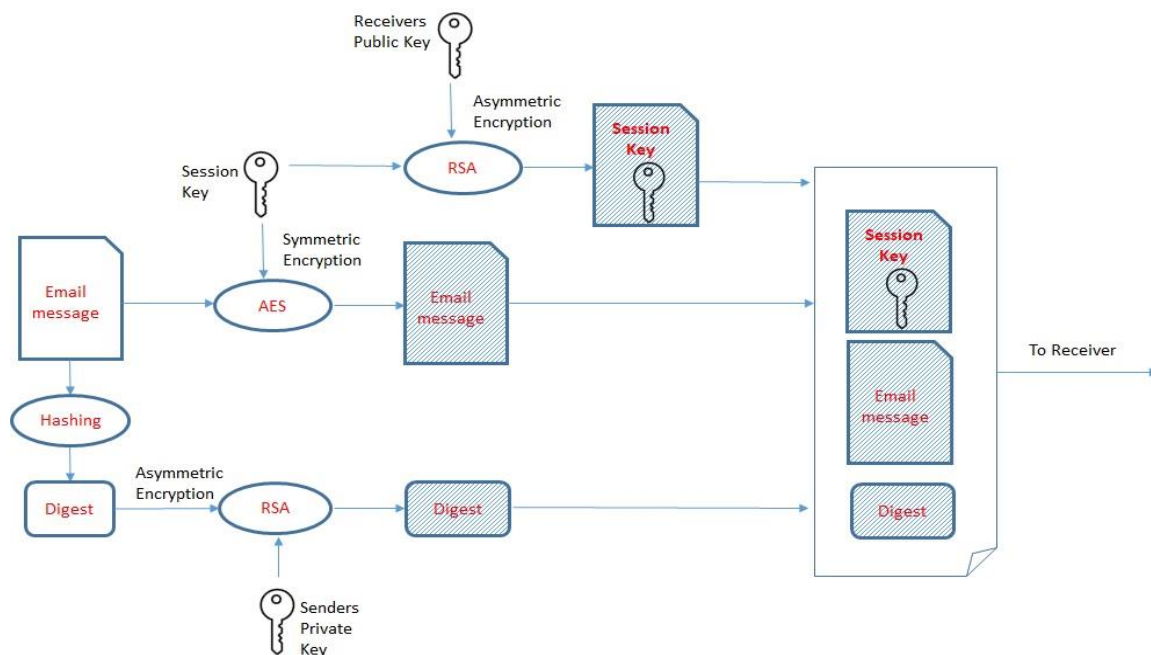
Pretty Good Privacy

Sender Site (Encryption Process):

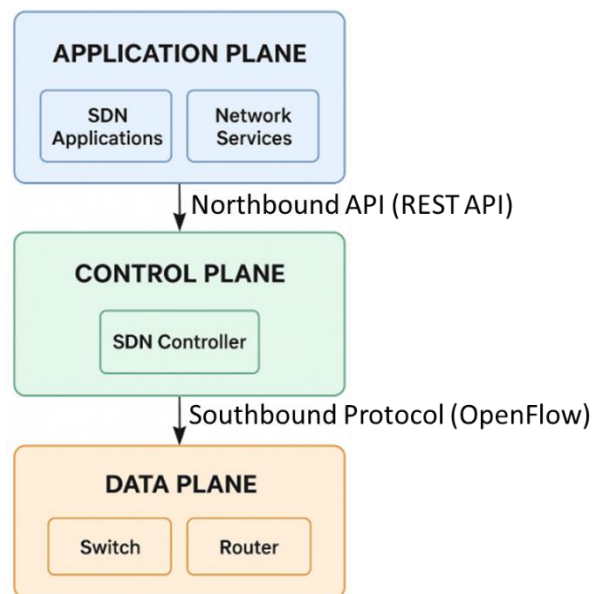
- The sender generates a random symmetric key (also called a session key) to encrypt the email using a fast symmetric encryption algorithm (e.g., AES).
- This session key is then encrypted with the recipient's public key using asymmetric encryption (e.g., RSA).
- Email is hashed and a digest is created, then email digest is encrypted using private key of sender.
- The encrypted session key, Email digest and the encrypted Email are sent to the recipient.

Receiver Site (Decryption Process)

- The recipient first uses their private key to decrypt the session key.
- The decrypted session key is then used to decrypt the actual email message. Recipient hash the email message to create email digest at receiver side.
- The recipient decrypt email digest received from sender using public key of sender, then compare email digest from sender with email digest generated at its own site, if match is found email is accepted otherwise email is discarded.



7. Explain SDN architecture.



SDN works by separating decision-making (control plane) from packet forwarding (data plane).

Applications tell the controller what they want, and the controller tells the switches, routers how to forward traffic, to complete requirements of Application plane

SDN architecture can be divided into three planes.

- Application Plane
- Control Plane
- Data Plane

Application plane

It doesn't directly control hardware — instead, it communicates with the SDN Controller (in the Control Plane) through northbound APIs.

The Application Plane talks to the Control Plane using Northbound APIs (usually REST APIs). These APIs are abstract, so the app developer doesn't need to know hardware details.

Example:

An app might just say:

“Ensure 10 Mbps guaranteed for Host A.”

The controller translates this into OpenFlow rules for the switches.

Control Plane

The control plane is responsible for making decisions about where and how packets should be forwarded.

In SDN, this logic is centralized in a controller (software), rather than being embedded in each network device.

Control plane installs flow rules into switches, routers and decides routing paths.

SDN Controller

A software platform (like ONOS, OpenDaylight, Ryu) that acts as the central decision-maker.

Example

Imagine a new device connects to the network. The switch doesn't know how to handle its packets, so it asks the SDN controller.

The controller decides, “Send these packets to Server A” and sends a flow rule to the switch.

Data Plane

The data plane (also called the forwarding plane) is responsible for moving packets based on rules given by the control plane.

Forwards, drops, or modifies packets according to flow table entries.

Example:

Once the controller installs a rule like

“Packets from Host X to Host Y → output port 3,”

the switch forwards those packets directly without asking the controller

6marks

1. Explain in detail TCP Flow Control with the help of Receiver Window and Sender Window diagrams.

Flow Control in TCP

Flow control in TCP means sender must send data with an amount that can be handled by receiver without overwhelming.

Receive Window

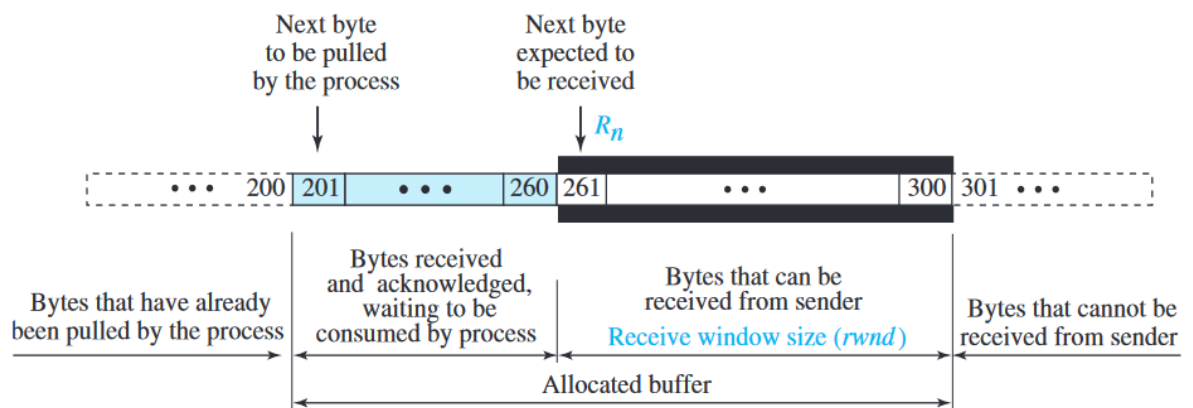


Fig. 4.12 Receive Window

The receiver advertises a window size i.e. **rwnd** (the amount of data receiver is ready to accept) to the sender. This size represents the free space in the receiver's buffer (*can be calculated by subtracting bytes not consumed the process from allocated buffer as shown in fig. 4.12*) and the sender must stay within this limit when transmitting data.

Send window

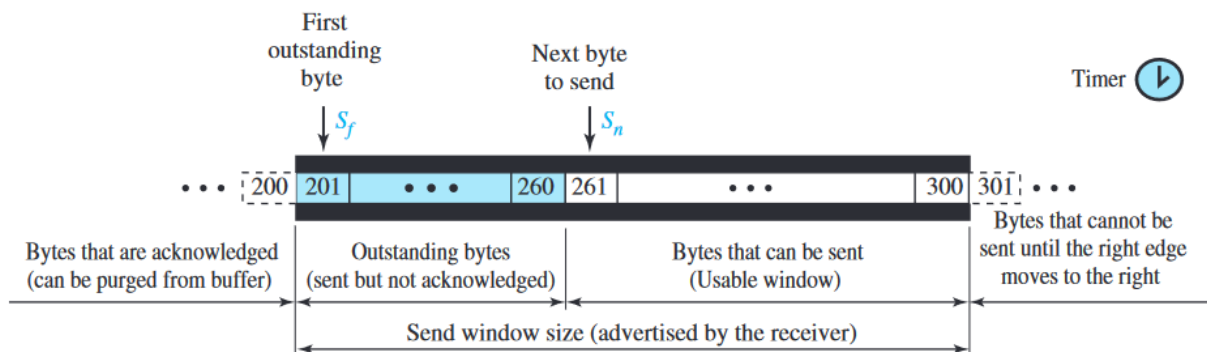


Fig. 4.13 Send Window

The sender adjusts its transmission rate based on the rwnd value received from the receiver.

If the *rwnd* is large, the sender can send more data at once. If it is small, the sender must slow down or stop until more space is available.

The sender keeps track of the amount of data that has been sent but not yet acknowledged (as shown in fig. 4.13) and ensures that this amount stays within the receiver's advertised window.

2. Define *cwnd*. Explain congestion control policies in TCP.

Congestion Window (*cwnd*)

- Use of *rwnd* guarantees that, there is no congestion at the other end, but there may be *congestion in the middle*. An intermediate router may receive data from more than one sender. No matter how large the buffers of a router may be, it may be overwhelmed with data, results congestion at middle.
- To control the number of segments to transmit, TCP uses another variable called a congestion window, *cwnd*, whose size is controlled by the congestion situation in the network (as we will explain shortly).
- The *cwnd* variable and the *rwnd* variable together define the size of the send window in TCP. The first is related to the congestion in the middle (network); the second is related to the congestion at the end. The actual size of the window is the minimum of these two.

Actual window size = minimum (*rwnd*, *cwnd*)

1. Slow Start – Exponential Increase

The Slow Start – Exponential Increase algorithm is based on the idea that the size of the congestion window (*cwnd*) starts with one maximum segment size (MSS), but it increases exponentially as follows.

Slow start: $\rightarrow cwnd = 2^0 = 1$

After Receiving **1** RTT $\rightarrow cwnd = 2^1 = 2$

After Receiving **2** RTT $\rightarrow cwnd = 2^2 = 4$

After Receiving **3** RTT $\rightarrow cwnd = 2^3 = 8$

3. Describe FTP. Explain communication over control connection and data connection in FTP.

File Transfer Protocol

- FTP uses the services of TCP. It needs two TCP connections.
- The well-known port **21** is used for the control connection and the well-known port **20** for the data connection.
- The control connection remains connected during the entire interactive FTP session.
- The data connection is opened and then closed for each file transferred.
- Control connection is there between client & server Control processes.
- Data connection is there between client & server Data transfer processes.

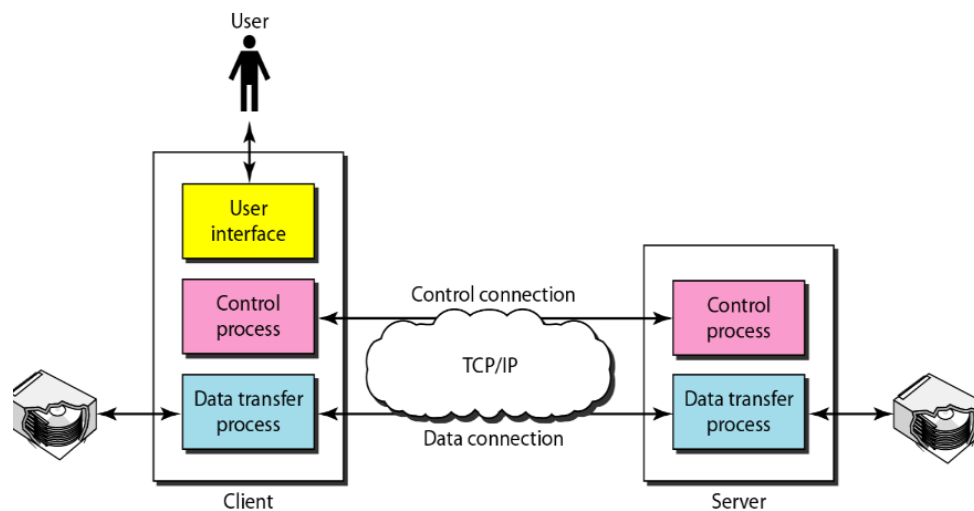


Fig. 5.8 File Transfer Protocol (FTP)

Communication over FTP Control Connection

- It uses the 7-bit ASCII character set (see Figure).
- Communication is achieved through commands and responses.
- Each command or response is only one short line, so we need not worry about file format or file structure.

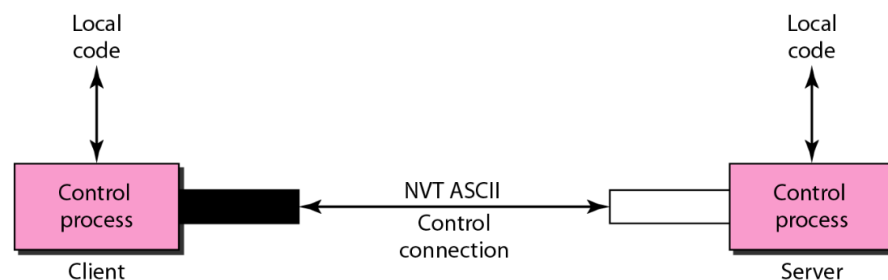
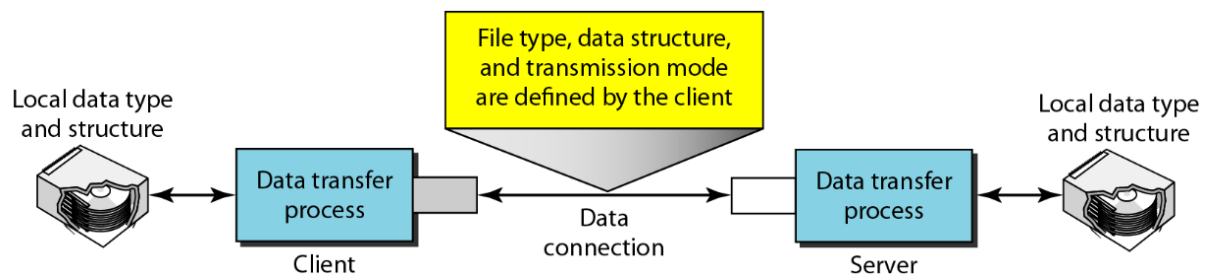


Fig. 5.9 FTP Control Connection

Communication over FTP Data Connection

- The purpose of the data connection is different from that of the control connection. It is used to transfer files.
- One of problem is that file structure at client and server may be differ, known as heterogeneity problem.
- The heterogeneity problem is resolved by defining three attributes of communication: **file type**, **data structure**, and **transmission mode**.



5.9 FTP Data Connection

4: List out SMTP commands. Use an example to show how a mail is sent using SMTP commands.

Command	Meaning / Purpose
HELO or EHLO	Greet the mail server and identify the client.
MAIL FROM:	Specifies the sender's email address.
RCPT TO:	Specifies the recipient's email address.
DATA	Indicates that the actual message (body + headers) will follow.
QUIT	Terminates the SMTP session.

SMTP uses a set of text-based commands between client and server to send emails in a reliable, step-by-step manner.

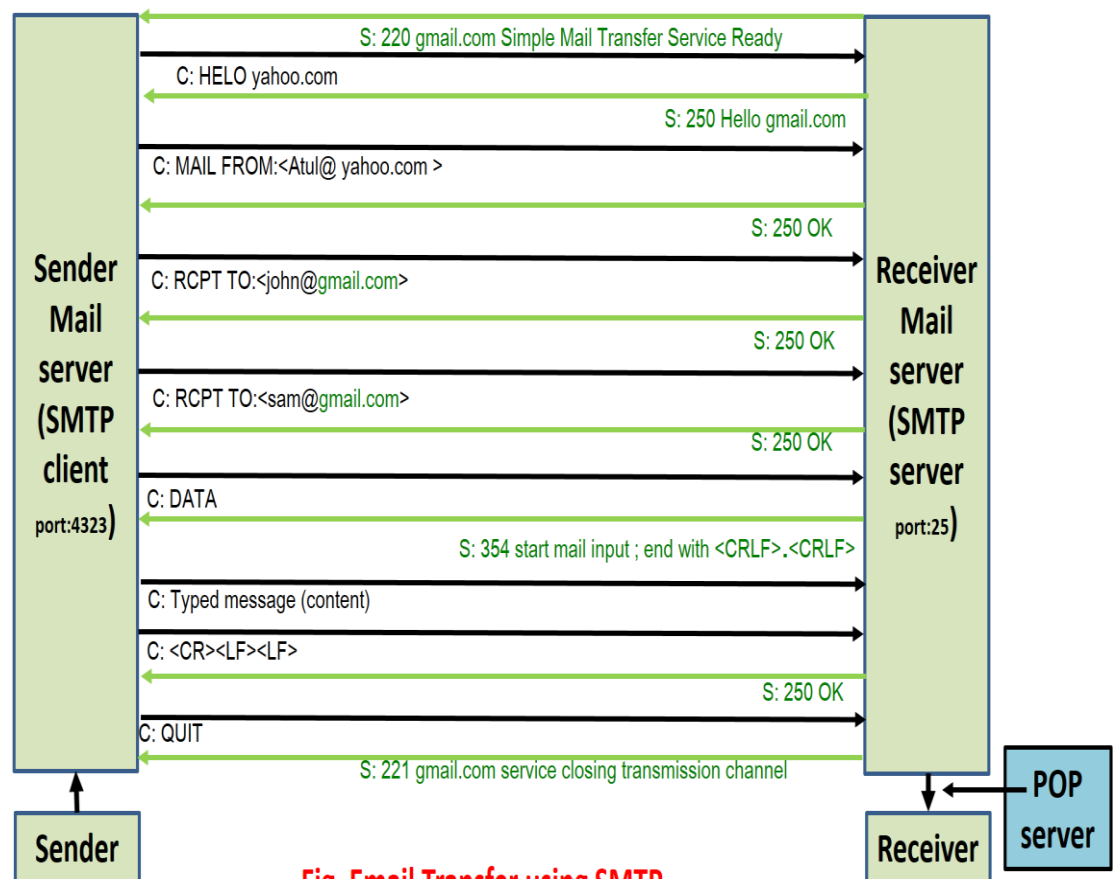


Fig. Email Transfer using SMTP

5. Explain NFV architecture with its Applications.

Network Function Virtualization (NFV) is a modern approach to networking where traditional network services (like firewalls, load balancers, intrusion detection systems, etc.) that used to run on dedicated hardware devices are now implemented as software functions running on standard servers, storage, and switches.

Instead of buying special hardware for each function, NFV lets operators use virtualized network functions (VNFs) that run in a virtual environment.

NFV Architecture:

The NFV architecture is consists of three main components:

i. NFV Infrastructure (NFVI):

This is the foundation layer that provides the hardware and virtualization environment for VNFs.

It includes:

- Compute: Standard servers (CPU, RAM).
- Storage: Hard disks, SSDs.
- Network: Physical network to interconnect servers.

Virtualization Layer: Hypervisors (like VMware, KVM, Xen) or container runtimes (like Docker).

Basically, NFVI supplies the resources that VNFs need to run.

ii. Virtual Network Functions (VNFs):

- These are software-based implementations of traditional network functions.
- Each VNF performs a specific task such as firewalling, routing, NAT, or intrusion detection.
- Example: Virtual Firewall, Virtual Router, Virtual Load Balancer.

iii. NFV Management and Orchestration (MANO):

This is the "brain" of the architecture.

Responsible for:

- Deploying VNFs.
- Scaling VNFs up/down when traffic changes.
- Monitoring performance and failures.
- Managing resources in NFVI.

NFV Management and Orchestration			
Orchestrator (NFVO)	VNF Manager (VNFM)	Infrastructure Manager (VIM)	
NFV Infrastructure (NFVI)			
Compute	Storage	Network	Virtualization Layer
Virtualized Network Functions (VNFs)			
<u>vRouter</u>	<u>vFirewall</u>	<u>vLoad Balancer</u>	<u>vIDS</u>

Fig. NFV Architecture

TS1- ACN – QB- SOLUTION – 2025-26

2marks

1. What is the difference between the Internet and an Intranet?

Internet:

The Internet is a global network that connects millions of computers and devices across the world. It allows users to access websites, communicate, share information, and use online services from anywhere.

Intranet:

An Intranet is a private network used within an organization. It is accessible only to employees or authorized users and is used to share resources, information, and applications securely within the organization.

2. Define the role of an Internet Service Provider (ISP).

- Provides internet access to users and businesses.
- Assigns public IP addresses.
- Offers services like email, domain hosting, and technical support.

3. What does ICANN stand for? List out the role of ICANN.

ICANN stands for Internet Corporation for Assigned Name and Numbers.

Roles:

- Manages domain names, IP address allocation, and protocol identifiers.
- Ensures uniqueness and coordination of DNS across the internet.

4. List any four fields of the IPv4 header.

- VER (4 bits):
- HLEN (4 bits):
- Total Length (16 bits):
- Service Type (8 bits):

5. What is the purpose of RARP in networking?

RARP is a network layer protocol used to map a physical address (MAC address) to a logical address (IP address). RARP is the reverse of ARP.

Mainly used by diskless computers (which don't have storage to save IP addresses).

It helps these computers automatically get their IP address from a RARP server at boot time.

6. What is the purpose behind dividing an autonomous system into areas in OSPF?

Compared with RIP, which is normally used in small ASs, OSPF was designed to be able to handle routing in a small or large autonomous system. However, the formation of shortest-path trees in OSPF requires that all routers flood the whole AS with their LSPs to create the global LSDB.

It may have created a huge volume of traffic in a large AS. To prevent this, the AS needs to be divided into small sections called areas.

7. List any two intra-domain routing algorithms with example protocols.

Algorithm	Protocol
Distance Vector Routing Algorithm	RIP
Link State Routing Algorithm	OSPF

8. List out any two types of ICMP error reporting messages.

ICMP Error Reporting Message

- Destination Unreachable
- Source Quench
- Time Exceeded
- Parameter problem
- Redirection

9. What is Autonomous system?

An autonomous system (AS) is a group of networks and routers under the authority of a single administration.

10. Give an example of an administrative authority for a real-time autonomous system.

An AS is a network or group of networks under a single administration.

Real-world examples include large companies like Google (AS15169), Facebook (AS32934), and Airtel (AS9498 in India).

11. List out types of links in OSPF.

Types of Links

- Point to Point
- Transient
- Stub
- Virtual

4marks

1. Explain the purpose of the “Time to Live (TTL)” and “Service Type” fields in the IPv4 header.

Time to Live (8 bits):

TTL field is used to prevent a packet from traveling forever in a loop. Senders sets a value, which is decremented at each hop. If it reaches zero, packet is discarded.

Service Type (8 bits):

IETF has changed the interpretation and name of this 8-bit field. This field, previously called service type, is now called differentiated services.

- First three bits denote PRECEDENCE (priority) of a packet. Ranging from 0 (000) to 7 (111) higher number have highest priority. Packets having lowest priority discarded first in case of congestion.
- Next 4 bits denotes service type. Description for these 4 bits shown in following table
- Last one bit is not used.

2. Explain the purpose of the “Next Header” and “flow label” fields in the IPv6 header.

Next Header (8-bits):

This field is used to indicate either the type of Extension Header, or if the Extension Header is not present then it indicates the Upper Layer PDU. The values for the type of Upper Layer PDU are same as IPv4's.

Flow Label (20-bits):

This label is used to maintain the sequential flow of the packets belonging to a communication. The source labels the sequence to help the router identify that a particular packet belongs to a specific flow of information. This field helps avoid re-ordering of data packets. It is designed for streaming/real-time media.

Example:

If a video conferencing app sends packets from one user to another, it can assign a flow label so that all related packets are routed through the same path, ensuring smooth, uninterrupted communication.

3. Compare IPv4 and IPv6 header formats with respect to Header Size, Security, Fragmentation and Packet Processing Time.

Parameter	IPv4	IPv6
Header Size	20 to 60 bytes (variable size).	Fixed size base header of 40 bytes.

Security	Security features are optional (IPSec is supported but not mandatory)	IPSec support is mandatory, ensuring better security by default
Fragmentation	Performed by both sender and routers	Performed only by the sender; routers do not fragment packets
Packet Processing Time	More time required due to optional fields, header length variations, and checksum calculations	Faster processing because of simplified header with flow label and no checksum field

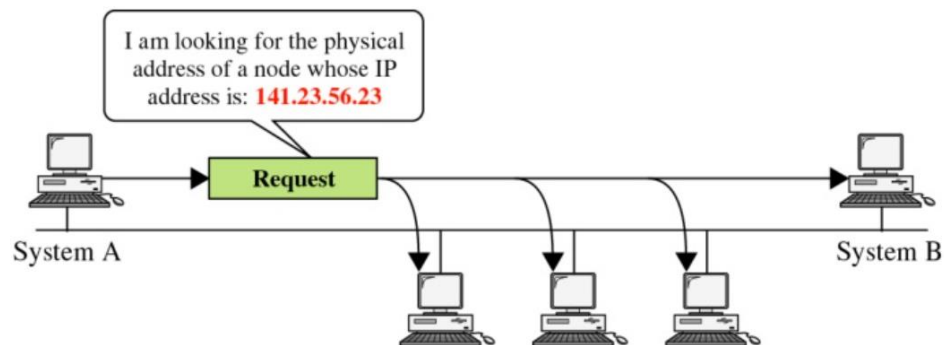
4. Explain how ARP maps a logical address to a physical address.

Working

1. ARP Query Packet Broadcast

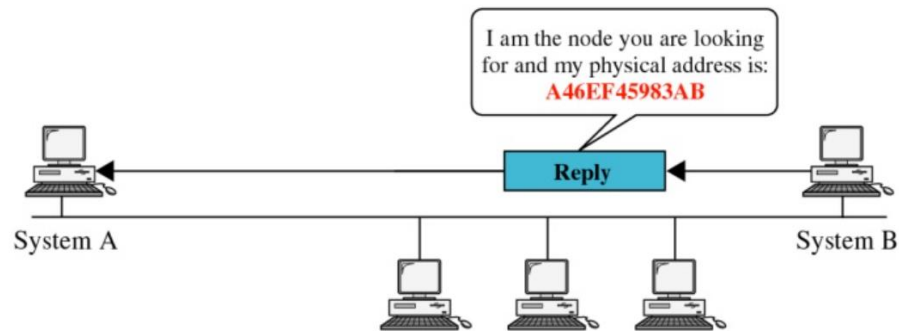
ARP Query Packet contains

- Senders IP Address
- Senders Physical Address
- Receivers IP Address (Whom Physical Address to be found)



ARP Request (Broadcast)

- Every host or router on the network receives and examines the ARP request packet. However, only the device with the matching IP address recognizes the request and replies by sending an ARP response packet.
- ARP Response Packet Unicast
The response packet includes the IP address and physical (MAC) address of the replying device. This packet is sent directly (unicast) to the original sender using the physical address provided in the ARP request.



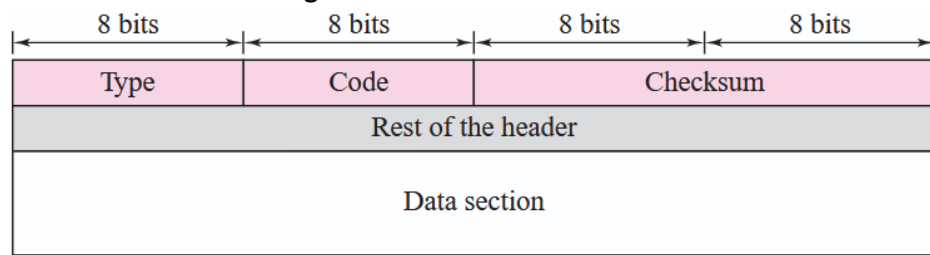
ARP Response (Unicast)

5. Explain the message format of ICMP.

- IPv4 has no mechanism for error reporting or error correcting.
- IPv4 protocol also lacks a mechanism for host and management queries.

The Internet Control Message Protocol version 4 (ICMPv4) has been designed to compensate for the above two deficiencies. It is a companion to the IP protocol.

General Format for ICMP messages



General format for ICMP message

The first field, ICMP **type**, defines the type of the message.

- e.g.
- Destination Unreachable – 3
 - Source Quench – 4
 - Time Exceed – 11
 - Parameter Problem – 12
 - Redirect – 5

Code field provides further details about the type of message. For example, in the “Destination Unreachable” message, the code indicates the reason for the failure.

The last common field is the **checksum** field, used to detect if any error in ICMP packet.

The **rest of the header** is specific for each message type.

Data field contains additional information like part of the original packet that caused the error or diagnostic data.

6. Describe four types of BGP messages.

Types of BGP packets/message

Open Message

To create a neighbourhood relationship, a router running BGP opens a TCP connection with a neighbour and sends an open message.

Update Message

The update message is the heart of the BGP protocol. It is used by a router to withdraw destinations that have been advertised previously, to announce a route to a new destination, or both. Note that BGP can withdraw several destinations that were advertised before, but it can only advertise one new destination (or multiple destinations with the same path attributes) in a single update message.

Keepalive Message

The BGP peers that are running exchange keepalive messages regularly (before their hold time expires) to tell each other that they are alive.

Notification

A notification message is sent by a router whenever an error condition is detected or a router wants to close the session.

7. Explain any four ICMP error reporting messages with examples.

ICMP Error Reporting Message

Destination Unreachable (type 3)

When a router cannot route a datagram or a host cannot deliver a datagram, the datagram is discarded and the router or the host sends a destination-unreachable message back to the source host that initiated the datagram. Note that destination unreachable messages can be created by either a router or the destination host.

Source Quench (type 4)

A source-quench message informs the source that a datagram has been discarded due to congestion in a router or the destination host. The source must slow down the sending of datagram's until the congestion is relieved.

Time Exceeded (type 11)

Whenever a router decrements a datagram with a time-to-live value to zero, it discards the datagram and sends a time-exceeded message to the original source. When the final destination does not receive all of the fragments in a set time, it discards the received fragments and sends a time-exceeded message to the original source.

Parameter Problem (type 12)

Any ambiguity in the header part of a datagram can create serious problems as the data-gram travels through the Internet. If a router or the destination host discovers an ambiguous or missing value in any field of the datagram, it discards the datagram and sends a parameter-problem message back to the source.

6 marks

1. Draw IPv4 datagram header format and explain its all fields.

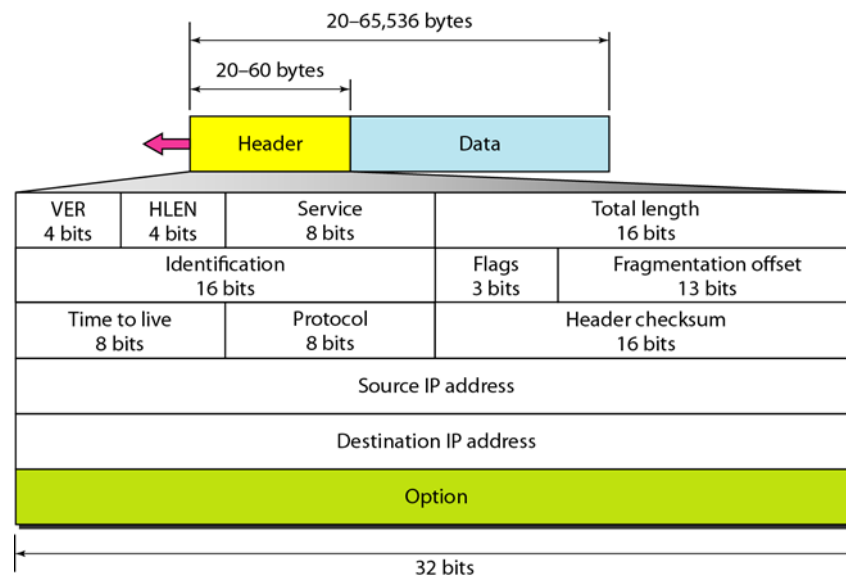


Fig. IPv4 Datagram Header

- Maximum size of IPv4 datagram : 65536B
- Maximum size of Header in IPv4: 60B
- Minimum size of Header in IPv4 : 20B

IPv4 DATAGRAM HEADER FIELDS

VER (4 bits):

This 4-bit field defines the version of the IPv4 protocol. Currently the version is 4. However, version 6 (or IPng) may totally replace version 4 in the future.

HLEN (4 bits):

This 4-bit field defines the total length of the datagram header in 4-byte words. This field is needed because the length of the header is variable (between 20 and 60 bytes).

When there are no options, the header length is 20 bytes, and the value of this field is 5

(5 x 4 = 20). When the option field is at its maximum size, the value of this field is 15 (15 x 4 = 60).

Total Length (16 bits):

Length in bytes of the datagram, including headers. Maximum datagram size is 65536 bytes.

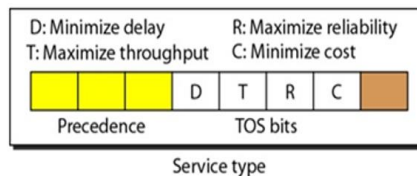
Service Type (8 bits):

IETF has changed the interpretation and name of this 8-bit field. This field, previously called *service type*, is now called *differentiated services*.

SERVICE TYPE:

- First three bits denote PRECEDENCE (priority) of a packet. Ranging from 0 (000) to 7 (111) higher number have highest priority. Packets having lowest priority discarded first in case of congestion.
- Next 4 bits denotes service type. Description for these 4 bits shown in following table
- Last one bit is not used.

TOS Bits	Description
0000	Normal (default)
0001	Minimize cost
0010	Maximize reliability
0100	Maximize throughput
1000	Minimize delay



value of 1111 (this might indicate low delay, high throughput, high reliability and minimized cost)

Time to Live (8 bits):

Prevents a packet from traveling forever in a loop. Senders sets a value, which is decremented at each hop. If it reaches zero, packet is discarded.

Protocol:

Defines the higher level protocol that uses the service of the IP layer

Header Checksum (16 bits):

Datagram is discarded in case of mismatch in checksum values

Source IP address (32 bits):

Internet address of the sender.

Destination IP address (32 bits):

Internet address of the destination.

Identification, Flags, Fragment Offset:

Used for handling fragmentation. (Discussed in detail later)

Options (variable width):

Can be used to provide more functionality to the IP datagram

2. Draw IPv6 packet format and explain its all fields.

IPV6 Packet Format

- The Ipv6 packet format is shown in figure-1.3. Each packet is composed of a mandatory base header followed by the payload.
- The payload consists of two parts: optional extension headers and data from an upper layer.
- The base header occupies 40 bytes, whereas the extension headers and data from upper layer contain up to 65,535 bytes of information.

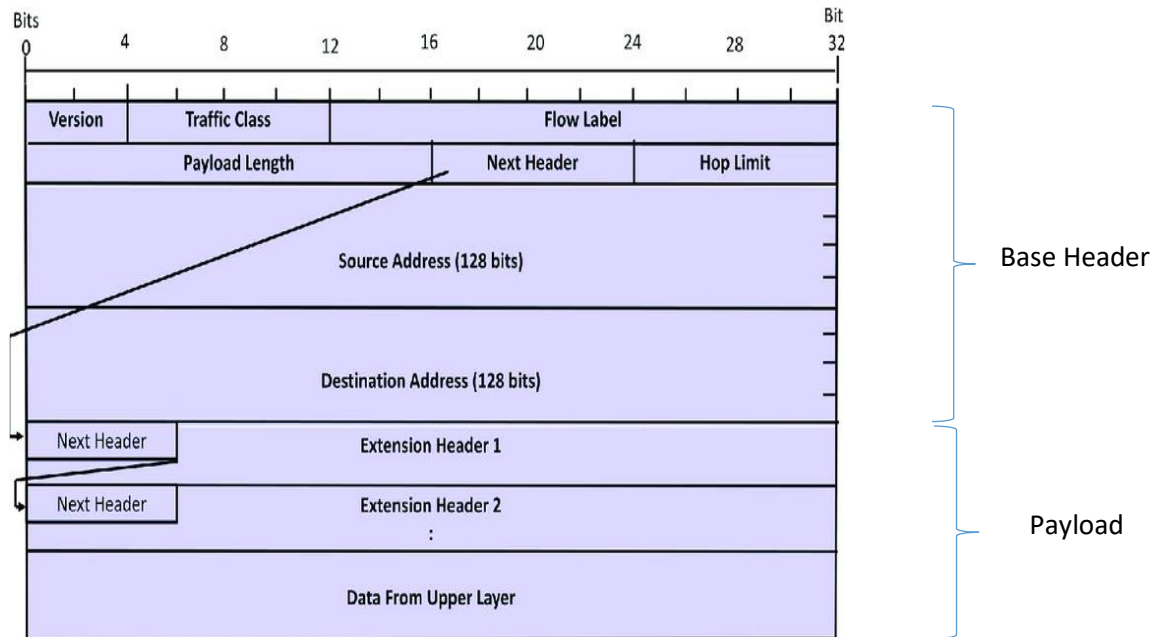


Fig. 1.3 IPv6 Packet Format

IPV6 Header Fields Description

Version (4-bits):

It represents the version of Internet Protocol, i.e. 0110.

Traffic Class (8-bits):

These 8 bits are divided into two parts. The most significant 6 bits are used for Type of Service to let the Router Known what services should be provided to this packet. The least significant 2 bits are used for Explicit Congestion Notification (ECN).

Flow Label (20-bits):

This label is used to maintain the sequential flow of the packets belonging to a communication. The source labels the sequence to help the router identify that a particular packet belongs to a specific flow of information. This field helps avoid re-ordering of data packets. It is designed for streaming/real-time media.

Payload Length (16-bits):

This field is used to tell the routers how much information a particular packet contains in its payload. Payload is composed of Extension Headers and Upper Layer data. With 16 bits, up to 65535 bytes can be indicated; but if the Extension Headers contain Hop-by-Hop Extension Header, then the payload may exceed 65535 bytes and this field is set to 0.

Next Header (8-bits):

This field is used to indicate either the type of Extension Header, or if the Extension Header is not present then it indicates the Upper Layer PDU. The values for the type of Upper Layer PDU are same as IPv4's.

Hop Limit (8-bits):

This field is used to stop packet to loop in the network infinitely. This is same as TTL in IPv4. The value of Hop Limit field is decremented by 1 as it passes a link (router/hop). When the field reaches 0 the packet is discarded.

Source Address (128-bits):

This field indicates the address of originator of the packet.

Destination Address (128-bits):

This field provides the address of intended recipient of the packet.

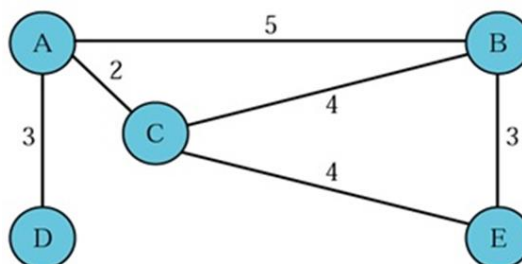
3. Describe the process of distance vector routing in detail with suitable examples.

Distance Vector Routing algorithm (Intra-domain Routing Algorithm)

In distance vector routing, the **least-cost route** between any two nodes is the route with minimum distance.

Steps to build Routing Table for each node.

- Initialization
- Sharing
- Updating



Consider above network in fig 2.2 of nodes (A,B,C,D,E) with cost. We will apply initialization step to this network.

Initialization

At the beginning, each node can know only the distance between itself and its immediate neighbours. Figure shows initial vector table for each node.

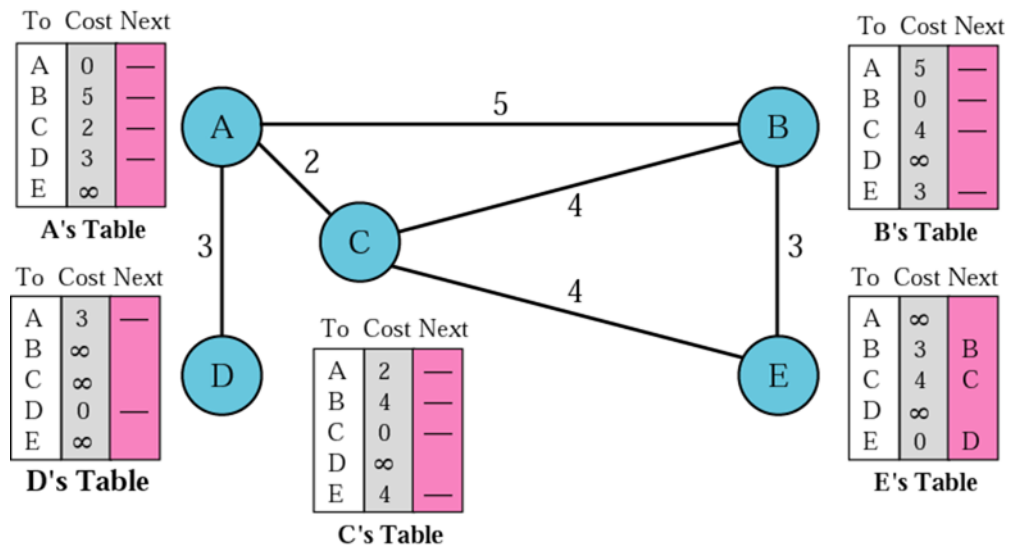


Fig. 2.3 Initial Vector tables for nodes

Sharing

In distance vector routing, each node shares its routing table with its immediate neighbours periodically and when there is a change.

Although node A does not know about node E, node C does. Therefore, if node C shares its routing table with A, node A can also know how to reach node E.

Updating in distance vector routing

Consider node C shares its routing table to node A.

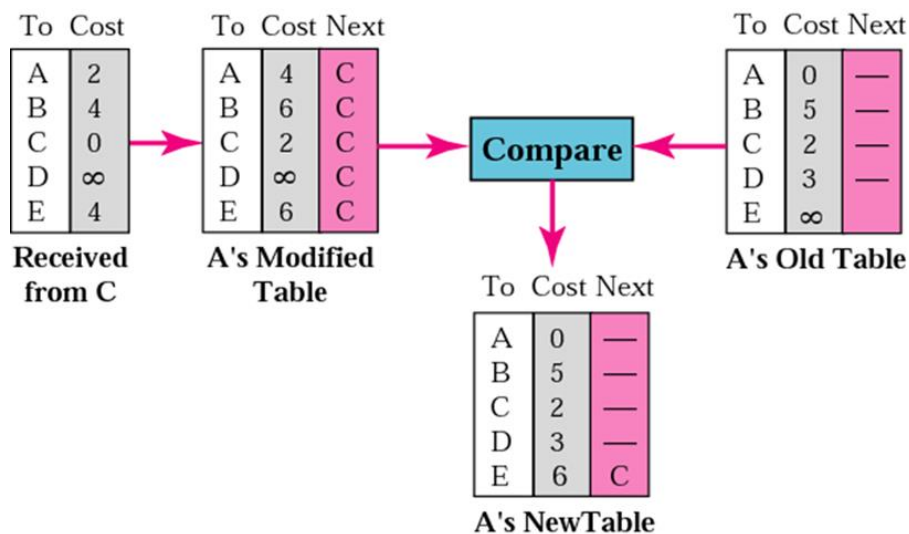


Fig. Example for Updating Vector Table for Node A

4. Explain the RIP protocol and draw the RIPv2 message format. Describe all the fields present in the message format.

RIP message format (RIP v2)

- RIP protocol is based on Distance vector routing algorithm.
- RIP is an Intra-domain routing protocol.
- RIP requires information about directly connected networks. We add this information and activate the RIP routing protocol on routers of the network.
- Once configured and activated, each router sends the routing update out of all active interfaces every 30 seconds.

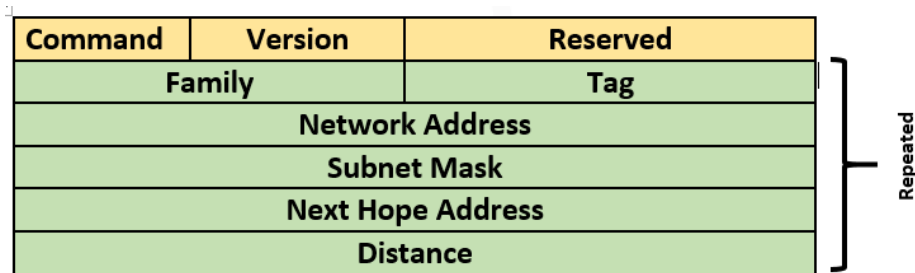


Fig. RIP Packet format version2

- Command (8bit) - type of message: request (1) or response (2).
- Version (8bit) - current Version 2
- Family (16bit) - Family of protocol, for TCP/IP value is 2
- Tag (16bit) - Information about autonomous system
- Network Address (32bit) - Destination address
- Subnet Mask (32bit) – To identify prefix bits
- Next Hop Address (32bit) – Defines via which interface packet can reach destination
- Distance (32bit) - Number of hops to the destination

Now consider an Example to design a RIP message to share a following routing table

Destination Network	Subnet Mask	Next Hop	Distance
192.168.1.0	255.255.255.0	10.0.0.2	1
172.16.0.0	255.255.0.0	10.0.0.3	2
10.1.1.0	255.255.255.128	10.0.0.4	3

Router will use RIP v2 format to create a RIP message as follows and then the designed RIP message is shared with immediate routers.

2	2	0
2		A202
192.168.1.0		
255.255.255.0		
10.0.0.2		
1		
2		A202
172.16.0.0		
255.255.0.0		
10.0.0.3		
2		
2		A202
10.1.1.0		
255.255.255.128		
10.0.0.4		
3		

Fig. Designed RIP message with entries of routing table.

5. Discuss link state routing in depth and explain how OSPF builds the network topology.

LINK STATE ROUTING - Building Routing Tables

In link state routing, four sets of actions are required to ensure that each node has the routing table showing the least-cost node to every other node.

1. Creation of the states of the links by each node, called the link state packet (LSP).
2. Dissemination of LSPs to every other router, called flooding, in an efficient and reliable way.
3. Formation of a shortest path tree for each node.
4. Calculation of a routing table based on the shortest path tree.

First two steps are used to avail a AS's topology to every node.

Creation of LINK STATE PACKET (LSP)

A link state packet can carry a large amount of information. For the moment, however, we assume that it carries a minimum amount of data:
the node identity, the list of links, a sequence number, and age.

The first two, node identity and the list of links, are needed to make the topology.
The third, sequence number, facilitates flooding and distinguishes new LSPs from old ones.
The fourth, age, prevents old LSPs from remaining in the domain for a long time.

Flooding of LSPs (Help to know complete Topology of N/W)

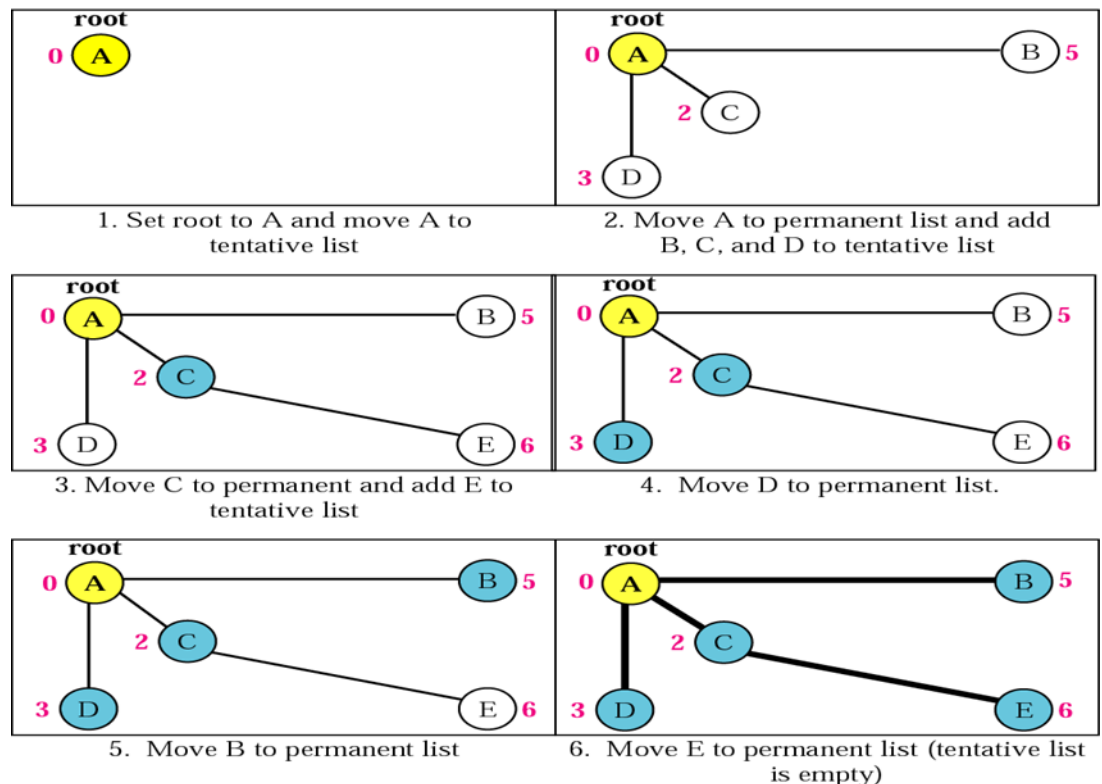
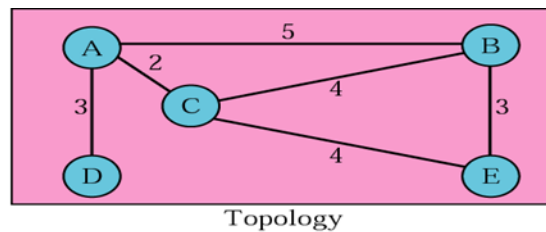
Flooding of LSPs After a node has prepared an LSP, it must be disseminated to all other nodes, not only to its neighbours. The process is called flooding and based on the following:

1. The creating node sends a copy of the LSP out of each interface.
2. A node that receives an LSP compares it with the copy it may already have. If the newly arrived LSP is older than the one it has (found by checking the sequence number), it discards the LSP. If it is newer, the node does the following:
 - a. It discards the old LSP and keeps the new one.
 - b. It sends a copy of it out of each interface except the one from which the packet arrived. This guarantees that flooding stops somewhere in the domain (where a node has only one interface).

Formation of Shortest path tree

A shortest path tree is a tree in which the path between the root and every other node is the shortest. What we need for each node is a shortest path tree with that node as the root.

The Dijkstra algorithm creates a shortest path tree from a graph. The algorithm divides the nodes into two sets: tentative and permanent.



Calculation of Routing Table

<i>Node</i>	<i>Cost</i>	<i>Next Router</i>
A	0	—
B	5	—
C	2	—
D	3	—
E	6	C